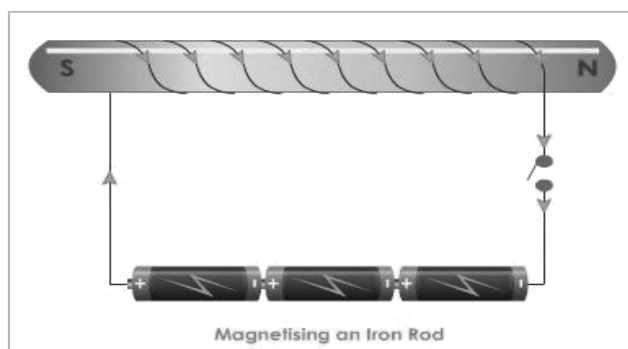


**CHEMICAL REACTIONS AND EQUATIONS**

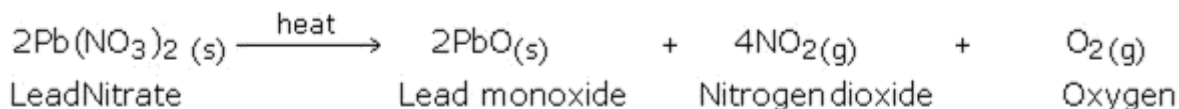
PHYSICAL CHANGE: A physical change is one in which the shape, size, appearance or state, of a substance may Change, but its chemical composition remains same. It is usually a change, which is reversible. By reversing the process, the original substance can be obtained.

Example: An iron rod can be converted into a magnet by passing an electric current through an insulated coil wound around. When the current flows, the iron bar gets magnetized. One end becomes North Pole and the other the South Pole. If the direction of the current is reversed, the North Pole changes to South Pole and the South Pole changes to North Pole.



CHEMICAL CHANGE: A change in which the composition of a substance is altered is called as chemical change. As a result, the original properties get changed and one or more new substances are formed.

Example: Effect of Heat on Lead Nitrate. Lead nitrate is a white, crystalline solid. When heated, it starts decomposing with a crackling sound, producing a reddish brown gas called nitrogen dioxide, and a colourless gas, oxygen. A yellow residue of lead monoxide is left behind in the test tube.

**DIFFERENCE BETWEEN PHYSICAL CHANGE & CHEMICAL CHANGE:**

Physical change	Chemical change
1. It is the change with respect to the shape, size, and appearance of an object.	1. It is the change with respect to the chemical composition of an object.
2. It is a temporary change.	2. It is a permanent change.
3. It is a reversible change.	3. It is an irreversible change.
4. In this change the object retains its identity.	4. In this change an object loses its identity permanently.
5. The chemical properties of an object remains same in this change, only physical properties changes	5. In this change both physical as well as chemical properties of an object changes.

CHEMICAL REACTION:

A chemical reaction is a process that leads to the transformation of one set of chemical substances to another. Classically, chemical reactions encompass changes that only involve the positions of electrons in the forming and breaking of chemical bonds between atoms, with no change to the nuclei (no change to the elements present), and can often be described by a chemical equation. Most reactions require heat, pressure, radiation or the presence of catalysts (accelerators). or Chemical reactions are the processes in which new substances with new properties are formed. e.g

- Formation of rust on a piece of iron with the help if moisture and oxygen present in air.
- Burning of fuels to produce heat and light and other harmful gases.

Historical Concept of Reactions:

During the beginning of the early Middle Ages, European and Persian philosophers had become awestruck with the way some substances seemed to “transmute” (or transform) into others. A lot of simple stones, for example, those containing sulfur, appeared to magically burn. Even the otherwise ordinary minerals got transformed like the ore cinnabar got converted into enchanting silvery liquid metal mercury after getting heated. Alchemists took inspiration from Aristotle’s ideas that claimed everything in the world was made up of four fundamental substances – air, earth, fire, and water.

They even planned and used up generations trying to establish that a lot of less expensive metals like copper and mercury do have the potential to turn into gold. In spite of their misguided approach, a lot of early alchemists performed foundational chemical experiments – transforming one substance into another, and therefore, it is tricky to point to an exact date or event as the birth of the idea of an ordered, quantifiable chemical reaction.

Niceties Of a Chemical Reaction:

1. The substance (or substances) initially involved in a chemical reaction are called *reactants or reagents*. Chemical reactions are usually characterized by a chemical change, and they yield **one or more products**, which usually have properties different from the reactants.
2. Reactions often consist of a sequence of individual sub-steps, the so-called **elementary reactions**, and the information on the precise course of action is part of the **reaction mechanism**.
3. Chemical reactions happen at a **characteristic reaction rate** at a given temperature and chemical concentration. Typically, reaction rates increase with increasing temperature because there is more thermal energy available to reach the activation energy necessary for breaking bonds between atoms.

Characteristics of a Chemical Reaction: In a chemical reaction the substances which take part in a chemical reaction are known as reactants and are converted into new substances known as products. The conversion of reactants into new products in a chemical reaction is often accompanied by some features which can be observed easily. The easily observable features which can be observed easily are known as characteristics of chemical reaction and are as:

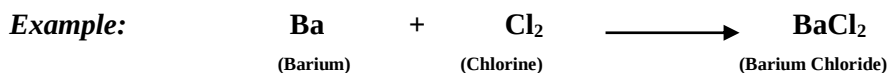
1. **Physical Change or Chemical Change:** One way to think about physical changes is that they are reversible since they do not affect a substance’s chemical identity. These include changes in size, shape, texture or state. A good example of physical change is ice melting into water. This change of state is a physical change since scooping up the water and returning it to the freezer reverses it. Chemical changes, on the other hand, can’t reverse since they involve breaking bonds between molecules and forming new bonds.
2. **Changes in Temperature:** While you can heat any substance, the spontaneous release of heat after mixing compounds together is a good indicator that a chemical reaction has taken place. Reactions that release heat are also called exothermic reactions. An example of a chemical change that releases heat is the thermite reaction, which is what causes fireworks to explode.
3. **Changes in Colour:** A spontaneous colour change is another a good clue that a chemical change has occurred. For example, products made of iron can develop rust, turning them from grey to reddish brown in colour. This colour change happens without human intervention as part of a chemical change called an oxidation reaction.
4. **A Marked Odour:** Any noticeable odour that occurs after a reaction begins (as opposed to the odours normally present for each compound) is a sign of a chemical reaction. For instance, a rotten egg gets its terrible smell from the irreversible decomposition reaction that occurs as it spoils.
5. **Formation of a Precipitate:** Sometimes when a chemical change occurs in a solution, a solid forms in that solution. This solid is a precipitate and is one of the most obvious ways to witness a chemical change since it can be very dramatic. A common household example is the appearance of soap scum, which is actually a precipitate that forms when minerals in hard water react with soap molecules.
6. **Release of Bubbles:** Bubbling that occurs during a reaction is an excellent sign that a chemical change is happening. Anyone who has made a volcano model erupt using vinegar and baking soda has seen this chemical change in action. The bubbles of carbon dioxide produced by mixing these common ingredients are actually the result of two reactions: an acid-base reaction followed by a decomposition reaction.

While telling the difference between a physical change and a chemical change can seem tricky at first, knowing the signs that indicate a chemical reaction has taken place can make this important scientific concept much more straightforward.

Chemical Equations:

Chemists use a shorthand notation called a chemical equation to indicate the reactants and the products. In this notation, the reactants are listed on the left side, the products are listed on the right side, and the reactants and products are separated by an arrow showing which direction the reaction proceeds. While many chemical equations show reactants forming products, in reality, the chemical reaction often proceeds in the other direction, too. In a chemical reaction and a chemical equation, no new atoms are created or lost (conservation of mass), but chemical bonds may be broken and formed between different atoms.

Chemical equations may be either unbalanced or balanced. An unbalanced chemical equation doesn't account for conservation of mass, but it's often a good starting point because it lists the products and reactants and the direction of the chemical reaction.

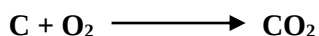


Writing of a chemical equation / Anatomy of a Chemical Equation: Chemical equations have always been associated with chemical reactions as they are basically the shorthand with the help of which chemical reactions are described. This simple fact alone makes equations really important. But we need to consider that equations play a critical role in unfolding the quantitative aspect of chemistry, something which is officially called stoichiometry.

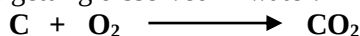
All chemical reactions take the help of the same, fundamental format. Therefore, the starting substances, or reactants, are listed with the help of their chemical formula to the left-hand side of an arrow, with multiple reactants separated with plus signs. In the example of a reaction between carbon and oxygen:



To the right hand of the arrow, we can find the chemical formulas of the new substance or substances (also called products) that are formed by the chemical reaction. In this case, carbon dioxide is the outcome of burning carbon in the presence of oxygen:



We know that a lot of reactions can result in both physical as well as chemical changes. But note that every substance is expressed as a state symbol written as a subscript to the right of the formula, which explains the physical form of the reactants and products. The most common state abbreviations are (s) for solids, (l) for liquids, (g) for gases and (aq) for any aqueous substances, i.e., those getting dissolved in water.



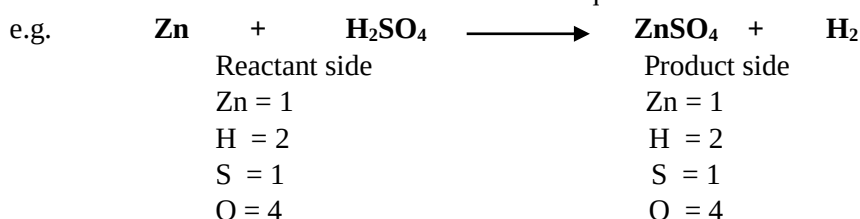
While writing a chemical equation following steps are to be followed.

- The symbols and formulae of reactants are to be written on left side of arrow with a (+) sign in between them.
- An arrow sign is to be put after the reactants to show the direction of chemical reaction pointing towards the right side.

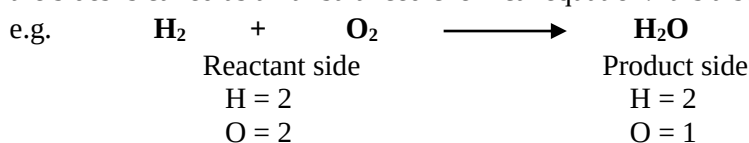
The symbols and formulae of products are to be written on the right side of the arrow.

Balanced and Unbalanced chemical equations:

- Balanced Chemical Equation:** A chemical equation which has equal number of atoms of different elements on both the sides is called a balanced chemical equation.



- Unbalanced Chemical Equation:** A chemical equation which does not have equal number of atoms on both the sides is called as an unbalanced chemical equation. It is also known as skeletal equation.



For ensuring that this illustration is as per the law of conservation of mass, the equation would need to be balanced by the addition of numbers in front of each species that create equal numbers of atoms of each element on each side of the equation. In a situation wherein there's the formation of carbon dioxide from carbon and oxygen, there is absolutely no need for the addition of such numbers (called the stoichiometric coefficients).

Balancing of a Chemical Equations: According to the Law of Conservation of Mass the number of atoms of all the participating elements should remain proportionately constant before and after the reaction.

The 8 Steps to Balancing Chemical Equations

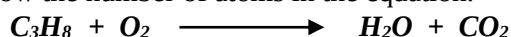
Step 1: Write down your given equation. For this example, you will use: Example Equation:



Reaction Occurs: Propane (C_3H_8) is burned in the presence of oxygen (O_2) to produce water (H_2O) and carbon dioxide (CO_2)

Step 2: Write down the number of atoms that you have on each side of the equation.

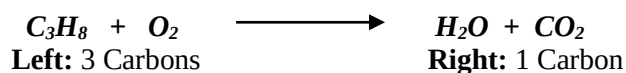
Subscripts next to each atom show the number of atoms in the equation.



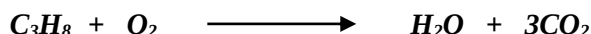
Left side: 3 carbon 8 hydrogen 2 oxygen

Right side: 1 carbon 2 hydrogen 3 oxygen

Step 3: Find the elements that you will be starting with. Always leave hydrogen and oxygen for last. This means that you will need to balance the carbon atoms first.

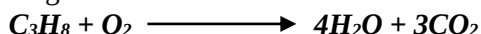


Step 4: Add a coefficient to the one carbon on the right of the equation. This balances it with the 3 carbon atoms on the left of the equation.



The coefficient 3 in front of carbon on the right side indicates 3 carbon atoms just as the subscript 3 on the left side indicates 3 carbon atoms.

Step 5: Begin balancing the Hydrogen atoms next. 8 are on the left side. This means 8 are needed on the right side.

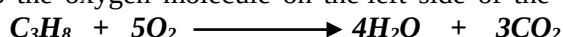


Left: 8 Hydrogen

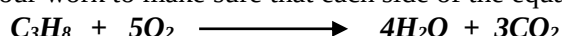
Right: 2 Hydrogen On the right side, 4 is added as the coefficient because the subscript showed that you already had 2 hydrogen atoms. When the coefficient 4 is multiplied by the subscript 2, 8 is the outcome.

Step 6: The other 6 atoms of Oxygen come from 3CO subscript (3CO₂ 3 × 2 = 6 Oxygen atoms) + 4 other Oxygen atoms of water molecules = 10 for the outcome

Step 7: Now balance the Oxygen atoms Since coefficients have been added to the molecules on the right side of the equation, the number of oxygen atoms has changed. Now there are 4 oxygen atoms in the water molecule. There are also 6 oxygen atoms in the carbon dioxide molecule. Which gives a total of 10 oxygen atoms. Add a coefficient of 5 to the oxygen molecule on the left side of the equation. Now the oxygen atoms are balanced



Step 8: Check all of your work to make sure that each side of the equations is balanced

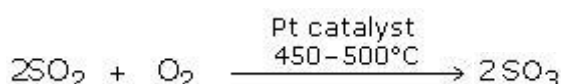


Limitation of a chemical equation:

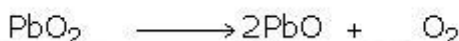
- i. It does not mention the state of the substances. Accordingly, the following symbols should be added to make it informative: (s) for solid, (l) for liquid, (aqueous) if the reactant or product is present as a solution in water, (g) for gas and (vapour) for vapour.



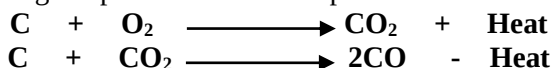
- ii. The reaction may or may not be complete. An equation does not reveal this.
- iii. It does not give any information regarding the speed of the reaction.
- iv. It does not give the concentration of the substances. In some cases, terms like diluted (dilute) and concentrated (concentrated) may be added.
- v. It does not give the conditions of temperature, pressure, catalyst, etc. This is overcome by mentioning these above or below the arrow e.g.



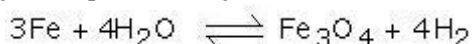
- vi. It does not give any idea about colour changes, which has to be mentioned separately.



- vii. It does not give any indication regarding the production or absorption of heat. This is mentioned separately.



Some reactions are reversible. They are represented by $\rightleftharpoons$ or $\leftrightarrow$



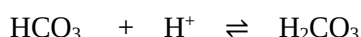
Reversibility Irreversibility and Equilibrium:

Most of the chemical reactions simply run in one direction till the time the reactants get used up. These reactions are **irreversible**. When it comes to other reactions, they are classified as reversible. Reversible reactions can travel in both the forward and backward directions.

($\leftrightarrow$)

Talking about reversible reactions, reactants get turned into products, but products can also turn back into reactants. In fact, it is said that both the forward reaction and its opposite take place at the same time. This circle of moving back and forth continues until a certain relative balance has been created between reactants and products are known as a state called equilibrium. And in equilibrium, all the forward and backward reactions still keep happening, and also the relative concentrations of products and reactants do not change. In a situation where a reaction can be classified as reversible, it is typically expressed with paired forward and backward arrows in order to show it can move both ways.

For instance, when it comes to human blood, excess hydrogen ions (H⁺) attach to bicarbonate ions (HCO₃⁻), eventually forming carbonic acid H₂CO₃:



Of course, this is a reversible reaction, so if carbonic acid gets added to the system, some of it would get turned into bicarbonate and hydrogen ions to restore equilibrium. As for a matter of fact, this buffer system has an important role in keeping your blood pH stable and healthy.

Types of Chemical Reactions:

Combination Reaction (Synthesis Reaction)

Decomposition reaction (Analysis Reaction)

Displacement reaction (Substitution Reaction)

Double Displacement reaction (Metathesis Reaction)

Oxidation & Reduction Reaction. (Redox Reaction)

Acid Base Reaction.

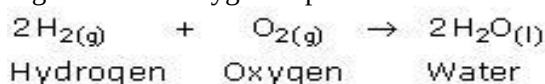
Cumbustion Reaction.

Isomerization Reaction

Hydrolysis Reaction.

(1) **COMBINATION REACTION:** Combination reaction are those reactions in which two or more substances combine to form a single substance. For example:

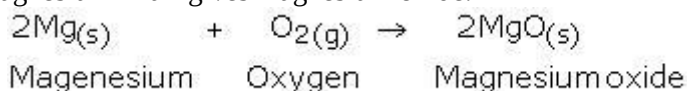
- i. Burning of hydrogen in air or oxygen to produce water.



- ii. Combination of ammonia and hydrogen chloride to produce ammonium chloride.

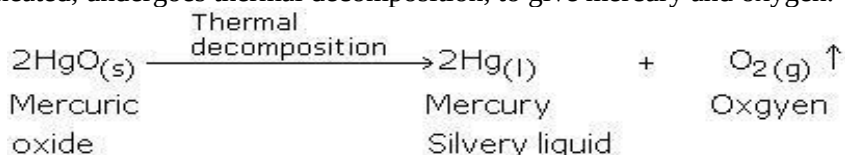


- iii. Burning of magnesium in air gives magnesium oxide.

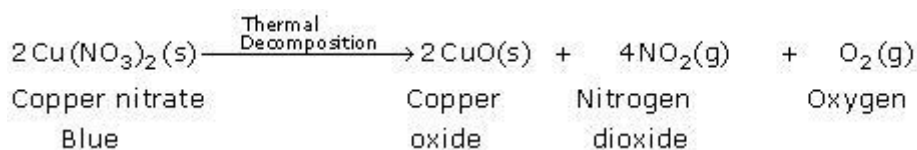


(2) **DECOMPOSITION REACTION:** Decomposition reactions are those reactions in which a compound splits up into two or more simpler substances are known as decomposition reactions. For Example

- (a) Mercuric oxide, when heated, undergoes thermal decomposition, to give mercury and oxygen.



- I. Similarly, if blue crystals of copper nitrate are heated, they undergo thermal decomposition to give black coloured copper oxide, reddish brown fumes of nitrogen dioxide, and a colourless gas of oxygen.

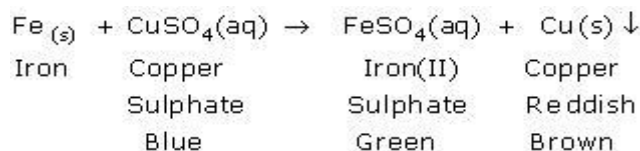


The four most common uses of decomposition reaction are given below.

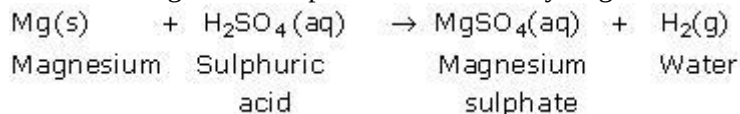
- This reaction is used in the process of extraction of various metals.
- In the stomach, this reaction helps in digestion of food.
- This reaction is used in the manufacture of cement and gypsum.
- This reaction is used in the process of welding.

(3) **DISPLACEMENT REACTIONS:** This reaction is also called as Substitution reaction. Displacement reactions are those reactions in which one element takes the place of another element in a compound are known as displacement reactions.

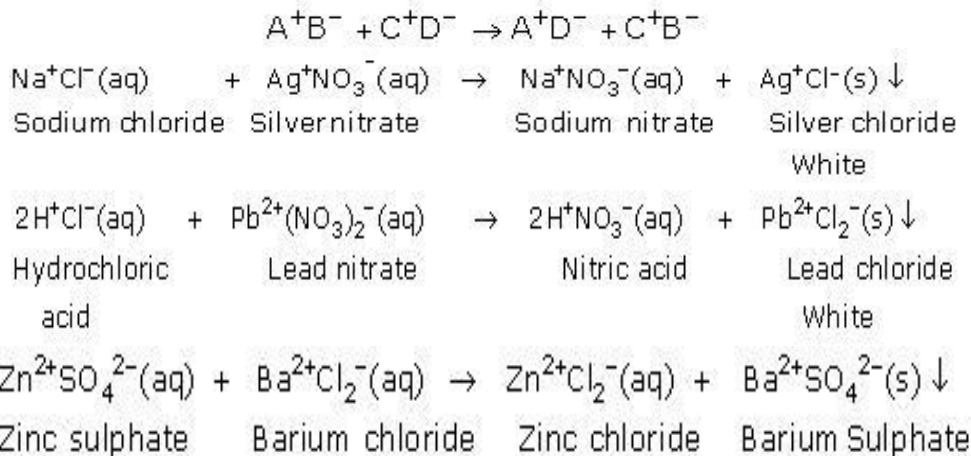
- i. When an iron knife is dipped in an aqueous solution of copper sulphate, a layer of reddish brown precipitate is formed on the iron knife. The iron from the knife being more reactive than copper, displaces it from the copper sulphate solution. The copper so separated gets collected on the surface of iron knife.



- i. When small clean pieces of magnesium are added into dilute sulphuric acid, magnesium sulphate and hydrogen are formed. Atoms of magnesium displace the atoms of hydrogen from sulphuric acid.

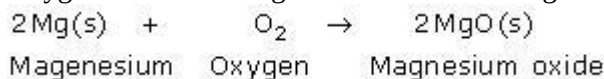


- (4) **DOUBLE DISPLACEMENT REACTION:** This reaction is also called as Metathesis reaction. This is a reaction in which the positive and negative ions of two substances in a solution are mutually interchanged.

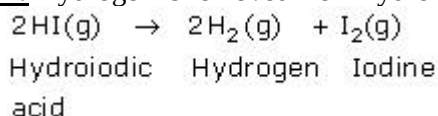


- (5) **OXIDATION - REDUCTION REACTIONS:** "Oxidation is a reaction in which oxygen is added or hydrogen is removed from a substance." Reduction is a reaction in which oxygen is removed from a substance or hydrogen is added to a substance."

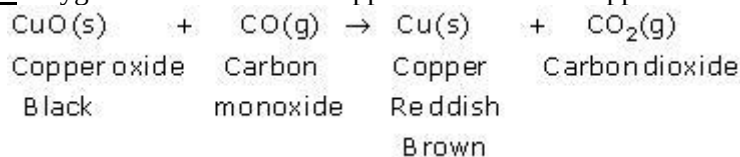
- i. **Addition of Oxygen:** Oxygen adds on to magnesium to become magnesium oxide.



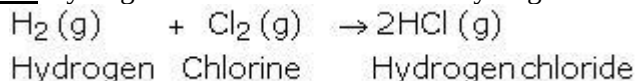
- ii. **Removal of Hydrogen:** Hydrogen is removed from hydroiodic acid to liberate free iodine.



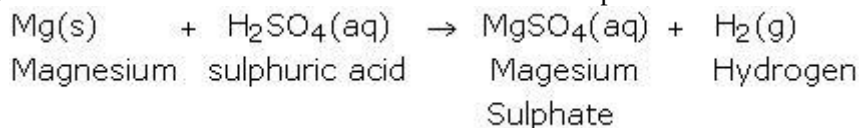
- iii. **Removal of Oxygen:** Oxygen is removed from copper oxide to form copper metal.



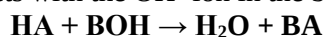
- iv. **Addition of Hydrogen:** Hydrogen adds to chlorine to form hydrogen chloride gas.



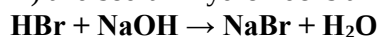
We can also explain oxidation- reduction in terms of 'electrons transfer'. Oxidation and reduction reactions occur simultaneously. In brief, we call these reactions as "redox" reactions. Example of redox reaction



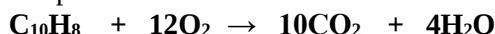
- (6) **ACID-BASE REACTION:** An acid-base reaction is a type of double displacement reaction that occurs between an acid and a base. The H^+ ion in the acid reacts with the OH^- ion in the base to form water and an ionic salt:



The reaction between a hydrobromic acid (HBr) and sodium hydroxide is an example of an acid-base reaction:



- (7) **COMBUSTION REACTION:** A combustion reaction is a type of redox reaction in which a combustible material combines with an oxidizer to form oxidized products and generate heat (exothermic reaction). Usually, in a combustion reaction oxygen combines with another compound to form carbon dioxide and water. An example of a combustion reaction is the burning of naphthalene:

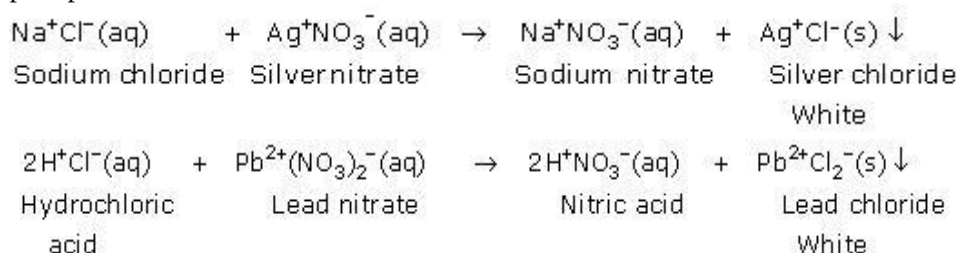


- (8) **ISOMERIZATION REACTION:** In an isomerization reaction, the structural arrangement of a compound is changed but its net atomic composition remains the same.

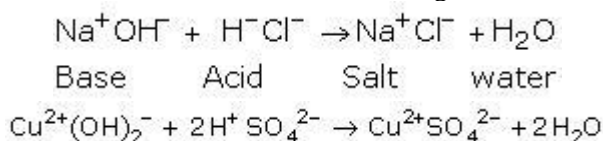
- (9) **HYDROLYSIS REACTION:** A hydrolysis reaction is a chemical process in which a molecule of water is added to a substance. Sometimes this addition causes both substance and water molecule to split into two parts. In such reactions, one fragment of the target molecule (or parent molecule) gains a hydrogen ion. The general form of a hydrolysis reaction is:



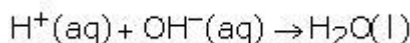
1. **Precipitation reactions:** All the reactions in which a white substance, which is insoluble in water, is formed are called precipitation reactions. This insoluble substance formed is known as a precipitate. A reaction that produces a precipitate is called a precipitation reaction.



2. **Neutralization reactions:** Neutralization is a type of double displacement reaction, in which, the reactants are a base and an acid, and the products are salt and water. The positive charge of the hydrogen ion of the acid, and the negative charge of hydroxyl ions or oxide ions of the base, lose their electrical charge, and become covalent molecule of water.

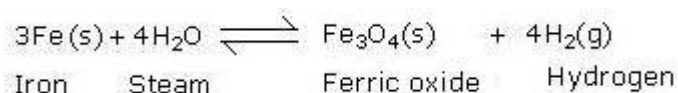


A neutralization reaction is basically a reaction between H^+ and OH^- ions i.e.



3. **Reversible Reactions:** Usually chemical reactions proceed only in one direction. But in some cases the reaction reverses itself.

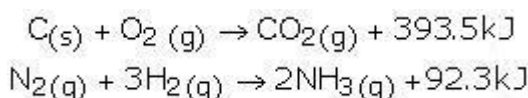
Example, when steam is passed over red hot iron, magnetic oxide of iron and hydrogen gas are formed. If hydrogen gas is passed over heated magnetic oxide of iron, the reverse reaction takes place, producing iron and steam.



Reversibility of a reaction is depicted by double arrow as $\rightleftharpoons$ or $\leftrightarrow$

4. **Exothermic reactions:** Those chemical reactions which are accompanied by the evolution of heat are known as exothermic reactions.

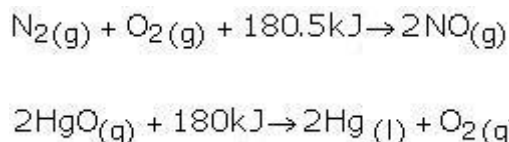
Examples:



The amount of heat (energy) produced is written along with the products. This indicates that heat is given out.

5. **Endothermic Reactions:** The chemical reactions, which proceed with the absorption of heat energy, are called endothermic reactions.

Examples:



The amount of heat absorbed is written along with the reactants since it is absorbed.

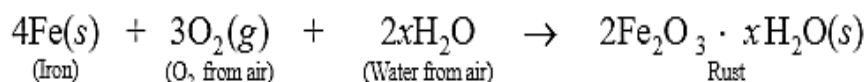
CORROSION OF METALS:

The Natural process of gradual destruction of pure metals by the action of oxygen, moisture or a chemical (such as an acid) on their surface and as a result of chemical reaction or electrochemical reaction between a metal and the surrounding environment to form a stable oxide is called corrosion. When a metal object is kept in damp air for a considerable time, then a powdery substance is formed on its surface. rate of deterioration of metal depends on the environmental conditions, particularly gasses that are in contact with the metal. All small group of metals, called the Noble Metals (rhodium, palladium, silver, platinum, and gold), are much less reactive than others. As a result, they corrode rarely.

The conditions necessary for the corrosion of metals are:

- i. The presence of atmospheric oxygen.
- ii. The presence of moisture (water).
- iii. The presence of a Pure Metal.

Corrosion of Iron Metal: Iron metal corrodes when kept in moist air for a considerable time. When an iron object is kept in moist air for a considerable time, then a red-brown substance called 'rust' ($Fe_2O_3 \cdot xH_2O$) is formed on its surface and the process is called rusting of iron. Rust is soft and porous and it gradually falls off from the surface of the iron object and then the iron below starts corroding. So, corrosion of iron is a continuous process that ultimately eats up the whole iron object. A large amount of metal is lost every year because of corrosion. The more reactive a metal is, the faster it is to corrode it.



Corrosion of Aluminium Metal: It can be observed that the aluminium utensils lose their shine and become dull very soon. This is due to aluminium metal's deterioration when exposed to damp air. When a gleaming aluminium metal is exposed to wet air, the oxygen in the air reacts with the metal to generate a thin, dull film of aluminium oxide all over it.

Corrosion of Copper Metal: It can be observed that the copper objects gradually lose their shine due to the formation of a copper oxide layer on them. When the copper metal is exposed to the atmosphere, copper oxidizes which causes the bright copper surfaces to tarnish. After some time, the tarnish turns dark brown or black, and then green.

Types of Corrosion: There are several types of corrosion and the science and understanding of these processes are constantly evolving. Here is a brief overview of some common types of corrosion:

1. **General Attack Corrosion:** This very common form of corrosion attacks the entire surface of a metal structure. It is caused by chemical or electrochemical reactions. While general attack corrosion can cause a metal to fail, it is also a known and predictable issue. As a result, it is possible to plan for and manage general attack corrosion.
2. **Localized Corrosion:** This corrosion attacks only portions of a metal structure.
3. **Galvanic Corrosion:** This can occur when two different metals are located together in a liquid electrolyte such as salt water. In essence, one metal's molecules are drawn toward the other metal, leading to corrosion in only one of the two metals.
4. **Environmental Cracking:** When environmental conditions are stressful enough, some metal can begin to crack, fatigue, or become brittle and weakened.
5. **Microbial corrosion:** Commonly referred to as microbiologically influenced corrosion (MIC) is caused by microorganisms. It applies to both metallic and non-metallic materials with or without oxygen.

How to Prevent Corrosion of Metals: Following are the various methods by which we can protect a metal from corrosion and they are Painting, Greasing or oiling, alloying, galvanizing, electroplating, etc.

Painting: Rusting of iron can be easily prevented by coating the surface with paint which protects iron from air and moisture,

Greasing or oiling: When some grease or oil is applied to the surface of an iron object, then the air and moisture cannot come in contact with it, and hence rusting is prevented.

Alloying: An alloy is a mixture of two or more metals. Alloying is a process where metals like iron or steel are mixed with a less reactive metal like chromium, magnesium, etc for protection against corrosion and to create non-rusting alloys. For e.g. Brass is an alloy which consists of copper is a cheap and non – reactive alloy. Another example of a non-rusting alloy is stainless steel, a mixture of iron and carbon.

Galvanizing: Coating iron or steel with Zinc to prevent corrosion is known as galvanizing. Dipping iron or steel in a liquid form of zinc and using it as the negative cathode zinc is coated on it, the layer is produced by electrolytic deposition. Zinc oxides or corrodes to create a zinc oxide layer that does not flake off like iron oxide rust.

Electroplating: Electroplating is a process where a metal is coated by electrolytic deposition with chromium, silver, or another metal. This process is generally held at room temperature from aqueous electrolytes. It is one of the most popular and common methods to prevent corrosion.

RANCIDITY

Rancidity is the complete or incomplete oxidation or hydrolysis of fats and oils when exposed to air, light, moisture or by bacterial action, resulting in unpleasant taste and odor. Specifically, it is the hydrolysis or autoxidation of fats, which are objectionable in taste and odor. When these processes occur in food, undesirable odors and flavors can result. In certain cases, however, the flavors can be desirable (as in aged cheeses). In processed meats, these flavors are collectively known as warmed-over flavor. Rancidification can also detract from the nutritional value of food, as some vitamins are sensitive to oxidation. Similar to rancidification, oxidative degradation also occurs in other hydrocarbons. Or we can also say that The process of oxidation of oils or fats in a food, resulting into a bad smell and bad taste is called as rancidity.

Rancidity of oils or fats or any food can be prevented by the following methods

- i. By adding antioxidants to packed food before packing.
- ii. By packing of food items.
- iii. By replacing the nitrogen with air present in the food packets before packing.
- iv. Refrigeration of the foods stuff under the temperature below zero degree celcius.

Some Important Key Words:

Combustion: An oxidation-reduction reaction that occurs so rapidly that noticeable heat and light are produced.

Corrosion: An oxidation-reduction reaction in which a metal is oxidized and oxygen is reduced, usually in the presence of moisture.

Oxidation: A process in which a chemical substance takes on oxygen or loses electrons.

Oxidizing agent: A chemical substance that gives up oxygen or takes on electrons from another substance.

Reducing agent: A chemical substance that takes on oxygen or gives up electrons to another substance.

Reduction: A process in which a chemical substance gives off oxygen or takes on electrons.

A reducing agent (in this case, carbon) is always oxidized.

TEXTUAL QUESTIONS

Q NO 1: Why should a magnesium ribbon be cleaned before it is burnt in air?

ANS: Magnesium is a very reactive metal. When stored, it reacts with oxygen to form a layer of magnesium oxide on its surface. This layer of magnesium oxide is quite stable and prevents further reaction of magnesium with oxygen. The magnesium ribbon is cleaned by sand paper for removing this layer so that the underlying metal can be exposed to air.

Q NO 2: Write the balanced equation for the following chemical reactions.

- (i) Hydrogen + Chlorine $\rightarrow$ Hydrogen chloride
- (ii) Barium chloride + Aluminium sulphate $\rightarrow$ Barium sulphate + Aluminium chloride
- (iii) Sodium + Water $\rightarrow$ Sodium hydroxide + Hydrogen

ANS:

- (i) $\text{H}_2 + \text{Cl}_2 \rightarrow 2\text{HCl}$
- (ii) $3\text{BaCl}_2 + \text{Al}_2(\text{SO}_4)_3 \rightarrow 3\text{BaSO}_4 + 2\text{AlCl}_3$
- (iii) $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$

Q NO 3: Write a balanced chemical equation with state symbols for the following reactions.

- (i) Solutions of barium chloride and sodium sulphate in water react to give insoluble barium sulphate and the solution of sodium chloride.
- (ii) Sodium hydroxide solution (in water) reacts with hydrochloric acid solution (in water) to produce sodium chloride solution and water.

ANS:

- (i) $\text{BaCl}_2 + \text{Na}_2\text{SO}_4 \rightarrow \text{BaSO}_4 + 2\text{NaCl}$
- (ii) $\text{NaOH} + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O}$

Q NO 4: A solution of a substance 'X' is used for white washing.

- (i) Name the substance 'X' and write its formula.
- (ii) Write the reaction of the substance 'X' named in (i) above with water.

ANS:

(i) The substance 'X' is calcium oxide. Its chemical formula is CaO.

(ii) Calcium oxide reacts vigorously with water to form calcium hydroxide (slaked lime).



Calcium Oxide (Quick Lime) + Water $\rightarrow$ Calcium Hydroxide (Slaked Lime)

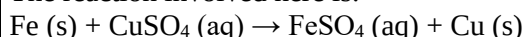
Q NO 5: Why is the amount of gas collected in one of the test tubes in Activity 1.7 double of the amount collected in the other? Name this gas.

ANS: Water contains two parts of hydrogen and one part oxygen. Therefore, during the electrolysis of water the amount of hydrogen gas collected in one of the test tubes is double than that of the oxygen produced and collected in the other test tube.

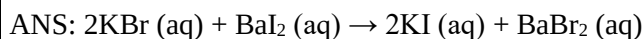
Q NO 6: Why does the colour of copper sulphate solution change when an iron nail is dipped in it?

ANS: When an iron nail dipped in the copper sulphate solution than iron displaces copper from the copper sulphate because iron is more reactive than copper. Therefore the colour of the copper sulphate solution changes.

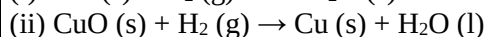
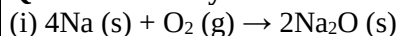
The reaction involved here is:



Q NO 7: Give an example of a double displacement reaction other than the one given in Activity 1.10.



Q NO 8: Identify the substances that are oxidised and the substances that are reduced in the following reactions.

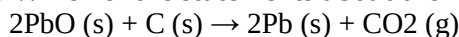


ANS:

(i) Sodium (Na) is oxidised as it gains oxygen and oxygen gets reduced.

(ii) Copper oxide (CuO) is reduced to copper (Cu) while hydrogen (H₂) gets oxidised to water (H₂O).

Q NO 9: Which of the statements about the reaction below are incorrect?



(a) Lead is getting reduced.

(b) Carbon dioxide is getting oxidised.

(c) Carbon is getting oxidised.

(d) Lead oxide is getting reduced.

(i) (a) and (b)

(ii) (a) and (c)

(iii) (a), (b) and (c)

(iv) all

► (i) (a) and (b)

Q NO 10: $\text{Fe}_2\text{O}_3 + 2\text{Al} \rightarrow \text{Al}_2\text{O}_3 + 2\text{Fe}$

The above reaction is an example of a

(a) combination reaction.

(b) double displacement reaction.

(c) decomposition reaction.

(d) displacement reaction.

► (d) displacement reaction.

Q NO 11: What happens when dilute hydrochloric acid is added to iron filings? Tick the correct answer.

(a) Hydrogen gas and iron chloride are produced.

(b) Chlorine gas and iron hydroxide are produced.

(c) No reaction takes place.

(d) Iron salt and water are produced.

► (a) Hydrogen gas and iron chloride are produced.

Q NO 12: What is a balanced chemical equation? Why should chemical equations be balanced?

ANS: A reaction which has an equal number of atoms of all the elements on both sides of the chemical equation is called a balanced chemical equation. Chemical reaction should be balanced to follow law of conservation of mass.

Q NO 13: Translate the following statements into chemical equations and then balance them.

- (a) Hydrogen gas combines with nitrogen to form ammonia.
- (b) Hydrogen sulphide gas burns in air to give water and sulphur dioxide.
- (c) Barium chloride reacts with aluminium sulphate to give aluminium chloride and a precipitate of barium sulphate.
- (d) Potassium metal reacts with water to give potassium hydroxide and hydrogen gas.

ANS:

- (a) $3\text{H}_2 (\text{g}) + \text{N}_2 (\text{g}) \rightarrow 2\text{NH}_3 (\text{g})$
- (b) $2\text{H}_2\text{S} (\text{g}) + 3\text{O}_2 (\text{g}) \rightarrow 2\text{H}_2\text{O} (\text{l}) + 2\text{SO}_2 (\text{g})$
- (c) $3\text{BaCl}_2 (\text{aq}) + \text{Al}_2(\text{SO}_4)_3 (\text{aq}) \rightarrow 2\text{AlCl}_3 (\text{aq}) + 3\text{BaSO}_4 (\text{s})$
- (d) $2\text{K} (\text{s}) + 2\text{H}_2\text{O} (\text{l}) \rightarrow 2\text{KOH} (\text{aq}) + \text{H}_2 (\text{g})$

Q NO 14: Balance the following chemical equations.

- (i) $\text{HNO}_3 + \text{Ca}(\text{OH})_2 \rightarrow \text{Ca}(\text{NO}_3)_2 + \text{H}_2\text{O}$
- (ii) $\text{NaOH} + \text{H}_2\text{SO}_4 \rightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$
- (iii) $\text{NaCl} + \text{AgNO}_3 \rightarrow \text{AgCl} + \text{NaNO}_3$
- (iv) $\text{BaCl}_2 + \text{H}_2\text{SO}_4 \rightarrow \text{BaSO}_4 + \text{HCl}$

ANS:

- (i) $2\text{HNO}_3 + \text{Ca}(\text{OH})_2 \rightarrow \text{Ca}(\text{NO}_3)_2 + 2\text{H}_2\text{O}$
- (ii) $2\text{NaOH} + \text{H}_2\text{SO}_4 \rightarrow \text{Na}_2\text{SO}_4 + 2\text{H}_2\text{O}$
- (iii) $\text{NaCl} + \text{AgNO}_3 \rightarrow \text{AgCl} + \text{NaNO}_3$
- (iv) $\text{BaCl}_2 + \text{H}_2\text{SO}_4 \rightarrow \text{BaSO}_4 + 2\text{HCl}$

Q NO 15: Write the balanced chemical equations for the following reactions.

- (a) Calcium hydroxide + Carbon dioxide $\rightarrow$ Calcium carbonate + Water
- (b) Zinc + Silver nitrate $\rightarrow$ Zinc nitrate + Silver
- (c) Aluminium + Copper chloride $\rightarrow$ Aluminium chloride + Copper
- (d) Barium chloride + Potassium sulphate $\rightarrow$ Barium sulphate + Potassium chloride

ANS:

- (a) $\text{Ca}(\text{OH})_2 + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O}$
- (b) $\text{Zn} + 2\text{AgNO}_3 \rightarrow \text{Zn}(\text{NO}_3)_2 + 2\text{Ag}$
- (c) $2\text{Al} + 3\text{CuCl}_2 \rightarrow 2\text{AlCl}_3 + 3\text{Cu}$
- (d) $\text{BaCl}_2 + \text{K}_2\text{SO}_4 \rightarrow \text{BaSO}_4 + 2\text{KCl}$

Q NO 16: Write the balanced chemical equation for the following and identify the type of reaction in each case.

- (a) Potassium bromide(aq) + Barium iodide(aq) $\rightarrow$ Potassium iodide(aq) + Barium bromide(s)
- (b) Zinc carbonate (s) $\rightarrow$ Zinc oxide (s) + Carbon dioxide (g)
- (c) Hydrogen(g) + Chlorine (g) $\rightarrow$ Hydrogen chloride(g)
- (d) Magnesium (s) + Hydrochloric acid (aq) $\rightarrow$ Magnesium chloride (aq) + Hydrogen (g)

ANS:

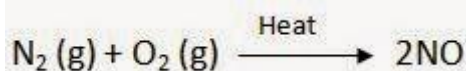
- (a) $2\text{KBr} (\text{aq}) + \text{BaI}_2 (\text{aq}) \rightarrow 2\text{KI} (\text{aq}) + \text{BaBr}_2 (\text{s})$: Double displacement reaction
- (b) $\text{ZnCO}_3 (\text{s}) \rightarrow \text{ZnO} (\text{s}) + \text{CO}_2 (\text{g})$: Decomposition reaction
- (c) $\text{H}_2 (\text{g}) + \text{Cl}_2 (\text{g}) \rightarrow 2\text{HCl} (\text{g})$: Combination reaction
- (d) $\text{Mg} (\text{s}) + 2\text{HCl} (\text{aq}) \rightarrow \text{MgCl}_2 (\text{aq}) + \text{H}_2 (\text{g})$: Displacement Reaction

Q NO 17: What does one mean by exothermic and endothermic reactions? Give examples.

ANS: Chemical reactions that release energy in the form of heat, light, or sound are called exothermic reactions.

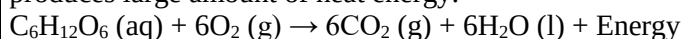
Example: $\text{C} (\text{g}) + \text{O}_2 (\text{g}) \rightarrow \text{CO}_2 + \text{Heat Energy}$

Reactions that absorb energy or require energy in order to proceed are called endothermic reactions. Example:



Q NO 18: Why is respiration considered an exothermic reaction? Explain.

ANS: Respiration is considered as an exothermic reaction because in respiration oxidation of glucose takes place which produces large amount of heat energy.

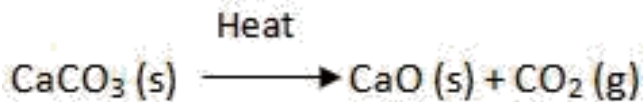


Q NO 19: Why are decomposition reactions called the opposite of combination reactions? Write equations for these reactions.

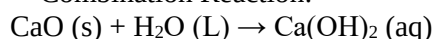
ANS: Decomposition reactions are those in which a compound breaks down to form two or more substances. These reactions require a source of energy to proceed. Thus, they are the exact opposite of combination reactions in which two or more substances combine to give a new substance with the release of energy.

For Example:

Decomposition Reaction:

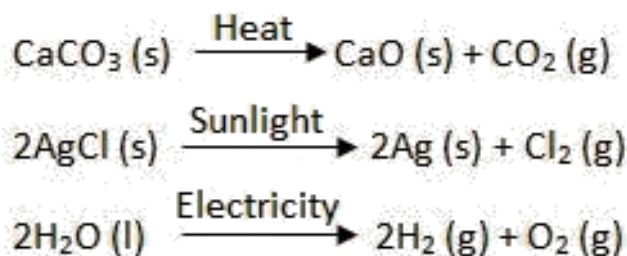


Combination Reaction:



Q NO 20: Write one equation each for decomposition reactions where energy is supplied in the form of heat, light or electricity.

ANS:



Q NO 21: What is the difference between displacement and double displacement reactions? Write equations for these reactions.

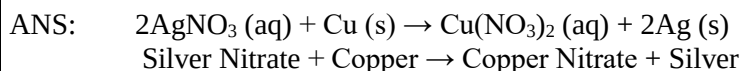
ANS: In a displacement reaction, a more reactive element replaces a less reactive element from a compound.

For Example: $\text{CuSO}_4 (\text{aq}) + \text{Zn} (\text{s}) \rightarrow \text{ZnSO}_4 (\text{aq}) + \text{Cu} (\text{s})$

In a double displacement reaction, two atoms or a group of atoms switch places to form new compounds.

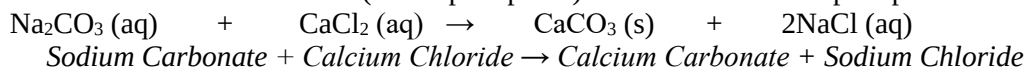
For Example: $\text{Na}_2\text{SO}_4 (\text{aq}) + \text{BaCl}_2 (\text{aq}) \rightarrow \text{BaSO}_4 (\text{s}) + 2\text{NaCl} (\text{aq})$

Q NO 22: In the refining of silver, the recovery of silver from silver nitrate solution involved displacement by copper metal. Write down the reaction involved.



Q NO 23: What do you mean by a precipitation reaction? Explain by giving examples.

ANS: A reaction in which an insoluble solid (called precipitate) is formed is called a precipitation reaction. For Example:



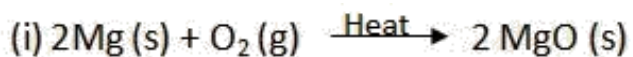
In this reaction, calcium carbonate is obtained as a precipitate. Hence, it is a precipitation reaction.

Q NO 24: Explain the following in terms of gain or loss of oxygen with two examples each.

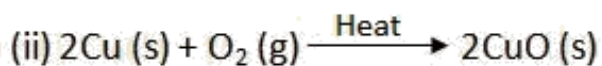
(a) Oxidation

(b) Reduction

ANS: Oxidation Reaction: It is a chemical reaction in which gain of oxygen or loss of hydrogen takes place.

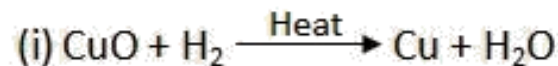


Here, Magnesium is oxidised to become Magnesium Oxide.

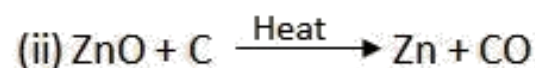


Here, Copper is oxidised to become Copper Oxide.

Reduction Reaction: It is a chemical reaction in which loss of oxygen or gain of hydrogen takes place.



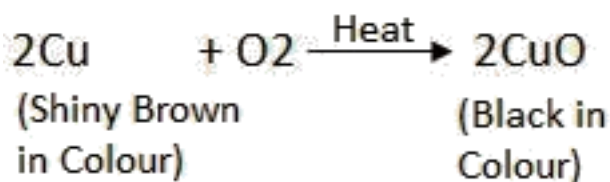
Here, Copper Oxide is reduced to become Copper.



Here, Zinc Oxide is reduced to become Zinc.

Q NO 25: A shiny brown-coloured element 'X' on heating in air becomes black in colour. Name the element 'X' and the black coloured compound formed.

ANS: 'X' is copper (Cu) and the black-coloured compound formed is copper oxide (CuO). The equation of the reaction involved on heating copper is given below.



Q NO 26: Why do we apply paint on iron articles?

ANS: Iron articles are painted because it prevents them from rusting. When painted, the contact of iron articles from moisture and air is cut off. Hence, rusting is prevented.

Q NO 27: Oil and fat containing food items are flushed with nitrogen. Why?

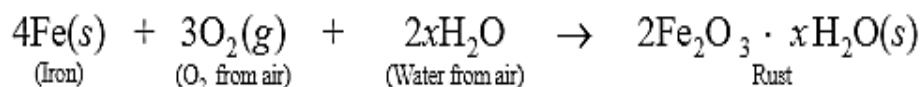
ANS: Oil and fat containing food items flushed with nitrogen because nitrogen acts as an antioxidant and it prevent them from being oxidised.

Q NO 28: Explain the following terms with one example each.

(a) *Corrosion*

(b) *Rancidity*

ANS: The Natural process of gradual destruction of pure metals by the action of oxygen, moisture or a chemical (such as an acid) on their surface and as a result of chemical reaction or electrochemical reaction between a metal and the surrounding environment to form a stable oxide is called corrosion.



Rancidity is the process of oxidation of fats and oils that can be easily noticed by the change in taste and smell is known as rancidity. For example, the taste and smell of butter changes when kept for long.

ACIDS, BASES AND SALTS

Till date one hundred and eighteen different elements are known. When these different elements combine they form (yield) a large number of chemical compounds which are classified or grouped into three types and are:

- I. Acids
- II. Bases
- III. Salts

In order to know whether a substance is an acid or a base, we should first know about acid-base indicators.

Indicators: Indicators are the chemicals that when added to solutions show whether the solution is acidic or basic, by a sudden change of their colour. The common indicators used are mentioned below:

- Litmus paper or litmus solution
- Methyl orange
- Phenolphthalein
- Turmeric (haldi)
- Red cabbage leaves.

The most common indicator used for testing acids and bases in school laboratories is litmus, which is a natural indicator. Litmus is extracted from a type of plant called lichen (thallophyta). Turmeric and red cabbage leaves are also natural indicators. Whereas methyl orange and phenolphthalein are synthetic indicators.

Litmus paper indicator is of two types;

- a) *Blue litmus paper*
 - b) *Red litmus paper*
- Blue litmus paper turns red on acidic solution where as red litmus paper turns blue on basic solution.
 - Methyl orange turns red colour in acidic solution & turns yellow colour in basic solution.
 - Phenolphthalein gives no colour (colourless) in acidic solutions but turns pink colour in basic solutions.
 - Turmeric turns red in basic solutions.
 - Red cabbage leaves turns red in acidic solution & turns green in basic solution.

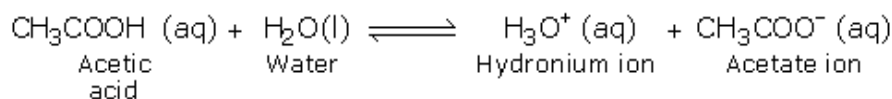
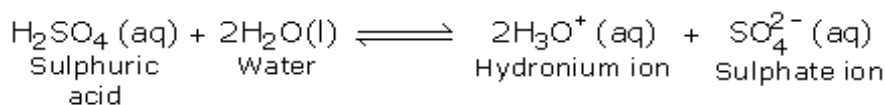
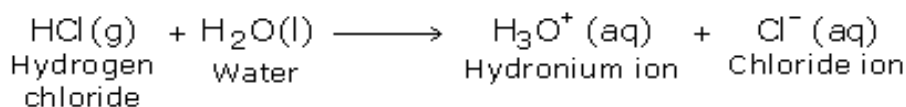
ACIDS: The word 'acid' is derived from a Latin word “acidus” which means "sour".

In 1887, Svante Arrhenius put forward his concept of acids, according to him, an acid is a hydrogen containing compound which gives free hydrogen ions (H^+) when dissolved in water & yields hydronium ions (H_3O^+) as the only positive ions. The characteristic property of an acid is due to the presence of these hydronium ions.

The sour taste of most of the fruits such as raw mangoes, lemon, raw grapes, orange & tamarind and vegetables is due to various types of acids present in them. The digestive juices of most of the animals and humans also contain acids. Acid turns blue litmus paper into red litmus paper. Some acids are naturally occurring present in plant materials & in animal bodies and they are called as organic acids. Some acids which are prepared in laboratories are called as man-made or synthetic or mineral acids. Some organic acids are as follows:

- Acetic acid found in vinegar (sirka).
- Citric acid present in citrus fruits like lemon, raw apples, raw mangoes & orange etc.
- Lactic acid is present in curd.
- Tartaric acid is present in unripened grapes & tamarind.
- Oxalic acid is present in tomatoes.
- Formic acid (methanoic acid) is present in sting of ant & nettle leaf sting.
- Organic acids or naturally occurring acids are not harmful to eat or drink.

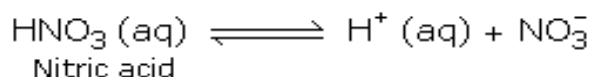
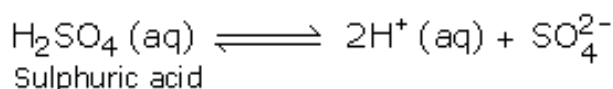
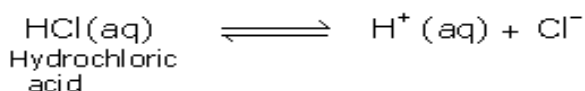
Formation of Hydronium ion by Acids: According to the classical concept all acids dissolve in water to give hydrogen ions (H^+). A hydrogen ion is a proton, since hydrogen atom has only one proton and one electron. These hydrogen ions can't exist independently in water and combine with neutral water molecules to form hydronium ions (H_3O^+).



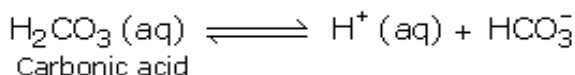
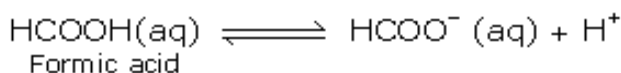
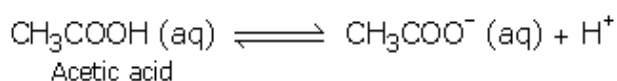
Classification of Acids: Acids can be classified in various ways, depending on the factors mentioned below.

- (1) **Classification Based on the Strength of the acid:** The strength of an acid depends on the concentration of the hydronium ions present in a solution. Greater the number of hydronium ions present, greater is the strength of acid. However, some acids do not dissociate to any appreciable extent in water such as carbonic acid. Therefore, these acids will have a low concentration of hydronium ions.

Strong acid: An acid, which dissociates completely in water to form positive and negative atoms separately, is classified as a strong acid. Or an acid which is completely ionized in water & thus produces a large number of hydrogen ions is called a strong acid. The word here strong refers to the degree of ionization. Strong acids are also called as strong electrolytes because for passage of electricity more hydrogen ions are present in strong acid solutions. All the mineral acids or man-made acids are strong acids except carbonic acid which is a weak acid. Strong acids are very dangerous to drink even in dilute form too. It must be noted that in these acids all the hydrogen ions combine with water molecule and exist as hydronium ions. Examples are: hydrochloric acid, sulphuric acid, nitric acid etc.



Weak acid: An acid that dissociates partially when dissolved in water and not separates hydrogen atoms completely, is classified as a weak acid. Or an acid which is partially ionized in water and thus produces a small amount of hydrogen ions is called a weak acid. Weak acids are also called as weak electrolytes because they produce less hydrogen ions in a solution. All the organic acids or naturally occurring acids are weak acids. Most of the molecules remain in solution in molecular form itself in such acid. Examples are: acetic acid, formic acid, carbonic acid etc.



- (2) **Classification Based on the Concentration of the Acid:** Concentration of an acid solution depends upon the percentage of the acid that is dissolved in water.

Concentrated acid: Concentrated acid is that acid in which minimum (less) amount of water is present in it. Or an acid that has a relatively high percentage of the acid in the aqueous solution is classified as a concentrated acid e.g. concentrated hydrochloric acid, sulphuric acid, nitric acid, acetic acid etc.

Dilute acid: Dilute acid is that acid in which maximum (more) amount of water is present in it. Or an acid, which has a relatively low percentage of the acid in the aqueous solution, is classified as a dilute acid e.g. dilute hydrochloric acid, sulphuric acid, nitric acid, acetic acid etc.

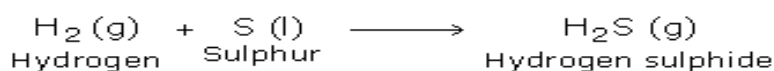
- (3) **Classification Based on the Presence or absence of Oxygen:**

Oxy-acids: Oxy-acids are those, which contain oxygen in their composition. Examples are: Sulphuric acid (H_2SO_4), Oxalic acid ($\text{H}_2\text{C}_2\text{O}_4$), Nitric acid (HNO_3), Acetic acid (CH_3COOH).

Hydracids: Acids, which contain hydrogen together with other elements and not any oxygen in their composition, are called Hydracids. Examples are: Hydrochloric acid (HCl), Hydroiodic acid (HI), Hydrobromic acid (HBr).

Preparation of Acids:

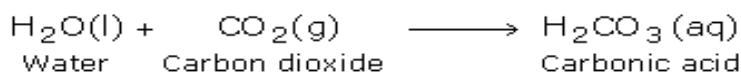
- (1). **Preparation by Direct Combination or Synthesis:** Some acids can be prepared by the direct combination of a non-metal with hydrogen. Example:



In the above reactions, the obtained gases like hydrogen chloride and hydrogen sulphide are also called the acid anhydrides. They exhibit acidic properties, only when dissolved in water. Therefore an acid anhydride is a substance that dissolves in water to form an acid. In other words, it is the acid minus water (acid - water = acid anhydride). CO_2 gas is

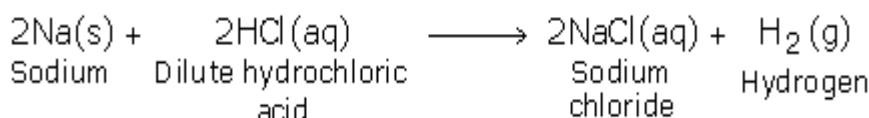
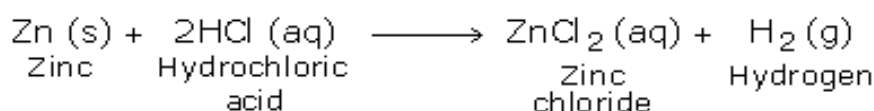
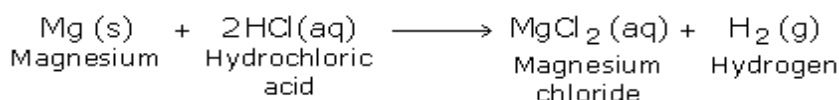
the acid anhydride of carbonic acid, SO_3 gas is the acid anhydride of carbonic acid, HCl gas is the acid anhydride of hydrochloric acid.

(2) **Preparation by Dissolving Acidic Oxide in Water:** Oxides that can add on hydrogen ions to their molecules are called acidic oxides. Thus when they dissolve in water they associate with hydrogen ions to form acids.

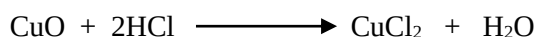
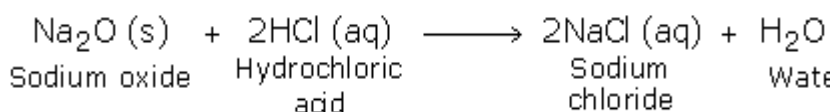
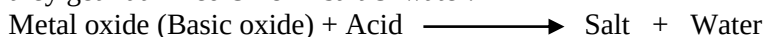


Properties of Acids:

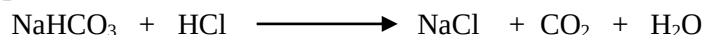
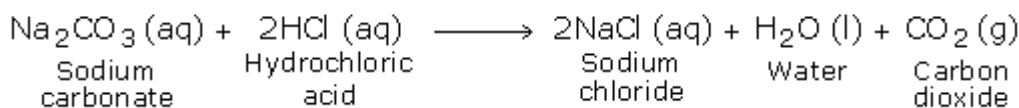
- **Taste:** Acids are generally sour in taste.
- **Action with indicators:** Acids turn blue litmus to red litmus.
- **Electrolyte:** Acid solution conduct electricity, that means they act as electrolytes because of presence of hydrogen ions present in solution.
- **Acids have corrosive nature:** Man made acids or mineral acids cause severe burns on the skin & attack and eat up materials like cloth, wood, metal structures. So, they are said to be corrosive. Acids are never stored in metal containers because they gradually corrode & eat up metal container by reacting with them.
- **Reaction of acids with Metals:** Metals above hydrogen in the metal activity series generally react with dilute acids to form their respective salt and hydrogen.



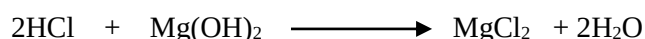
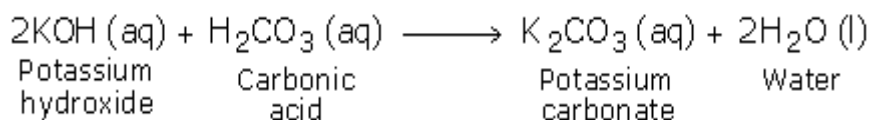
- **Reaction of acids with Metal Oxides or Basic oxides:** Whenever acids react with basic oxides (metal oxides) they get neutralized & form salt & water.



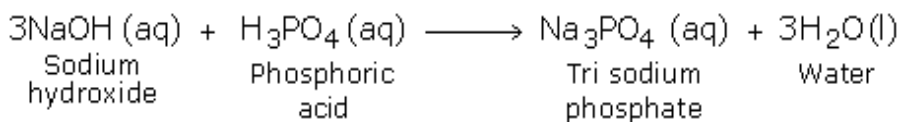
- **Reaction of acids with metal Carbonates & metal Hydrogen carbonates:** When an acid reacts with metal carbonates or with metal hydrogen carbonates they form their respective salt, water and carbon dioxide.



- **Action with Bases (alkalis):** When an acid reacts with a base, neutralization reaction takes place & salt & water are formed.



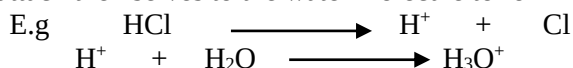
The reaction between an acid and a base (metal hydroxide) is also a kind of neutralization reaction. The antacid called milk of magnesia which is used to remove indigestion caused by excess release of HCl inside stomach is a metal hydroxide. Milk of magnesia is basic in nature. It reacts with the excess hydrochloric acid & neutralizes it.



- **BASICITY OF ACIDS:** Basicity means the number of hydrogen ions in one molecule of acid e.g. HCl has one replaceable hydrogen ion, hence, its basicity is one and is called monobasic acid. Similarly, the basicity of H₂SO₄ and H₂SO₃ is two and are called dibasic acids. Whereas, phosphoric acid (H₃PO₄) has three replaceable hydrogen ions and is called tribasic acid.

What do all acids have in common:

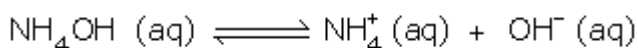
When any acid is dissolved in water, it separated out as positively charged hydrogen ion (H⁺). The hydrogen ion don't exist as H⁺ ions in solution but they attach themselves to the water molecule to form hydronium ion (H₃O⁺).



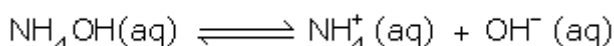
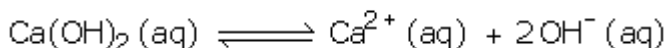
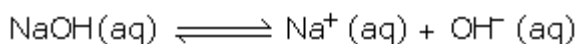
A common substance in all acids is that they produce hydrogen ions (H⁺) when dissolved in water. Thus, the acidic behaviour of an acid solution is due to the presence of hydrogen ions in it (H⁺).

The compounds such as glucose (C₆H₁₂O₆) & alcohol (C₂H₅OH) also contain hydrogen atoms but they don't lose hydrogen atoms in water, hence they don't show acidic character. From this we learned that though all the acids contain hydrogen, but all the hydrogen containing compounds are not acids.

BASE: A base is a compound, which on dissolving in water yields hydroxyl ions (OH⁻) as the only negative ions. The characteristic property of a base is due to the presence of these negative hydroxyl ions. A base may be an oxide or a hydroxide of a metal. If a base is soluble in water, it dissociates to form a metal ion and the only negative hydroxyl ion. Substances that combine with acids to form salt and water only are also called bases. It may be noted that ammonium hydroxide is also a base. In water, it dissociates to form ammonium ion and a hydroxyl ion.



Alkalis: All alkalis are bases that dissociate in water to yield hydroxyl ion(OH⁻) as the only Negative ions. Sodium hydroxide, potassium hydroxide, calcium hydroxide and ammonium hydroxide are the common alkalis.



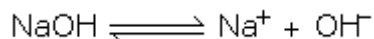
Classification of Bases: Bases can be classified in various ways, depending on the following factors:

Classification Based on the Strength of the Base: The strength of a base depends on the concentration of the hydroxyl ions when it is dissolved in water.

Strong base: A base that dissociates completely or almost completely in water is classified as a strong base. The greater the number of hydroxyl ions the base produces, the stronger is the base.

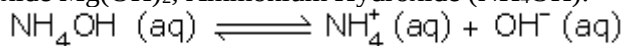
Example: Sodium hydroxide (NaOH), Potassium hydroxide (KOH),

Calcium Hydroxide Ca(OH)₂



Weak base: A base that dissociates in water only partially is known as a weak base.

Example: Magnesium Hydroxide Mg(OH)₂, Ammonium Hydroxide (NH₄OH).

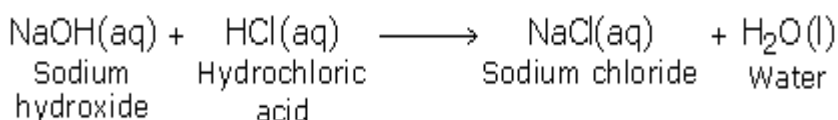


Classification based on the Acidity of Bases: The acidity of a base is equal to the number of hydroxyl ions it has. It is also the number of hydrogen ions with which a base can combine as one hydrogen ion combines with one hydroxyl ion

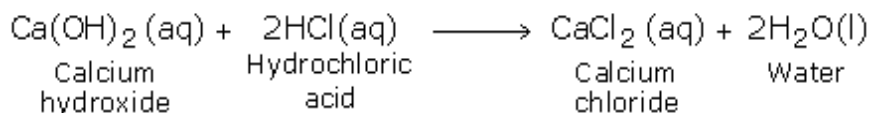
Monoacidic base: It is a base that has only one hydroxyl ion. It combines with only one hydrogen ion.

Example: Sodium hydroxide: NaOH,

Potassium hydroxide: KOH, Lithium hydroxide: LiOH, Ammonium hydroxide: NH₄OH etc.

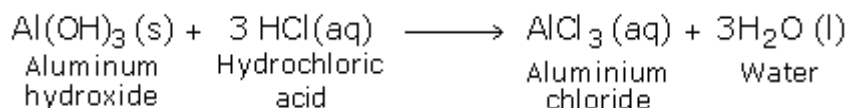


Diacidic base: It is a base that has two hydroxyl groups and combines with two hydrogen ions. Example: Calcium hydroxide: Ca(OH)_2 , Iron (II) hydroxide: Fe(OH)_2 , Magnesium hydroxide: Mg(OH)_2 , Copper hydroxide: Cu(OH)_2 , Zinc hydroxide: Zn(OH)_2 , Lead hydroxide: Pb(OH)_2



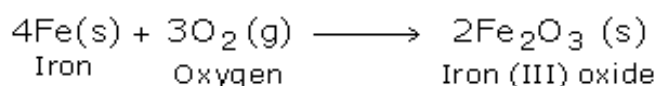
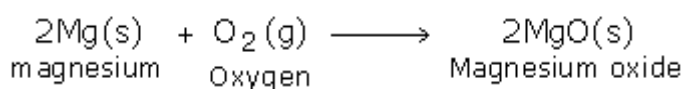
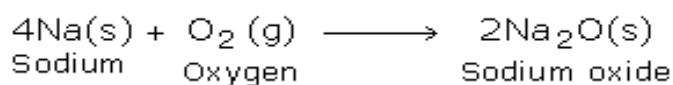
Triacidic base: A base that has three hydroxyl ions and can combine with three hydrogen ions, is known as a triacidic base. Example:

Aluminium hydroxide: Al(OH)_3 , Iron (III) hydroxide: Fe(OH)_3 .

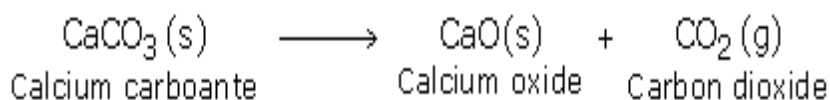
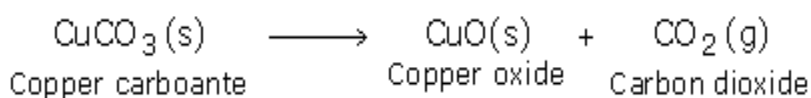


Methods of Preparation of Bases:

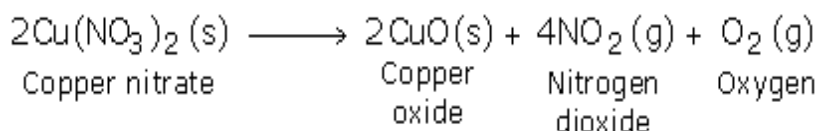
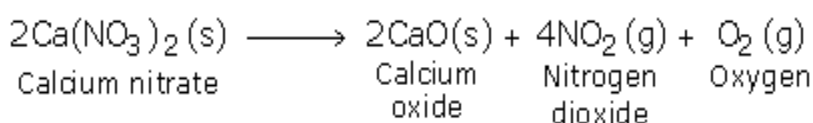
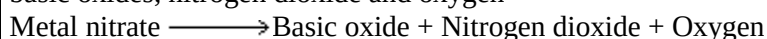
(1) Preparation by Direct Combination with Oxygen: Some bases can be prepared by the direct combination of a metal with oxygen.



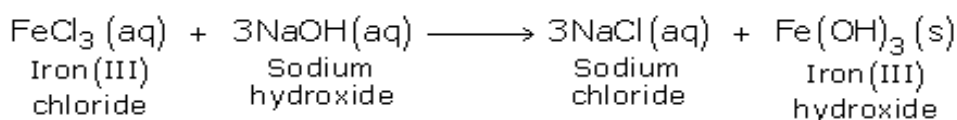
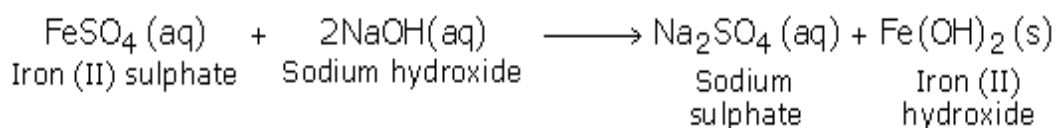
(2) Preparation by Thermal Decomposition of Carbonates: The thermal decomposition of metal carbonates produces basic oxides and carbon dioxide.

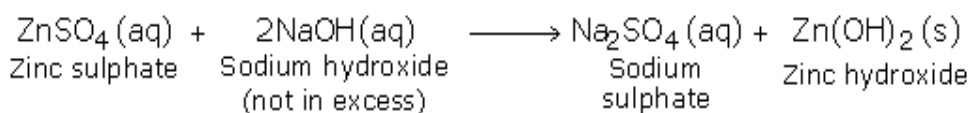


(3) Preparation by Thermal Decomposition of Metal Nitrates: The thermal decomposition of metal nitrates produces basic oxides, nitrogen dioxide and oxygen

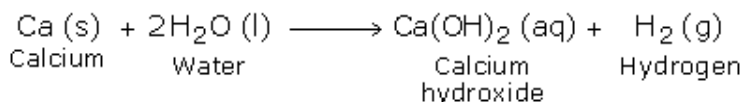
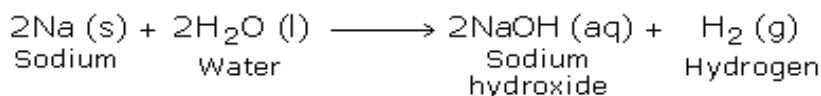


(4) Preparation by Double Decomposition: Certain basic hydroxides can be obtained through precipitation of the hydroxides, by treating aqueous solutions of certain salts, with other stronger bases.

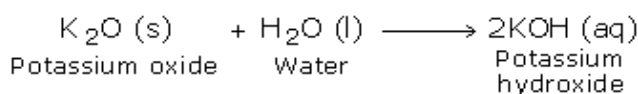
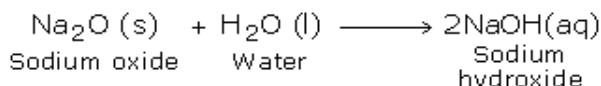




(5) Preparation by the Action of very Active Metals with Water: Some active metals react spontaneously with water to form their respective bases and release the hydrogen gas.



(6) Preparation by Dissolving Basic Oxides in Water: When basic oxides dissolve in water they associate with hydroxyl ions to form bases.



Properties of Bases:

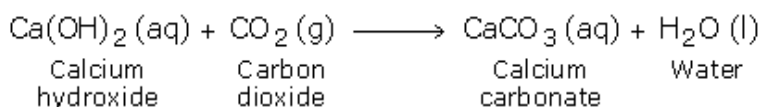
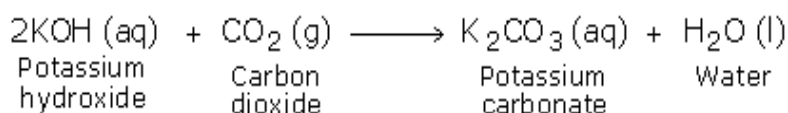
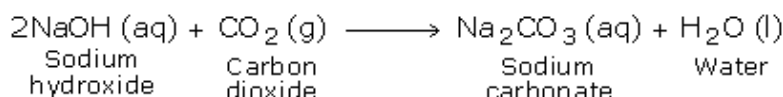
Taste: Bases are bitter to taste.

Feel: They are soapy and slippery to touch.

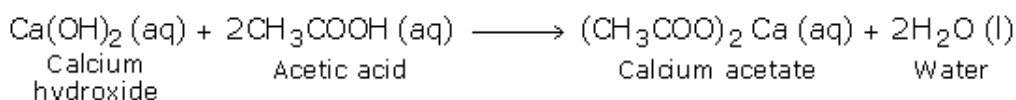
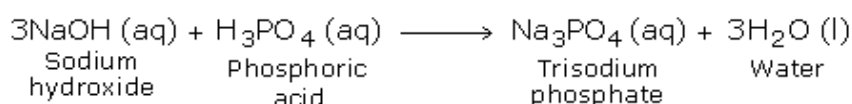
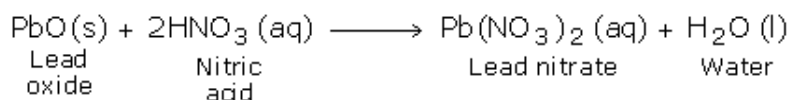
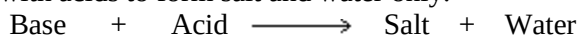
Electrical Conductivity: Some bases are good conductors of electricity. e.g. soluble bases like sodium hydroxide and potassium hydroxide are used as electrolytes.

Corrosive Action: Strong alkalis like sodium hydroxide and potassium hydroxide are highly corrosive or caustic in nature. Sodium hydroxide and potassium hydroxide are commonly called caustic soda and caustic potash respectively. Organic tissues like skin, etc. get completely corroded by these two alkalis. However, the other alkalis are only mildly corrosive.

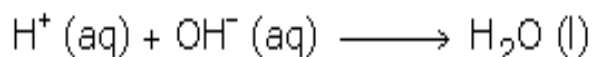
Action with Air: When exposed to air, alkalis combine with the carbon dioxide of the air, to form the respective carbonate.



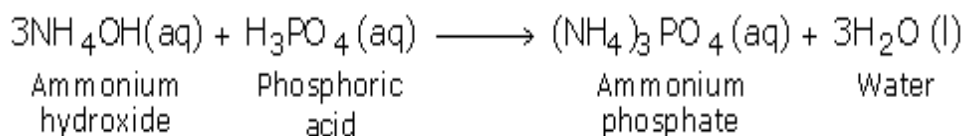
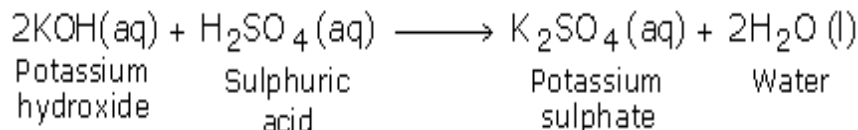
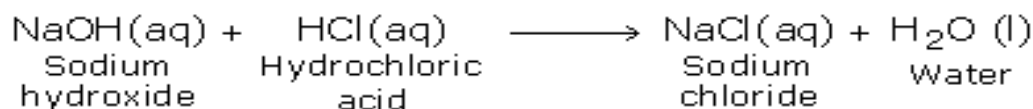
Action with Acids: Bases combine with acids to form salt and water only.



Neutralization: The acidic property of an acid is due to the presence of hydrogen ions (H^+) while that of a base or alkali, is due to the presence of hydroxyl (OH^-) ions in them. When an acid and base (alkali) combine, the positively charged hydrogen ion of the acid combines with the negatively charged hydroxyl ion of the base to form a molecule of water. Hence, the water molecule formed does not have any charge because the positive and negative charges of the hydrogen ions and hydroxyl ions get neutralized.



Neutralization is a reaction in which an acid combines with a base, neutralizing the positively charged hydrogen ion and the negatively charged hydroxyl ion, forming a molecule of water and the respective salt.

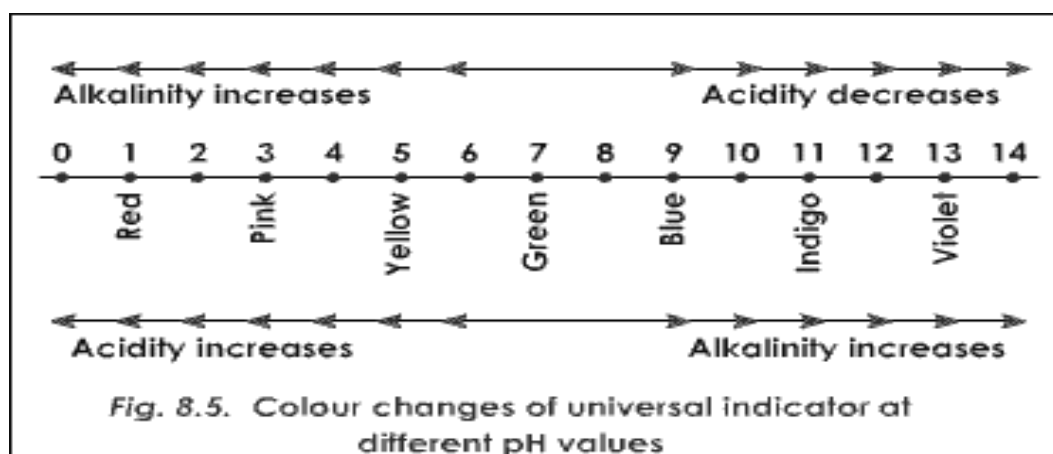
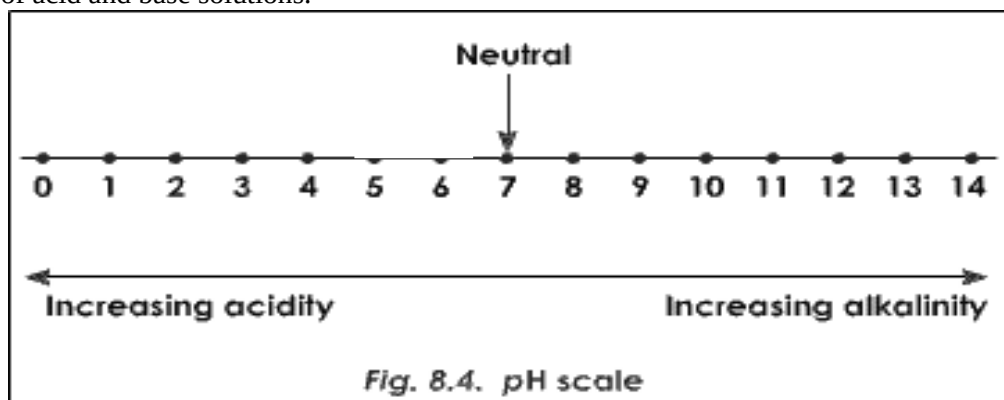


The strength of acids and bases can be expressed in terms of the concentration of the hydronium ions or hydroxyl ions in a solution. The concentration of hydronium ions in a solution can be expressed as moles of hydronium ions per litre of solution. This is expressed by a term called 'pH' which represents the 'power of hydrogen ion concentration' in solution.

Define P^H: In 1909, S.P. Sorensen proposed that the acidity of a solution be expressed in terms of a parameter called P^H. The notation pH has been derived from the French word *potenz de hydrogen* (meaning power of hydrogen). The indicators ordinarily used in the laboratory only indicate whether the solution is acidic, basic or neutral. They cannot give any idea as to the comparative strength of various solutions. However a 'Universal indicator' is used, which is a mixed indicator. It not only shows whether the solution is acidic or basic, but also shows the approximate pH values by giving a wider range of colours for different values of P^H.

The pH is defined as: 'the negative logarithm (to the base 10) of the hydrogen ion concentration, expressed in moles per litre'; i.e. $\text{pH} = -\log_{10} [\text{H}^+]$, where $[\text{H}^+]$ is the hydrogen ion concentration in molecule/Litre.

The strength of various acids and bases is expressed on the pH scale. The pH scale is a 14 point scale, which expresses the strength of an acid or a base in terms of the hydronium ion concentration. The range from 0 to 14 gives a measurement of comparative strength of acid and base solutions.



Pure water and other neutral solutions have a pH value of 7. A pH value less than 7 indicates that the solution is acidic, and a pH value greater than 7 indicates that the solution is basic.

Approximate pH Values of Some Common Substance:

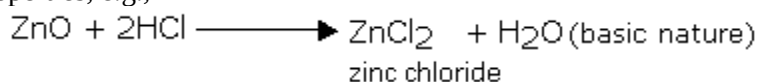
Substance	pH value	Substance	pH value
Hydrochloric acid	1.0	Bread	5.5
Sulphuric acid	1.2	Potatoes	5.8
Gastric juice	2.0	Rain water	6.2
Lemon	2.3	Milk	6.5
Vinegar (Acetic acid)	2.8	Pure water	7.0
Soft drink	3.0	Egg	7.8
Apple	3.1	Sea water	8.5
Grape	3.1	Ammonium hydroxide	11.1
Tomato	4.2	Sodium hydroxide	13.0
Banana	4.6		

Importance of pH in everyday life:

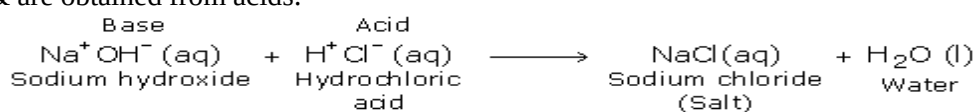
- Most of reactions taking place in human body are in the narrow pH range of 7.0 to 7.8. e.g. human body, tears, saliva have pH value nearly 7.4, if it falls below 7.0 or rises above 7.8 the survival of living organisms become difficult.
- Plants can have a healthy growth only when the soil has a specific pH range which should be neither alkaline nor highly acidic.
- Hydrochloric acid produced in the stomach helps in the process of digestion. However, if excess of acid is produced due to spicy food, it causes indigestion and we have to take antacids to prevent acidity. The most commonly used antacid is magnesium hydroxide (milk of magnesia).
- Acid is also produced in our mouth due to degradation of sugar, sweets and left out food and hence causes tooth decay. To overcome this problem we can use the tooth paste with suitable pH value as these tooth pastes neutralizes excess acid present in our mouth.
- Animals and plants have their self defence through chemicals. The sting by honey bee or yellow ant or leaves of nettle plant injects methanoic or formic acid into our body and makes us to feel irritating. To overcome this problem we usually apply the solution of mild base baking soda or rub the area of our skin with the leaf of a dock plant to neutralize the acid of the sting.
- The tarnished surface of a copper vessel due to copper oxide layer (which is basic) can be cleaned by rubbing with lemon (which is acidic).

ACIDITY OF BASES: The number of replaceable hydroxyl (OH⁻) ions in a base is called the acidity of the base. For example, NaOH has one replicable hydroxyl ion, hence its acidity is one and is called monoacidic base. The acidity of Ca(OH)₂ is two, and is called diacidic base. The aluminium hydroxide Al(OH)₃ has three replicable hydroxyl ions, so its acidity is three and is called triacidic base.

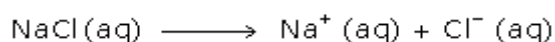
Amphoteric oxides: Amphoteric oxides are metallic oxides, which show both basic as well as acidic properties. When they react with an acid, they produce salt and water, showing basic properties. While reacting with alkalies they form salt and water showing acidic properties, e.g.,



SALT: Generally, salts are obtained by treating an acid with a base. Salts consist of both positive ions or cations, and negative ions or anions. The cations are called basic radicals and are obtained from metallic ions, while the anions are called acidic radicals. Salts have high melting & boiling points, usually soluble in water & conduct electricity because they dissociate as positive & negative ions on dissolving in water. Salts are ionic compounds because of presence of metal & non-metal ions & are obtained from acids.



Salt is a compound, which on dissociation in water yields positive ions other than a hydrogen ion or hydronium ion, and a negative ion other than hydroxyl ion.



- The salts of hydrochloric acid are called chlorides.
- The salts of sulphuric acid are called sulphates.
- The salts of nitric acid are called nitrates.

- The salts of carbonic acid are called carbonates.
- The salts of acetic acid are called acetates.

Family of Salts: Family of salts mean those salts having the same positive ions or negative ions are said to belong to a family of salt.

E.g: NaCl & Na₂SO₄ belong to the same family called sodium salts because they have same positive ions.

E.g: NaCl & KCl belong to the same family called chloride salts because they have same negative ions.

Classification of Salts: Salts can be classified into the following types:

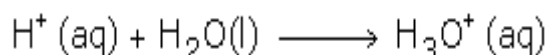
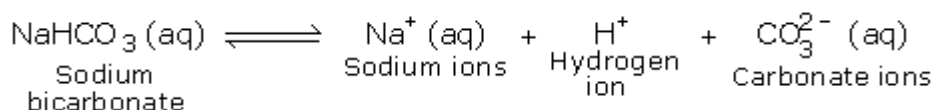
a) Normal salt: A salt that does not contain any replaceable hydrogen ions is a normal salt. It is obtained by replacing all the hydrogen ions of an acid by metal ions or ammonium ions.

Examples: NaCl, Na₂SO₄, Na₃PO₄, NH₄Cl, K₂CO₃, CuSO₄ etc.

b) Bisalts or Acid salts: A bisalt, or an acid salt contains replaceable hydrogen ions in association with the acidic radicals. It is obtained by the incomplete replacement of hydrogen ions by metal ions or ammonium ions.

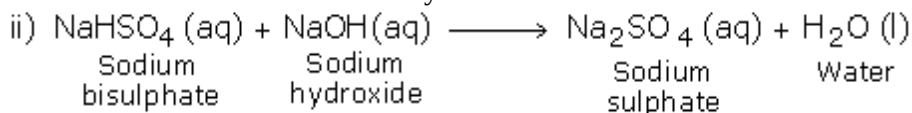
Examples: NaHCO₃, NaH₂PO₄ and Na₂HPO₄.

Since such salts also contain hydrogen ion, the term 'bi' or acid can be substituted by hydrogen, e.g. sodium bicarbonate or sodium acid carbonate is also called sodium hydrogen carbonate. When bisalts or acid salts dissociate in water, they yield hydrogen or hydronium ions.

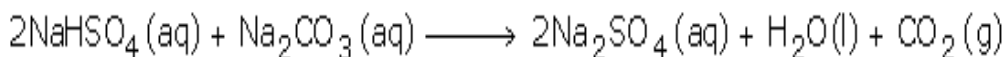


Since an acid salt liberates hydrogen or hydronium ions, it exhibits some acidic properties. Hence it behaves like an acid.

i) An acid salt reacts with bases to form salt and water only.



iii) They react with carbonates to yield carbon dioxide.



Some Common Acid Salts are NaHCO₃, NaHSO₃, NaHSO₄, Na₂HPO₄, Ca(HCO₃)₂, Ca(HSO₄)₂, KHCO₃, KHSO₃, KHSO₄, NaH₂PO₄ and Mg(HCO₃)₂

c) Basic Salts: A basic salt is formed by the action of an acid with higher proportion of the base, than is necessary for the formation of a normal salt. Example: Basic copper carbonate CuCO₃.Cu(OH)₂, Basic lead carbonate (white lead) PbCO₃.Pb(OH)₂.

A basic salt may also be formed by the partial replacement of the hydroxyl groups of a diacidic or triacidic base, by an acid radical. Basic salts are usually insoluble in water.



d) Double Salts: Such salts are formed by the intimate association of two simple salts, when crystallized from a solution of a mixture of the two.

Some Common Double Salts	
Potassium alum	K ₂ SO ₄ . Al ₂ (SO ₄) ₃ . 24H ₂ O
Ammonium alum	(NH ₄) ₂ SO ₄ . Al ₂ (SO ₄) ₃ . 24H ₂ O
Ferric ammonium alum	(NH ₄) ₂ SO ₄ . Fe ₂ (SO ₄) ₃ . 24H ₂ O
Potassium chromium alum	K ₂ SO ₄ . Cr ₂ (SO ₄) ₃ . 24H ₂ O

e) Mixed Salts: Salts that contain more than one acid radical or basic radical are called as mixed salts.

Some Common Mixed Salts	
Potassium calcium phosphate	KCaPO ₄
Bleaching powder	Ca(OCl)Cl
Sodium potassium chloride	NaKCO ₃
Sodium potassium sulphate	NaKSO ₄

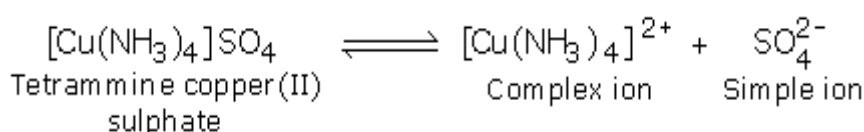
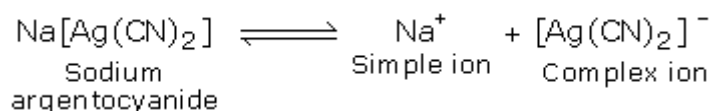
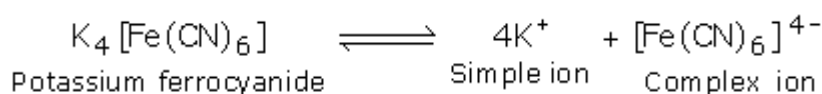
f) **Binary Salts:** Salts that contain atoms of two elements only are termed as binary salts.

Some Binary Salts	
Copper Chloride: CuCl ₂	Sodium chloride: NaCl
Iron (II) sulphide: FeS	Aluminium chloride: AlCl ₃
Zinc sulphide: ZnS	Magnesium chloride: MgCl ₂

g) **Ternary Salts:** These salts contain atoms of three elements only.

Some Ternary Salts	
Copper sulphate: CuSO ₄	Sodium carbonate: Na ₂ CO ₃
Silver nitrate: AgNO ₃	Zinc sulphite: ZnSO ₃
Calcium phosphate: Ca ₃ (PO ₄) ₂	Lead nitrate: Pb(NO ₃) ₂

h) **Complex Salts:** Salts that dissociate in water to give one simple ion and one complex ion are called complex salts.



The PH of Salt Solution:

A salt is formed only by the reaction of an acid with a base. Most of the salt solutions are neutral but some show acidic PH & some show basic PH.

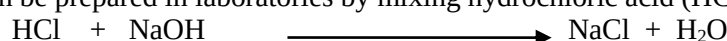
(i) Those salts which are made up of strong acid & strong base are neutral solutions (having PH = 7). E.g: NaCl is a salt having PH = 7 & is formed from HCl (strong acid) & NaOH (strong base).

(ii) Those salts which are made up of strong acid and weak base are acidic salts (having PH less than 7). E.g: NH₄Cl is a salt having PH less than 7 & is formed from HCl (strong acid) & ammonium hydroxide (weak base).

(iii) Those salts which are made up of weak acids & strong base are basic salts (having PH more than 7). E.g: Na₂CO₃ is a salt having PH more than 7 & is formed from carbonic acid (weak acid) & NaOH (strong base).

COMMON SALT (SODIUM CHLORIDE, NaCl):

Chemically common salt is a compound of sodium and chlorine. Its colour is white & is used in preparing food especially vegetables so, also known as table salt. Its chemical name is sodium chloride (NaCl) & is neutral in nature. Sodium chloride can be prepared in laboratories by mixing hydrochloric acid (HCl) with sodium hydroxide (NaOH).



Uses of common salt: The important uses of common salt are.

- WASHING SODA (SODIUM CARBONATE, $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$):**

Preparation of washing soda ($\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$): It is prepared by the Solvay process. Solvay process requires three steps to synthesize washing soda.

$$\text{CO}_2 + \text{NaCl} + \text{H}_2\text{O} + \text{NH}_3 \longrightarrow \text{NaHCO}_3 + \text{NH}_4\text{Cl}$$
$$2\text{NaHCO}_3 \xrightarrow[\text{Sodium hydrogen carbonate}]{\text{Heat}} \text{Na}_2\text{CO}_3 + \text{CO}_2 + \text{H}_2\text{O}$$

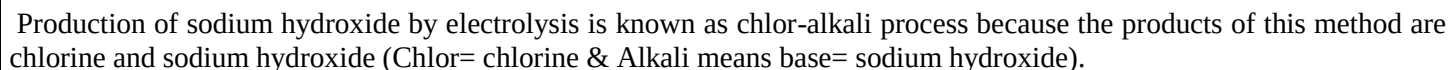
sodium carbonate

$$\text{Na}_2\text{CO}_3 + 10\text{H}_2\text{O} \xrightarrow{\text{g}} \text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$$

sodium carbonate Washing soda

- i. It is used in glass, soap and paper industry.
- ii. It is used in manufacture of sodium compounds such as borax.
- iii. Sodium carbonate can be used as a cleaning agent for domestic purposes
- iv. It is used for removing permanent hardness of water.

Sodium hydroxide commonly known as Caustic soda is a strong base. Its chemical formula is NaOH. Sodium hydroxide is used as a starting material for making many other chemical compounds, so it is produced on a large scale. Sodium hydroxide or caustic soda is manufactured by the electrolysis of an aqueous solution of sodium chloride (brine). When electricity is passed through brine or concentrated solution of sodium chloride, it decomposes to form sodium hydroxide (Caustic soda) , Chlorine gas & Hydrogen gas. Chlorine gas is given off at the anode and the hydrogen gas at the cathode. Sodium hydroxide solution is formed near the cathode. All these products (sodium hydroxide, chlorine & hydrogen) are collected separately.

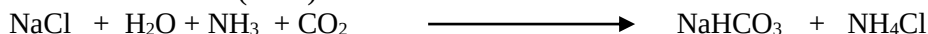


- i. It is used in the manufacture of soaps and detergents.
- ii. It is used for de-greasing metals.
- iii. It is used in making of artificial fibres like rayon etc.

- iv. It is used in paper industry.
- v. It is used in the process of petroleum refining.
- vi. It is used for mercerizing of cotton i.e. making cotton unshrinkable.
- vii. It is also used as a laboratory reagent for various blood tests.
- viii.

BAKING SODA (Sodium hydrogen carbonate, NaHCO₃):

Baking soda or sodium bicarbonate or sodium hydrogen carbonate (NaHCO₃) is obtained as intermediate product in Solvay's process for the manufacture of sodium carbonate. It can also be prepared by on large scale by reacting concentrated solution of sodium chloride (brine) with ammonia & carbon dioxide.



Sodium hydrogen carbonate or baking soda is soluble in water & is a non-corrosive base. When baking soda is heated, then it decomposes to give sodium carbonate, carbon dioxide gas & water.

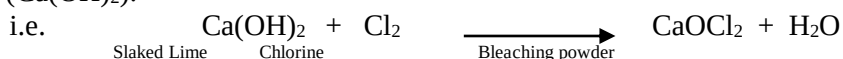


Uses of Baking Soda:

- i. It is used as a component of baking powder used in making breads & cakes etc.
- ii. It is used in fire extinguishers with sulphuric acid to produce carbon dioxide gas.
- iii. It is used as an antacid in medicine to remove acidity of the stomach.
- iv. It acts as a mild antiseptic for skin infections.
- v. It is also used as an additive in foods and drinks.
- vi. Baking soda is sometimes added for faster cooking of foods such as gram (*chana*).

BLEACHING POWDER (calcium oxychloride, CaOCl₂):

Bleaching powder or chloride of lime or calcium oxychloride (CaOCl₂) is a yellowish white powder which gives strong smell of chlorine. It is soluble in cold water leaving behind a small residue of lime. When exposed to air bleaching powder deteriorates giving off chlorine. Bleaching powder is prepared on industrial scale by passing chlorine gas through dry slaked lime (Ca(OH)₂).

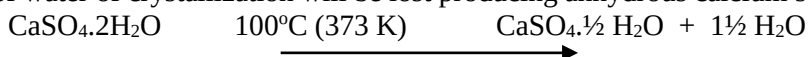


Uses of bleaching powder:

- i. It is used chiefly for bleaching cotton textiles, wood and paper pulp. Delicate articles like silk, wool, straw, etc. are not bleached by it, as these are likely to be damaged.
- ii. It is a strong disinfectant and is therefore, used for sterilization of water.
- iii. It is employed for making wool unshrinkable.
- iv. It is used as oxidising agent and in manufacture of chloroform (CHCl₃).

PLASTER OF PARIS / Calcium sulphate Hemi-hydrate (CaSO₄ · ½ H₂O): Plaster of Paris or Calcium sulphate Hemi-hydrate. The name plaster of Paris came from the fact, that it was first of all made by heating gypsum which was mainly found in Paris. It is commonly known as P.O.P.

is prepared by heating gypsum in a kiln to a temperature of 100°C (373 K) at this temperature, it loses three fourth of its water of crystallization forming plaster of Paris. The temperature should not be allowed to rise beyond 100°C as otherwise whole of water of crystallization will be lost producing anhydrous calcium sulphate which is called as dead burnt plaster.



It is a white powder, which when mixed with water sets to a hard due to rehydration of plaster of Paris to gypsum and is accompanied by a slight increase in volume (about 1%).

Uses of Plaster of Paris:

- i. It is used to make casts in dentistry and for setting the fractured bones in the right position.
- ii. It is used in making of toys, casts of statues, decorative materials, jewellery and cosmetics.
- iii. It is used for decorative and plastering purposes in the building construction.
- iv. It is used for making chalks for writing on the black board.
- v. It is also used for making the laboratory apparatus air tight by sealing the gaps while performing experiments.
- vi. It is used for making fire proof materials.

General Properties of Salts:

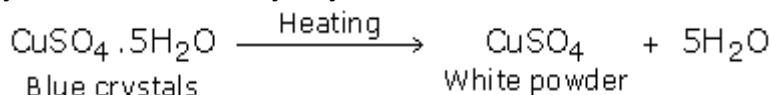
a) Water of Crystallisation: When crystals of certain salts are formed, they do so with a definite number of molecules of water, chemically combined in a definite proportion. Water of crystallisation is the number of water molecules, chemically combined in a definite molecular proportion, with the salt in the crystalline state. This water is responsible for the geometric shape and colour of the crystals.

A substance containing water of crystallization is called a hydrous substance or a hydrate. This water can be expelled, by heating, and then the salt is said to have become anhydrous. Some examples of hydrous salts are

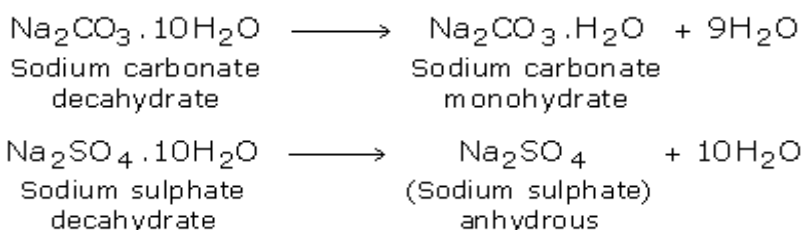
Crystalline substances without water of crystallization: Certain substances like sodium chloride do not require the help of water to form their crystalline shape. These salts are:

- i) Chlorides of potassium, sodium and lead.
- ii) Bromides of potassium, sodium, lead and silver.
- iii) Iodides of silver and lead.
- iv) Nitrates of potassium, sodium, lead, silver and ammonia.
- v) Sulphate of ammonia.

b) Effect of heat on hydrous salt: On heating, hydrous crystals lose their water of crystallization and turn into a powder. They are then said to be anhydrous. Sometimes they may also lose their colour.



c) Efflorescence: It is the property by which certain hydrous crystals when exposed to air, lose their water of crystallisation, fully or partially, and crumbles into a powder. Such substances are termed as efflorescent substances



d) Hygroscopic Substances: Hygroscopic substances are those that absorb water vapour from the air. The process is called hygroscopicity.

Concentrated sulphuric acid	H ₂ SO ₄	Magnesium chloride	MgCl ₂
Anhydrous calcium Chloride	CaCl ₂	Ferric chloride	FeCl ₃
Solid potassium hydroxide	KOH	Copper nitrate	Cu(NO ₃) ₂

e) Deliquescence: Certain hygroscopic substances, not only absorb moisture from the air, but they themselves dissolve in that moisture, undergoing a change of state. Such substances are called deliquescent substances. The process is called deliquescence.

TEXTUAL QUESTIONS

Q NO 1: You have been provided with three test tubes. One of them contains distilled water and the other two contain an acidic solution and a basic solution, respectively. If you are given only red litmus paper, how will you identify the contents of each test tube?

ANS: The acidic and basic nature of the solution can be identified by using litmus paper. A basic solution turns red litmus to blue and an acidic solution turns the blue litmus to red. The solution which changes the red litmus to blue is basic solution. Now, between two solutions, which turn the litmus which turned blue because of basic solution, to red, is acidic solution. The last one solution is distilled water.

Q NO 2: Why should curd and sour substances not be kept in brass and copper vessels?

ANS: Curd and other sour substances reacts with brass and copper vessels and forms hydrogen and other harmful substances due to presence of acid in them. That's why curd and other sour substances should not be kept in brass and copper vessels.

Q NO 3:-Which gas is usually liberated when an acid reacts with a metal? Illustrate with an example. How will you test for the presence of this gas?

ANS:-When acid reacts with metal, it liberates hydrogen gas usually. This can be shown by following

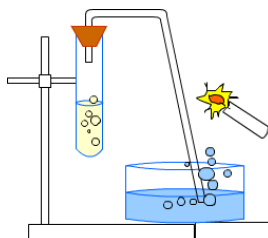
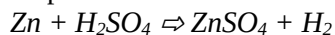


Fig: Reaction of metal and acid

- Take some zinc granules in the test tube and pour sulphuric acid in it.
- Allow the gas pass through the solution of soap.
- When gas starts coming out from the soap solution bubbles are formed.
- Bring a burning candle near the bubbles of gas it burns with a pop sound.
- Burning of gas with pop sound proves that it is hydrogen gas.

Reaction involved in this process can be written as follows:



Q NO 4: Metal compound A reacts with dilute hydrochloric acid to produce effervescence. The gas evolved extinguishes a burning candle. Write a balanced chemical equation for the reaction if one of the compounds formed is calcium chloride.

ANS: When calcium carbonate reacts with hydrochloric acid, it produces carbon dioxide gas with effervescence. Carbon dioxide used as fire extinguisher. Hence, it extinguishes a burning candle. Therefore, the metal compound A is calcium carbonate.



Q NO 5: Why do HCl, HNO₃, etc., show acidic characters in aqueous solutions while solutions of compounds like alcohol and glucose do not show acidic character?

ANS: HCl, HNO₃, etc. form hydronium ions in aqueous solutions while alcohol and glucose do not dissociates hydronium ion in the solution instead they contain hydrogen. Hence, HCl, HNO₃, etc show acidic characters in aqueous solution while solutions of compounds like alcohol and glucose do not.

Q NO6: Why does an aqueous solution of an acid conduct electricity?

ANS: Acid dissociates ions in aqueous solution, which conducts electricity.

Q NO 7: Why does dry HCl gas not change the colour of the dry litmus paper?

ANS: Because dry HCl gas does not contain hydronium ions. Hence, it does not changes the colour of dry litmus paper.

Q NO 8: While diluting an acid, why is it recommended that the acid should be added to water and not water to the acid?

ANS: When acid is mixed with water it evolves heat. The reaction takes place in this process is exothermic. If water is mixed with acid, then because of presence of large amount of acid, more heat would be evolved, which may splashes out and can be resulted into accident due to burning. Therefore, it is recommended that the acid should be added to water and not water to the acid.

Q NO 9: How is the concentration of hydronium ions (H₃O⁺) affected when a solution of an acid is diluted?

ANS: The concentration of hydronium ions (H₃O⁺) decreases when a solution of an acid is diluted.

Q NO 10: How is the concentration of hydroxide ions (OH⁻) affected when excess base is dissolved in a solution of sodium hydroxide?

ANS: The concentration of hydroxide ions increase when excess base is dissolved in a solution of sodium hydroxide.

Q NO 11: You have two solutions, A and B. The pH of solution A is 6 and pH of solution B is 8. Which solution has more hydrogen ion concentration? Which of this is acidic and which one is basic?

ANS: An acidic solution has pH value less than 7 and a basic solution has pH value more than 7. Hence, the solution A having pH value is 6 is acidic consequently has more hydrogen ion concentration. The solution B which has pH value is 8 is basic solution.

Q NO 12: What effect does the concentration of H⁺(aq) ions have on the nature of the solution?

ANS: The concentration of H⁺ (aq) ions determines the acidic nature of solution. Hence, acidity of a solution increases with increases in concentration of H⁺ (aq) ions and vice versa.

Q NO 13: Do basic solutions also have H⁺(aq) ions? If yes, then why are these basic?

ANS: Yes basic solutions also have H⁺(aq) ions. They are basic because the concentration of hydroxide (OH⁻) is more than the H⁺(aq) ions.

Q NO 14: Under what soil condition do you think a farmer would treat the soil of his fields with quick lime (calcium oxide) or slaked lime (calcium hydroxide) or chalk (calcium carbonate)?

ANS: If the soil is acidic then it should be treated with quick lime or slaked lime to make it neutralize so that soil can be made proper for cultivation

Q NO 15: What is the common name of the compound CaOCl_2 ?

ANS: Common name of CaOCl_2 is bleaching powder.

Q NO 16: Name the substance which on treatment with chlorine yields bleaching powder.

ANS: Calcium hydroxide $[\text{Ca}(\text{OH})_2]$ yields bleaching powder when treated with chlorine.

Q NO 17: Name the sodium compound which is used for softening hard water.

ANS: Sodium carbonate (Washing soda) is used for softening the hard water.

Q NO 18: What will happen if a solution of sodium hydrocarbonate is heated? Give the equation of the reaction involved.

ANS: When solution of sodium hydrocarbonate is heated, sodium carbonate, water and carbon dioxide gas is formed.



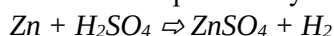
Q NO 19: Write an equation to show the reaction between Plaster of Paris and water.

ANS: $\text{CaSO}_4 \cdot (0.5)\text{H}_2\text{O} + (1.5)\text{H}_2\text{O} \Rightarrow \text{CaSO}_4 \cdot 2\text{H}_2\text{O}$

Q NO 20: Write word equations and then balanced equations for the reaction taking place when

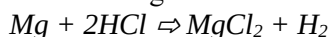
a. Dilute sulphuric acid reacts with zinc granules.

ANS: Sulphuric acid + Zinc $\rightarrow$ Zinc sulphate + Hydrogen



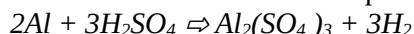
b. Dilute hydrochloric acid reacts with magnesium ribbon.

ANS: Hydrochloric acid + Magnesium $\rightarrow$ Magnesium chloride + Hydrogen



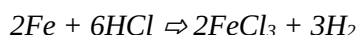
c. Dilute sulphuric acid reacts with aluminium powder.

ANS: Sulphuric acid + Aluminium $\rightarrow$ Aluminium sulphate + Hydrogen



d. Dilute hydrochloric acid reacts with iron filings.

ANS: Sulphuric acid + Iron $\rightarrow$ Ferric chloride + Hydrogen



1. A solution turns red litmus blue, its pH is likely to be

(a). 1

(b). 4

(c). 5

(d). 10

ANS: (d) 10

Q NO 21: A solution reacts with crushed egg-shells to give a gas that turns lime-water milky. The solution contains:

(a). NaCl

(b). HCl

(c). LiCl

(d). KCl

ANS: (b) HCl

Q NO 22: 10 mL of a solution of NaOH is found to be completely neutralized by 8 mL of a given solution of HCl. If we take 20 mL of the same solution of NaOH, the amount HCl solution (the same solution as before) required to neutralise it will be

(a). 4 mL

(b). 8 mL

(c). 12 mL

(d). 16 mL

ANS: (d) 16 mL

Q NO 23: Which one of the following types of medicines is used for treating indigestion?

(a). Antibiotic

(b). Analgesic

(c). Antacid

(d). Antiseptic

ANS: (c) Antacid

Q NO 24: Compounds such as alcohols and glucose also contain hydrogen but are not categorized as acids. Describe an Activity to prove it.

ANS: Following activity can be done for this:

- Take solutions of glucose, alcohol, hydrochloric acid, sulphuric acid, etc.
- Fix two nails on a cork, and place the cork in a 100 mL beaker.
- Connect the nails to the two terminals of a 6 volt battery through a bulb and a switch, as shown in figure.
- Now pour some glucose solution in the beaker and switch on the current.
- Repeat with alcohol solution and hydrochloric acid.

It is observed that bulb does not glow using glucose or alcohol solution. While if a solution of acid is used instead of glucose or alcohol solution, bulb glows.

Bulb glows when the solution of hydrochloric acid is used, because it dissociates hydronium ion in aqueous solution. But glucose and alcohol do not dissociate hydronium ion in the water, which is responsible for the conduction of electricity. Hence solution of glucose and alcohol do not conduct electricity.

Q NO 25: Why does distilled water not conduct electricity, whereas rain water does?

ANS: Rain water contains acids and many other substances as impurities, which produce hydronium ion, while distilled water does not contain any ionic substance that can dissociate hydronium ion. That's why rain water conduct electricity and distilled water does not.

Q NO

26: Why do acids not show acidic behaviour in the absence of water?

ANS: Hydronium ions are responsible for behavior as acid. Acids do not dissociate hydronium ion in the absence of water, hence they do not show acidic behavior in absence of water.

Q NO 27: Five solutions A, B, C, D and E when tested with universal indicator showed pH as 4, 1, 11, 7 and 9, respectively. Which solution is

a. neutral?

ANS: D is neutral, as its pH value is equal to 7

b. strongly alkaline?

ANS: C is strongly alkaline, as its pH value is 11.

c. strongly acidic?

ANS: B is strongly acidic, as its pH value is equal to 1

d. weakly acidic?

ANS: A is weakly acidic, as its pH value is equal to 4 and more than solution B.

e. weakly alkaline?

ANS: E is weakly alkaline, as its pH value is equal to 9 and less than C.

pH value of the given solution in Increasing order of hydrogen- ion concentration:

$11 < 9 < 7 < 4 < 1$

Q NO 28: Equal lengths of magnesium ribbons are taken in test tubes A and B. Hydrochloric acid (HCl) is added to test tube A, while acetic acid (CH_3COOH) is added to test tube B. Amount and concentration taken for both the acids are same. In which test tube will the fizzing occur more vigorously and why?

ANS: In test tube A fizzing occurs more vigorously. Because HCl is stronger acid than acetic acid. Hence, HCl liberates hydrogen gas more vigorously, which causes fizzing more vigorously.

Q NO 29: Fresh milk has a pH of 6. How do you think the pH will change as it turns into curd? Explain your answer.

ANS: The value of pH of milk decreases, when it turns into curd. Curd is acidic in nature. Because an acidic substance has less value of pH.

Q NO 29: A milkman adds a very small amount of baking soda to fresh milk.

(a). Why does he shift the pH of the fresh milk from 6 to slightly alkaline?

ANS: By making the milk more alkaline, it is prevent for more time to turn into curd. That's why milkman shifts the pH of fresh milk to slightly alkaline by adding a very small amount of baking soda to it.

(b). Why does this milk take a long time to set as curd?

ANS: As this milk is slightly more alkaline than other, therefore, acid produced to set into curd will be neutralized by baking soda added by milkman. Hence, this milk takes a longer time to set as curd.

Q NO 30: Plaster of Paris should be stored in a moisture-proof container. Explain why?

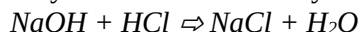
ANS: Because plaster of Paris turns into Gypsum after reacting with moisture present in air. That's why Plaster of Paris should be stored in a moisture-proof container.

Q

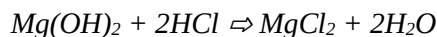
NO 31: What is a neutralization reaction? Give two examples.

ANS: When acid is added to base, it produces salt and water after neutralizing each other, such reactions are known as neutralization reaction.

Example 1: When Sodium hydroxide is added to hydrochloric acid. Sodium chloride and water is formed.



Example 2: Milk of magnesia, which is a base, is given as antacid in the case of indigestion, to neutralize the more acid produced in stomach.



Q NO 32: Give two important uses of washing soda and baking soda.

ANS: Two important uses of washing soda and baking soda are as follows:

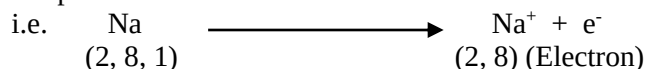
- Washing soda is used to remove the permanent hardness of water.
- Washing soda is used in making soap.
- Baking soda is used as baking powder, in the making of bread, and other many items.
- Baking soda is used in fire-extinguisher.

METALS & NON-METALS

METALS: Metals are defined as the elements which loose the electrons from their outermost shell to form the positive ions, hence called as electropositive elements (except hydrogen which losses electron but is a non-metal. The electropositive behaviour of metals is due to low ionisation energies. It is this electropositive behaviour of metals which produces their characteristic chemical properties.

The most abundant metal in the earth's crust is aluminium, which constitutes about 7% of the earths crust. The second most abundant metal in the earths crust is iron, which constitutes about 4% of the earths crust.

For example: Sodium is a metal with atomic number 11, i.e. it has 11 electrons revolving around its nucleus with the following electronic configuration 2, 8, 1 with only one electron its outermost shell which it will donate to some other element to form the positive sodium ion.



Some examples of metals are sodium(Na), Potassium(K), Magnesium(Mg), Calcium(Ca), Aluminium(Al), Copper(Cu), Gold(Au), Silver(Ag) etc.

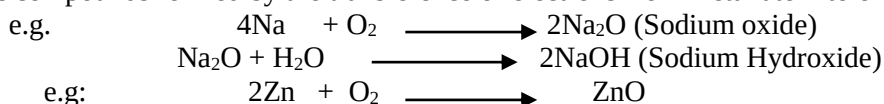
Physical properties of metals: The important physical properties of the metals are

- Metals in their pure state possess shiny surface known as metallic lustre. Most of the metals lose their brightness when exposed to air for a long time and acquire dullness due to the corrosion. It is the shiny appearance that makes metals good reflectors of light.
- Metals are highly malleable, i.e. metals can be drawn into sheets when beaten (hammered) without breaking. Gold and silver are the best malleable metals.
- Metals are highly ductile, i.e. metals can be drawn into thin wires. All metals are not equally ductile Gold is one of the best ductile metals, 1g of gold can be drawn into a thin wire of about 2km.
- Metals are good conductors of heat, i.e. metals can allow heat to pass through them. Silver is the best conductor of heat. The poorest conductor of heat among the metals is lead.
- Metals are good conductors of electricity, i.e. metals allow the electric current to pass through them and offer less resistance to the flow of current. Silver and copper are the best electrical conductors.
- The metals are solid at room temperature. The only exception is mercury. Gallium also has melting point (30°C) at the room temperature. It melts when held in the hand.
- The metals possess high tensile strength. That is metals can be bent to a very large extent and heavy weights can be suspended from metallic wires without breaking the wire.
- Metals possess high densities. Among metals osmium is the heaviest metal with density 22 g cm⁻³ and lithium is the lightest metal with density 0.5 gcm⁻³.
- Most of the metals are hard except some alkali metals like sodium and potassium which can be cut easily with a knife.
- All metals possess high melting and boiling points except mercury, gallium and alkali metals. Among metals tungsten has the highest melting point and mercury has the lowest melting point.e.g: Iron metal has 1535°C. There are some exceptions like Sodium (98°C), Potassium (64°C), Gallium (30°C) & Cesium (28°C).
- Metals form homogenous mixtures with each other and also with non metals to form alloys. e.g. Brass is an alloy of copper and zinc.
- Metals are generally sonorous, i.e. metals produce sound when hit with some other solid objects.
- Metals usually have a silver or grey colour except copper and gold which have reddish brown and yellowish colour respectively.

- xiv. Metals are lustrous or having shining appearance and are used in decoration like in making jewellery. The metals lose their shine or brightness on keeping in air for a long time and acquire a dull appearance due to the formation of a thin layer of oxide, carbonate or sulphide on their surface.
- xv. Metals are strong except sodium and potassium metals which are not strong.

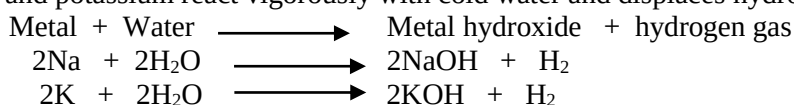
Chemical properties of metals: Some of the important chemical properties of metals are:

- i. **Reaction of metals with oxygen (air):** All metals combine with oxygen to form their respective metal oxides. Metal oxides are basic in nature. Some of the metal oxides are insoluble in water but some metal oxides combine with water to form alkalis. Due to their basic nature metal oxides turns red litmus paper red. Metal oxides are ionic compounds formed by the transference of electrons from metal atom to oxygen atom.

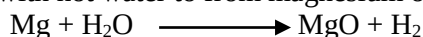


- ii. **Reaction of metals with water:** Metals react with water to form respective metal hydroxide (or metal oxide) and liberates the hydrogen gas. Less reactive metals like copper, silver & gold do not displace hydrogen from the water. Some metals react with water vigorously and some metals do not react with water under ordinary conditions.

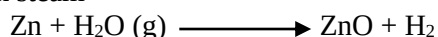
e.g. Sodium and potassium react vigorously with cold water and displaces hydrogen gas.



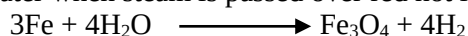
Magnesium reacts very rapidly with hot water to form magnesium oxide and it does not react with cold water.



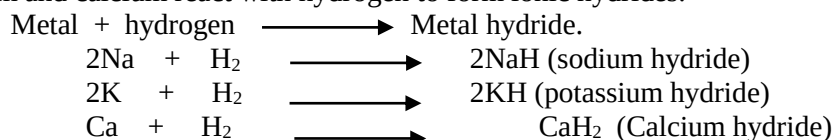
Zinc reacts rapidly but only with steam



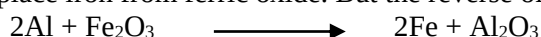
Iron displaces hydrogen from water when steam is passed over red hot iron. The reaction is reversible in nature.



- iii. **Reaction of metals with hydrogen:** Most of the metals do not react with hydrogen. A few reactive metals such as sodium, potassium and calcium react with hydrogen to form ionic hydrides.



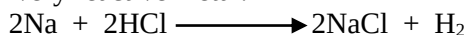
- iv. **Reaction of metals with metal oxides:** A more reactive metal can displace a less reactive metal from its oxide. e.g. Aluminium can displace iron from ferric oxide. But the reverse of such reactions is not possible.



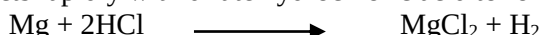
- v. **Reaction of metals with acids:** Metals usually displace hydrogen gas from dilute acids. Only less reactive metals like copper, silver and gold do not react with dilute acids. Metals on reacting with acids produce salt and displace hydrogen gas.



e.g: Sodium reacts with dilute hydrochloric acid with an explosive violence and hence show that sodium very reactive metal.



Magnesium reacts rapidly with dilute hydrochloric acid to form magnesium chloride and hydrogen gas.

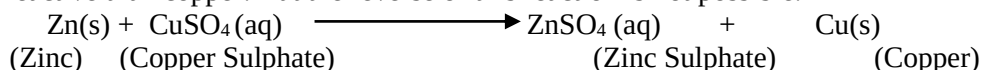


Zinc combines with dilute hydrochloric acid, but less rapidly than magnesium. This shows zinc is less reactive than magnesium.

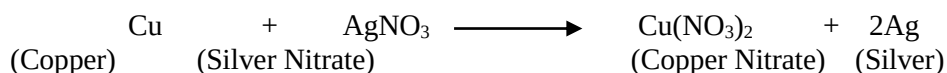


- vi. **Reaction of metal with metal salts (Displacement reaction):** The basic principle of displacement reactions is that a more reactive metal can displace a less reactive metal from the solution of its salt.

e.g. Zinc when combines with copper sulphate produce zinc sulphate solution and displaces copper, shows that zinc is more reactive than copper. But the reverse of this reaction is not possible.



Copper when combines with silver nitrate replaces silver to form copper nitrate. This shows that copper is more reactive than silver.



vii. **Reaction of metal with Chlorine:** When ever metals react with chlorine they form ionic chlorides. Metals chlorides are usually solid and conduct electricity hence are electrolytes.

Metal + Chlorine $\longrightarrow$ Metal chloride

2Na + Cl₂ $\longrightarrow$ 2NaCl

Mg + Cl₂ $\longrightarrow$ MgCl₂

Cu + Cl₂ $\longrightarrow$ CuCl₂

AQUA REGIA: Aqua regia is a freshly prepared mixture of one part of concentrated nitric acid and three parts of concentrated hydrochloric acid. thus the ratio of concentrated HNO₃ and HCl in aqua regia is 1:3. Aqua regia is highly corrosive and can dissolve all metals.

REACTIVITY SERIES OF METALS (ACTIVITY SERIES): The arrangement of metals in a vertical column in order of their decreasing reactivity downwards is called as the activity series or reactivity series of metals. The arrangement of metals in such a order or series is made on the basis of their chemical reactivity.

Although most metals are usually electropositive in nature and lose electrons in a chemical reaction they do not react with the same vigour or speed. Metals display different reactions towards different substances. The greater the ease with which an element loses its electrons and acquires a positive charge, the greater is its reactivity. Further, the greater the number of shells and lesser the number of valence electrons, the greater is the reactivity of the metal.

The activity series of metals, arranges all metals in order of their decreasing chemical activity. As we go down the activity series from potassium to gold the ease with which a metal loses electrons, and forms positive ions in solutions, decreases. The most active metal, potassium, is at the top of the list and the least reactive metal, gold, is at the bottom of the list. Although hydrogen is a non-metal it is included in the activity series due to the fact that it behaves like a metal in most chemical reactions i.e., the hydrogen ion has a positive charge [H⁺] like other metals.

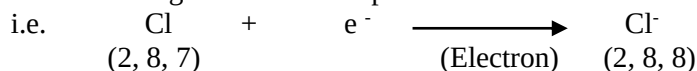
	<u>Element</u>	<u>Symbol</u>		<u>Element</u>	<u>Symbol</u>	<u>Group Number</u>
These metals are More reactive than Hydrogen	Potassium	K (Most reactive metal)		Potassium	K	1
	Barium	Ba		Sodium	Na	1
	Calcium	Ca		Lithium	Li	1
	Sodium	Na		Calcium	Ca	2
	Magnesium	Mg		Magnesium	Mg	2
	Aluminium	Al		aluminium	Al	3
	Zinc	Zn		Carbon	C	4 (non-metal)
	Iron	Fe		Zinc	Zn	Transition metal
	Nickel	Ni		Iron	Fe	Transition metal
	Tin	Sn		Tin	Sn	4
These metals are Less reactive than hydrogen	Lead	Pb		Lead	Pb	4
	Hydrogen	H		Hydrogen	H	Non-metal
	Copper	Cu		Copper	Cu	Transition metal
	Mercury	Hg		Silver	Ag	Transition metal
	Silver	Ag		Gold	Au	Transition metal
	Platinum	Pt		Platinum	Pt	Transition metal
	Gold	Au (Less reactive metal)				

Metals occupying higher position in the series have more tendency to lose electrons and are more reactive. The metals at the bottom of the series are least reactive.

NON-METALS: Non-metals are defined as the elements which gain the electrons to form the negative ions, hence are called as electronegative elements. Since large amount of energy is needed to remove one or more electrons from non metals, so non metals do not form positively charged ions. On the other hand, energy is released when a non metals accepts one or more electrons. Thus non metals readily form negatively charged ions by the gain of electrons. Therefore non-metals are electronegative elements.

The most abundant non-metal in the earths crust is oxygen, Which constitutes about 50% of the earths crust. The second most abundant non-metal in the crust is silicon, which constitutes about 26% of the crust. All the non-metal are 22 in number. Out of these, 10 non-metals are solids, 1 non-metal is a liquid where as the remaining 11 non-metals are gases.

For example: Chlorine is a non metal with atomic number 17, i.e. it has 17 electrons revolving round the nucleus in their orbits with the following electronic configuration 2,8,7 with 7 electrons in its outermost shell. It has to gain an electron from some other combining element to complete its outermost shell to form the negative chlorine ion.



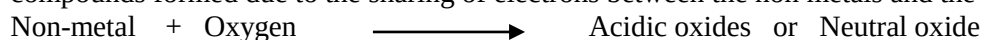
Some examples of non-metals are Hydrogen (H), Oxygen (O), Nitrogen (N), Chlorine (Cl), Bromine (Br), Sulphur(S), Phosphorous (P), Carbon(C) etc.

Physical properties of non metals:- Some of important physical properties of non metals are:

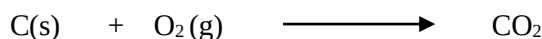
- Non-metals do not possess any lustre (except iodine which is a non metallic solid but has lustre).
- Non-metals are soft and brittle; non-metals can break into pieces when hammered except diamond which is the hardest known substance.
- Non-metals are neither malleable nor ductile, i.e. non metals cannot be drawn into sheets and thin wires.
- Non-metals are generally bad conductors of heat and electricity except graphite an allotropic form of carbon which is a good conductor of electricity due to the presence of free electrons in it.
- Non-metals are non sonorous substances. i.e. non metals cannot produce sounds when hit with a hard object.
- Non-metals generally have low melting and boiling points except boron diamond, and graphite which have high melting and boiling points.
- Non-metals have generally low densities, i.e. non metals are generally lighter than metals. Sulphur has the lowest density of 2.0 g cm^{-3} .
- Non-metals have low tensile strength. i.e. non metals can be broken easily and cannot have the resistance to bear the force as that of the metals.
- Non-metals can exist in solid, liquid, and gaseous states at room temperature. Carbon sulphur, phosphorous, and iodine are solid, bromine is the only non-metal in liquid state while as hydrogen, oxygen chlorine etc. are gaseous in state.

Chemical properties of non-metals: Some of the important chemical properties of non-metals are

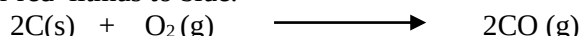
- Reaction non-metals with oxygen:** Non-metals when heated react with oxygen to form their respective oxides which are either acidic or neutral in nature and are never basic in nature. The non-metallic oxides are covalent compounds formed due to the sharing of electrons between the non metals and the oxygen atom.



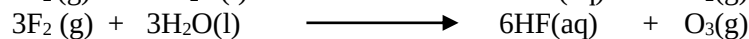
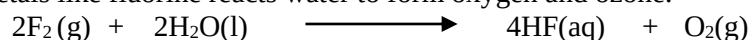
The non-metallic oxides which react dissolve in water to form acids are called as acidic oxides. These oxides turn blue litmus paper red.



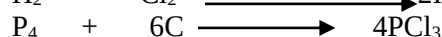
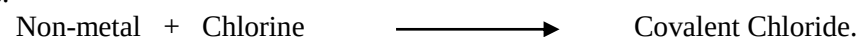
The non-metallic oxides which neither react with acids nor with bases are called as neutral oxides. These neither turn blue litmus paper red nor red litmus to blue.



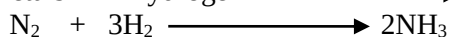
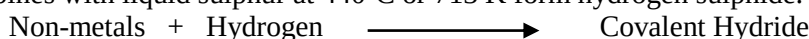
- Reaction of non-metals with water:** Non-metals usually do not react with water (or steam) to produce H_2 gas. It is because non-metals cannot reduce the hydrogen (H^+) ions of water to hydrogen gas. However some highly reactive non-metals like fluorine reacts water to form oxygen and ozone.



- Reaction of non-metals with dilute acids:** Non-metals do not react with acids to displace hydrogen gas, it is because non-metals are electron acceptors and cannot supply electrons to H^+ ions to reduce them to hydrogen gas.
- Reaction of non-metals with chlorine:** Non-metals react with chlorine to form covalent chlorides which are non-electrolytes. It is because non-metals combine with chlorine to form covalent chlorides by sharing electrons, and they cannot give electrons to chlorine atoms to form chloride ions. These covalent chlorides are generally volatile liquids or gases.

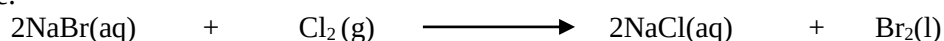


- Reaction of non-metals with hydrogen:** Non-metals combine with hydrogen to form covalent hydrides. These hydrides are generally gases or liquids, but these hydrides do not conduct electricity. e.g. Hydrogen when combines with liquid sulphur at 440°C or 713 K form hydrogen sulphide.



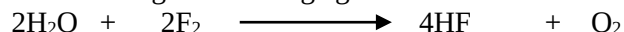
- Reaction of non-metals with salts:** A more reactive non-metal displaces a less reactive non-metal from the solution of its salt.

e.g. when chlorine is passed through the solution of sodium bromide, bromine is displaced with the formation of sodium chloride.



- vii. **Oxidising behaviour of non-metals:** Non-metals have a strong tendency to accept electron to form negative ions. Therefore non-metals behave as oxidising agents.

e.g. fluorine is the strongest oxidising agent. It oxidises water to oxygen and hydrogen fluoride.



❖ **Difference between metals and non-metals:**

METALS	NON-METALS
1. Metals show metallic lustre (shiny appearance). 2. Metals are generally hard except sodium and potassium. 3. Metals are malleable and ductile. (can be drawn into sheets and wires). 4. Metals are good conductors of heat and electricity. 5. Metals generally have high melting and boiling points, except mercury and gallium. 6. Metals are sonorous (can produce sound when hit with a solid). 7. Metals generally have high density. 8. Metals generally have high tensile strength and cannot be broken easily. 9. Metals are generally solid (except mercury gallium which are liquids) at room temperature. 10. Metals are electropositive elements and can lose electrons. 11. Metals form basic oxides 12. Metals which are more reactive than hydrogen displace hydrogen from dilute acids. 13. Metals have a strong tendency to lose electrons and hence behave as reducing agents. 14. Metals generally combine with chlorine to form ionic chlorides.	1. Non-metals do not show metallic lustre (Shiny appearance). 2. Non-metals are generally soft except diamond and boron. 3. Non-metals are neither malleable nor ductile. They are brittle. 4. Non-metals are generally bad conductors of heat and electricity except graphite and gas carbon. 5. Non-metals generally have low melting and boiling points. 6. Non-metals are generally non sonorous (cannot produce sound when hit with a solid). 7. Non-metals have generally low densities. 8. Non-metals generally have low tensile strength and can be broken easily. 9. Non-metals are either solids or gases at room temperature, only bromine is in liquid state at room temperature. 10. Non-metals are electronegative elements and hence gain electrons to form negative ions. 11. Non-metals form either acidic or neutral oxides. 12. Non metals do not react with dilute acids and hence do not displace hydrogen from dilute acids. 13. Non metals have a strong tendency to accept electrons and hence behave as oxidising agents. 14. Non-metals combine with chlorine to form covalent chlorides.

USES OF METALS:

Some general uses of metals are described below:

- **Copper and Aluminium:** - Copper and aluminium are very good conductors of electricity. So, copper and aluminium are used for making electrical wires and cables. Aluminium is also used for making cooking utensils. Aluminium foils are used for packaging of medicines, food materials, etc.
- **Iron:** Iron is the most widely used metal. Its uses, however, mainly depend upon its carbon content. For example,
- Steels are used for the manufacture of permanent magnets, engine parts, utensils, surgical equipments, springs, gears, drive shafts, armour plates, etc.
- Wrought iron is used for making anchors, wire ropes, bolts, chains and agricultural appliances.
- Zinc is used for the galvanization of iron.
- Nickel and chromium are mainly used for electroplating and for the manufacture of stainless steel.
- Gold and silver are used for making jewellery and for decorative purposes.
- Sodium, titanium and zirconium are used in atomic energy and space science projects.
- Mercury is used in thermometers and barometers.
- Titanium is used in
 - i. Aerospace
 - ii. Aircraft frames and engine
 - iii. Military hardware
 - iii. Atomic energy and space science projects.
- Titanium due to its special applications is considered as a strategic metal. These applications of titanium are due to its

- i. High tensile strength
- ii. Lightness
- iii. Resistance to corrosion
- iv. High melting and boiling points.

USES OF NON-METALS:

Some general uses of Non Metals are:

Uses of silicon

- Silicon is extensively used in the electronic and computer industries to make devices, such as transistors, microchips and solar cells.
- It is used to manufacture silicones, which are used for making waterproof clothes, greases and polishes, and insulating material for electrical appliances.

Uses of Sulphur

- It is used in the manufacture of sulphuric acid, which is used in several industrial processes.
- It is used to skin ointments because of its fungicidal properties.
- Being an insecticide and fungicide, it is used to spray fruit trees against pests and diseases.

Uses of phosphorous

- Phosphorous is used in making match boxes and in the fireworks industries.
- It is used in the manufacture of phosphate fertilizers.

Uses of Graphite

- Graphite is used in making electrodes in dry cells and electric arcs. It is also used as a lubricant.

Uses of Hydrogen

- It is used for the manufacture of ammonia gas, which is used to manufacture fertilizers such as urea and ammonium sulphate.
- It is used for hydrogenation of vegetable oils to manufacture vanaspati.
- It is used in the extraction of metals such as copper, lead and tin.

HOW DO METAL & NON-METALS REACT: As for the chemical properties of metals and non-metals are concerned it is being observed that both react with a number of reagents to form new compounds or new molecules. The most common compounds are found between metals and non-metals like Sodium Chloride (NaCl), Magnesium Chloride (MgCl₂) etc. or between non-metal atoms itself like Hydrochloric acid (HCl), Water (H₂O).

The force which holds a number of atoms together within the compound or the molecule is known as a chemical bond.

If we observe the electronic configuration of noble gases, metals and non-metals, it is clear that noble gases have their outermost shell completely filled with require 8 electrons except helium which has only 2 electrons in its outermost shell as its first shell is its last shell, and hence they are stable due to which they does not require to lose or gain electrons from the outermost shells and hence does not react with any other element. On the other hand the outer most shells of almost all metals and non-metals are having less than eight (8) electrons, so to obtain the stable electronic configuration an element either has to lose or gain electron in its outermost shell by reacting with other elements.

Since metals and non-metals also have to acquire the stability in their electronic configuration by combining with other elements in order to lose or gain electrons to complete their outer most shell.

Thus, the cause of chemical combination is the tendency of the atoms to complete their octet, (i.e. outermost shell with 8 electrons) or duplet (i.e. outermost shell with 2 electrons in case of elements having K shell as their first and last shell) so that they acquire the stable nearest noble gas configuration. The atoms combine with one another to achieve the inert gas electronic arrangement and become more stable. An atom can achieve the inert gas electronic configuration in three ways:

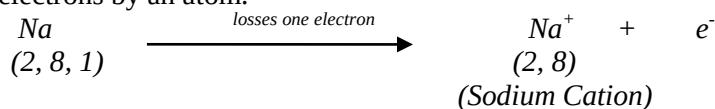
1. By losing one or more electrons towards another atom.
2. By gaining one or more electrons from another atom.
3. By sharing one or more electrons with another atom.

The chemical reaction in which the inert gas electronic arrangement is achieved by the loss and gain of electrons between atoms, take place between metals and non-metals. On the other hand the chemical reactions in which the inert gas electron configuration is achieved by the sharing of electrons between atoms, take place between non-metals and non-metals. In order to know the reaction between metals and non-metals it is important to know the term like Ion, and how they are formed.

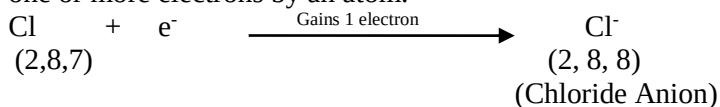
- ❖ **ION:** An ion is an electrically charged atom whether positive charge or negative charge. e.g: Mg²⁺, H⁺, Cl⁻, Na⁺
Ion is formed by the loss of electron or electrons by an atom, so it contains an unequal number of electrons and protons. There are two types of ions and are

- Cation

- Anion
- **Cation:** A positively charged atom is known as cation. E.g:- Na^+ , Mg^{2+} . A cation is formed by the loss of one or more electrons by an atom.



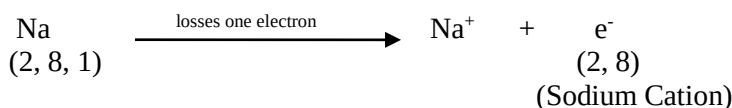
- **ANION:** A negatively charged atom is known as anion. e.g:- Cl^- , O^{2-} . An anion is formed by the gain of one or more electrons by an atom.



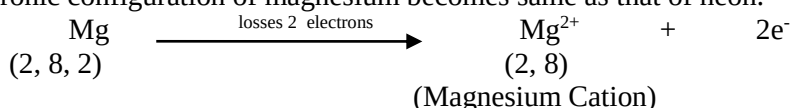
FORMATION OF POSITIVE IONS (CATIONS):

If an element has 1, 2 or 3 electrons in the outermost shell of its atom, then it loses these electrons to achieve the inert gas electronic configuration and forms positively charged atom or cation. The metal atoms lose electrons to form cations or positive atoms.

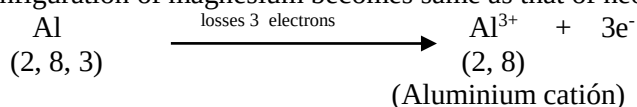
Example 1: Formation of Sodium ion:- Sodium atom (Na) with atomic number 11 and valence electron is only one, so, it can lose this valence electron towards non-metal and forms a positive charge. After losing the electron, the electronic configuration of sodium becomes same as that of neon & due to one more proton than electron, a sodium ion has 1 unit positive charge.



Example 2: Formation of Magnesium Ion:- Magnesium (Mg) with atomic number of 12 and valence electrons are 2, so, it can lose these two electrons towards any non-metal and acquires two positive charges. After losing these two electrons, The electronic configuration of magnesium becomes same as that of neon.



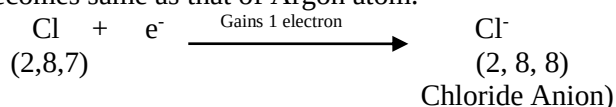
Example 3: Formation of Aluminium Ion:- Aluminium (Al) with atomic number of 13 and valence electrons are 3, so, it can lose these three electrons towards any non-metal and acquires three positive charges. After losing these electrons, the electronic configuration of magnesium becomes same as that of neon atom.



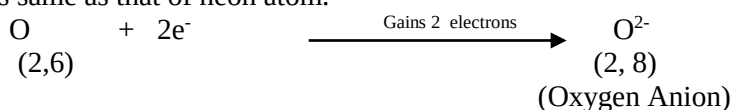
FORMATION OF NEGATIVE IONS (ANIONS):

If an element has 5, 6 or 7 electrons in the outermost shell of its atom, then it gains electron / electrons to achieve the stable noble gas configuration and forms negative charged atoms called anions by losing electrons (it is not possible to lose 5, 6 or 7 electrons from an atom due to requirement of very high energy). All Non-metals acquire negative charge by gaining electrons except hydrogen.

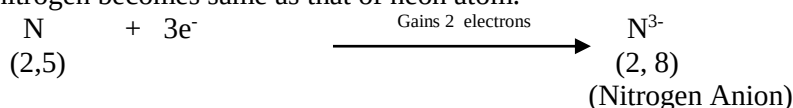
Example 1: Formation of Chlorine ion: Chlorine (Cl) with atomic number of 17 and valence electrons are 7, so, it can gain one electron from any metal and acquires a positive charge. After gaining one electron, the electronic configuration of magnesium becomes same as that of Argon atom.



Example 2: Formation of Oxygen ion: Oxygen (O) with atomic number of 8 and valence electrons are 6, so, it can gain two electrons from any metal and acquires two positive charges. After gaining two electrons, the electronic configuration of oxygen becomes same as that of neon atom.



Example 3: Formation of Nitrogen ion: Nitrogen (N) with atomic number of 7 and valence electrons are 5, so, it can gain three electrons from any metal and acquires three positive charges. After gaining three electrons, the electronic configuration of nitrogen becomes same as that of neon atom.

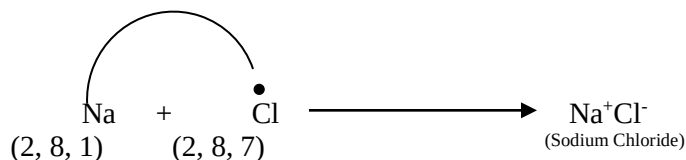


TYPES OF BONDS: There are two types of chemical bonds:

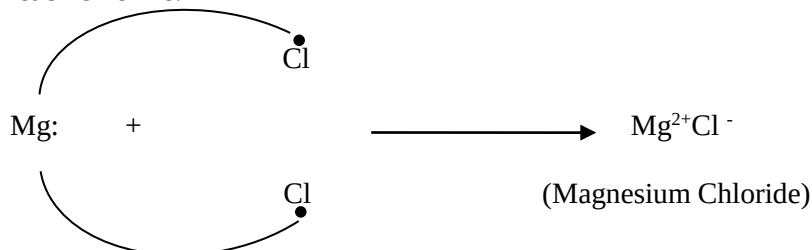
1. Ionic bond
2. Covalent bond

1. **Ionic bond:** Ionic bonds are formed by the transfer of electrons from one atom to another atom or transfer of electrons between metal and non-metal. An ionic bond is formed when one of the atoms can donate electrons to achieve the inert gas configuration, and the other atom needs electrons to achieve the inert gas configuration. Metal atoms have usually 1, 2 or 3 electrons and can donate these electrons. On the other hand non-metal atoms have usually 5, 6 or 7 valence electrons. So they need electrons. Thus a metal reacts with a non-metal and forms ionic bond. Thus ionic bonds are formed between metal and non-metal.

e.g: Formation of Sodium chloride: Sodium atom donates one electron to chlorine and forms an ionic bond.



e.g: Formation of Magnesium chloride: Magnesium atom donates two electrons to two chlorine atoms and forms two ionic bonds with each chlorine.



Thus, the compounds formed in this manner by the transference of electrons from a metal to a non-metal are known as ionic compounds or electrovalent compounds. (Ionic bond or electrovalent bond).

Ionic compounds are basically defined as being compounds where two or more ions are held next to each other by electrical attraction. One of the ions has a positive charge (called a cation) and the other has a negative charge (anion). Cations are usually metal atoms and anions are either non-metals or polyatomic ions (ions with more than one atom).

When a complete transfer of one or more valence electrons from the atom of a metal to that of a non-metal takes place, an ionic (or electrovalent) bond is formed. As a result of this electron transfer following changes occur in the reacting atoms.

Properties of Ionic Compounds:-

- Ionic compounds do not exist in the form of independent molecules but exist in the form of ions in crystalline solids,
- Ionic compounds have high melting and boiling points. A large amount of energy is required to break the ionic molecule, because the ions are held together by strong electrostatic forces of attraction. For example NaCl has a high melting point of 800°C and boiling point of 1413°C
- Ionic compounds are generally soluble in water and other polar solvents having high dielectric constants and are insoluble in organic solvents like ether, Acetone, benzene, kerosene etc. This is due to larger electrostatic interactions between polar solvents and ions. However ionic compounds are insoluble in non-polar solvents.
- Ionic compounds are good conductors of electricity in these states. However, these do not conduct electricity in their solid state because of strong electrostatic forces between the ions, which does not make them mobile.
- In ionic-compounds each ion is surrounded by oppositely charged ions uniformly distributed all around the ion, thus resulting in a three dimensional solid aggregate called an ionic crystal.

2. **COVALENT BOND:** The chemical bond formed by the sharing of electrons between two non metal atoms known as chemical bond. After sharing of electrons each atom achieves noble gas electronic configuration. Non-metals usually have 5, 6 or 7 electrons in the valence shell, so, all the non metal needs electrons to achieve noble gas configuration. They get these electrons by mutual sharing. So, whenever a non-metal combines with another or same type of non-metal sharing of electrons take place between their atoms and a covalent bond is formed. Covalent bonds are of three types and are:-

- **SINGLE BOND**
- **DOUBLE BOND**
- **TRIPLE BOND**

SINGLE BOND: A single covalent bond is formed of two electrons shared between two atoms. A single covalent bond is denoted by putting a line (-) between the atoms.

Example: Formation of chlorine molecule (Cl_2): A chlorine atom is very reactive non-metal and can't exist in free state because it does not have the stable electron arrangement of an inert gas. Each molecule of chlorine atoms joined by a single covalent bond. Valence electrons of chlorine atom are 7 and needs only one

electron to complete its octet and becomes stable. It gets this electron by sharing of one electron. So, two chlorine atoms share one electron each to form a chlorine molecule.

(a). Cl-Cl (b). H-Cl (c). CH_4 (d). CCl_4 .

DOUBLE BOND: A double covalent bond is formed of four electrons shared between two atoms and each atom contributes two electrons. A double covalent bond is denoted by putting two lines (=) between the atoms.

Example: Formation of oxygen molecule (O_2):- Oxygen atom is very reactive and can't exist free because it does not have the stable inert gas electronic configuration. So, oxygen can't be a single atom. It can be as O_2 . Oxygen has 6 valence electrons and needs only 2 electrons to complete its octet. The oxygen atom gets these 2 electrons by sharing of its two electrons with the other oxygen atom. So, two oxygen atoms share 2 electrons each to become stable.

(a). O_2 (b). CO_2 (c). C_2H_4

TRIPLE BOND: A triple covalent bond is formed of six electrons shared between two atoms and each atom contributes three electrons. A triple covalent bond is denoted by putting three lines ($\equiv$) between the atoms.

Example: Formation of Nitrogen molecule (N_2):- A nitrogen atom is very reactive and can't exist free because it does not have the stable electronic configuration. So, it does not consist of a single atom, it consists of a more stable N_2 molecule. Nitrogen atom has 5 valence electrons and needs 3 more electrons to become stable. So, two nitrogen atoms combine together by sharing 3 electrons each to form a molecule of nitrogen gas.

(a). N_2 (b). C_2H_2

properties covalent compounds:

- Covalent compounds are usually liquids or gases. Only some of them are solids. E.g: Alcohol, ether, benzene, carbon disulphide and bromine are liquids.
- Covalent compounds have usually low melting points and low boiling points. E.g: Naphthalene has a low melting point of 80°C and carbon tetrachloride has a boiling point of 77°C .
- Covalent compounds are usually insoluble in water but they are soluble in organic solvents. E.g: Naphthalene is insoluble in water but dissolves in organic solvents like ether.
- Covalent compounds do not conduct electricity. This means that covalent compounds are non-electrolytes and are electrically neutral molecules.

OCCURENCE OF METALS:

The earth's crust is the major source of metals. Most of the metals are quite reactive and hence they don't occur as free elements in nature. So, most of the Metals are found in the form of their compounds called combined state elements. Any natural materials in which the metals or their compounds are found are found in earth known as minerals. Some minerals may contain a large number of metals in them where as some minerals contain a less percent of metal in them. Thus all the minerals can't be used to extract metals. Those minerals from which metals can be extracted profitably and conveniently are known as ores. Ores contain metal compounds with a lower percentage of impurities. All the ores are minerals, but all minerals are not necessarily ores. Metals occur in nature in the free as well as in the combined states. The most less reactive metals are generally found in the Free State like silver, gold and platinum. Most metals however, are found in the combined form as minerals because they are highly reactive. Medium reactive metals are also found in combined state.

In the Free State: Very few metals exist in the free state or native state. Only metals like gold, platinum, copper, Mercury and Silver are occasionally found in the Free State i.e., in the pure form. Sometimes, copper and silver are found in the combined State. Such metals are not acted upon by air or water.

Native metals: Ag, Au, Pt, Cu, Hg.

In the Combined State: The rest of the metals occur in the combined form as compounds such as oxides, carbonates, sulphides, sulphates, silicates, chlorides, nitrates, phosphates etc. Copper and silver are two metals which occur in free as well as combined state as sulphide, oxide or halide ores. Metals at the top of the activity series (K, Na, Ca, Mg and Al) are so reactive that they are never found in nature as free elements. The metals in the middle of the activity series (Zn, Fe, Pb, etc.) are moderately reactive. They are found in the earth's crust mainly as oxides, sulphides or carbonates.

Combined form metals are:

As Phosphates: P

As nitrates: Na, K

As sulphates: Ba, Ca, Pb etc.

As carbonates: Cu, Zn, Pb, Ca, Na, Mg

As halides: Ag, Na, K, Mg (as chlorides) Al, Fe, Ca, (as fluorides)

As sulphides: Cu, Zn, Ag, Pb, Fe, Hg, As, etc.

As oxides: Cu, Zn, Al, Fe

Minerals: Natural materials found inside the earth containing metals in their combined states (as a single compound or as a mixture of compounds) mixed with non-metallic impurities of Earth and rock (called gangue), are termed minerals.

Ore: All the metals do not occur in the native form and need to be extracted for use. All minerals are not suitable for the extraction of metals. Minerals from which the metal can be extracted easily and economically are called ores. Thus, all ores are minerals, but all minerals are not ores.

Gangue (matrix): The impurity in the mineral like earth, rock, and sandy impurities are called as gangue or matrix.

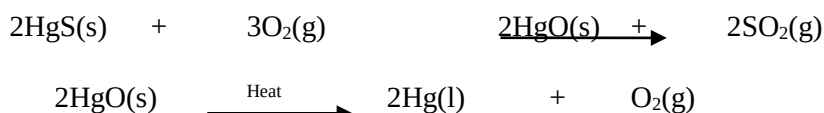
Extraction of metals (Metallurgy): An ore may contain the metal in the elementary state or in the form of its compounds along with gangue. Getting metal out of the ore is called as extraction of the metal. The various steps involved in the extraction of the metal its ores followed by refining of the metal is called as metallurgy. The steps involved in the extraction of a metal are

- i. Enrichment or Concentration of the ore.
- ii. Extraction of metal from the concentrated ore.
- iii. Refining of the impure metal.

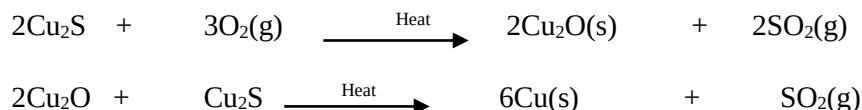
➤ **CONCENTRATION (ENRICHMENT) OF ORE:** concentration of ore means to separate the gangue and metal from an ore. Ore is an impure compound of a metal and a large amount of sand, mud, rocky substances & water. Before extracting the metals from the ore, it is necessary to remove these impurities (gangue). By removing the gangue, we get a concentrated ore containing much percentage of metal. The method concentration of the ore of a metal depends upon the physical and chemical properties of ore and the physical and the chemical properties of the gangue present in the ore. The various methods of concentration of the ores are: Hydraulic washing, Electromagnetic separation & Froth floatation process etc.

➤ **CONVERSION OF CONCENTRATED ORE INTO METAL:** Different methods are used for extracting metals from its concentrated ore and it belongs to the reactivity of the metal.

Extraction of metals which are low in the reactivity series (Cu, Hg, Ag, Au, Pt): Metals in the lower part of the activity series are very unreactive. The oxides of these metals can be reduced to metals by heating alone. Example: Cinnabar (HgS) is an ore of mercury, when it is heated in air, it is first converted into mercuric oxide (HgO). Mercuric oxide is then reduced to mercury on further heating.



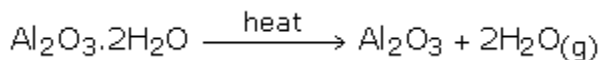
Similarly, copper glance(Cu_2S) can be reduced from its ore by just heating in air.



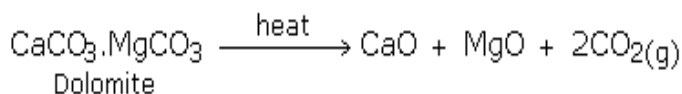
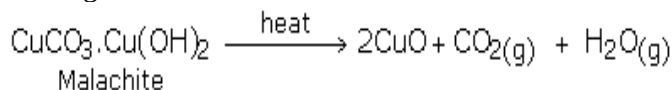
Extraction of metals which are in the middle of reactivity series (Fe, Zn, Pb, etc.): The metals present in the middle of the reactivity series are usually found in nature in the form of sulphides or carbonates. The concentrated ores can be converted into metal oxide by the process of Calcination and Roasting. It is easier to reduce oxides than sulphides or carbonates, so the sulphides and carbonate ores are first converted into the corresponding metal oxides.

i. **Calcination:** The concentrated ore is converted into oxide by Calcination i.e, heating it strongly in absence of air or roasting (heating it strongly in presence of air). This helps in removing volatile impurities like CO_2 , SO_2 , organic matter, and moisture from the ore. For example:

It removes moisture from bauxite.

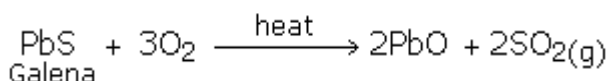


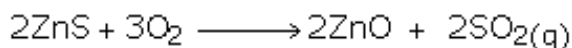
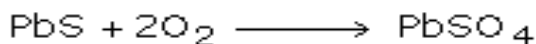
It removes CO_2 from carbonate ores e.g.



ii. **Roasting:** In this process the ore (usually sulphide) is heated strongly, in the presence of excess of air but below its melting temperature. The result is:

It removes moisture, CO_2 , SO_2 and organic matter. The sulphide ore is converted partly into its oxide or sulphate i.e.

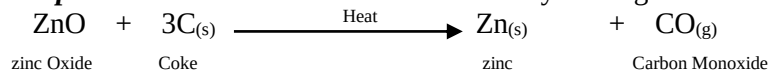




The roasting converts metal sulphide into metal oxide. The metal oxides obtained by calcinations and roasting are converted to the free metal by using reducing agents like carbon, aluminium, sodium or calcium.

- **Reduction of metal oxide to free metal by carbon (reducing agent):** Some metal oxide ore can be reduced by heating with carbon in the form of coke, the coke combines with the oxygen of the metal oxide forming carbon monoxide. As a result, metal oxide is reduced to metal.

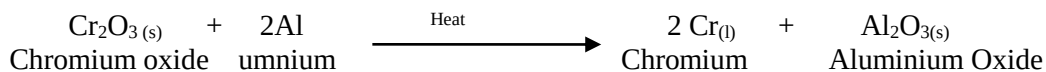
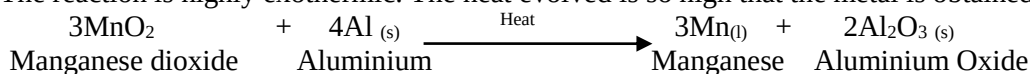
Example: reduction of zinc oxide to its metal by heating with coke.



The process of reduction of metal oxide by heating it with coke is called as smelting.

Some metal oxide ores can be reduced by heating with aluminium powder to produce large heat (exothermic reaction) is called thermite reaction or aluminothermy (thermite process).

Example: manganese oxide and chromium oxide can be reduced by this process as aluminium is more reactive than manganese or chromium. As a result, aluminium is converted into aluminium oxide whereas the metal oxide is reduced to the metal. The reaction is highly exothermic. The heat evolved is so high that the metal is obtained in the molten state.



The process of reduction of metal oxides to metal using aluminium as the reducing agent is known as aluminothermy.

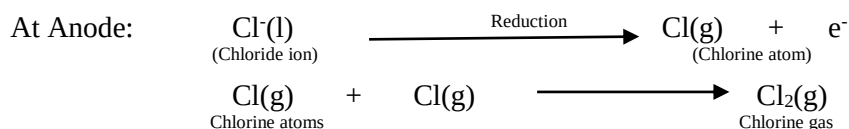
Extraction of metals which are at the top of reactivity series (K, Na, Ca, Mg, Al): Metals such as sodium, magnesium, calcium, aluminium high up in the reactivity series are very reactive and cannot be obtained from their compounds by heating with carbon.

This is because these metals have more affinity for oxygen than carbon. These metals are obtained by electrolytic reduction. Metals like K, Na, Ca, Mg, AL cannot be reduced by coke or carbon monoxide. Electricity is used to reduce these metals.

Example: Electrolysis of molten sodium chloride. When sodium chloride melts, it splits into sodium ion (Na^+) and chlorine ion (Cl^-)



When electricity is passed through the molten sodium chloride, Na^+ ions go to the cathode whereas Cl^- ions are liberated at the anode, Na^+ ions gain electrons at the cathode and are thus reduced to sodium atoms. Cl^- ions lose electrons at the anode and are thus oxidized to chlorine atoms. These chlorine atoms then combine with each other to form chlorine (Cl_2) gas.



Thus, sodium metal is obtained at the cathode whereas chlorine gas is liberated at the anode.

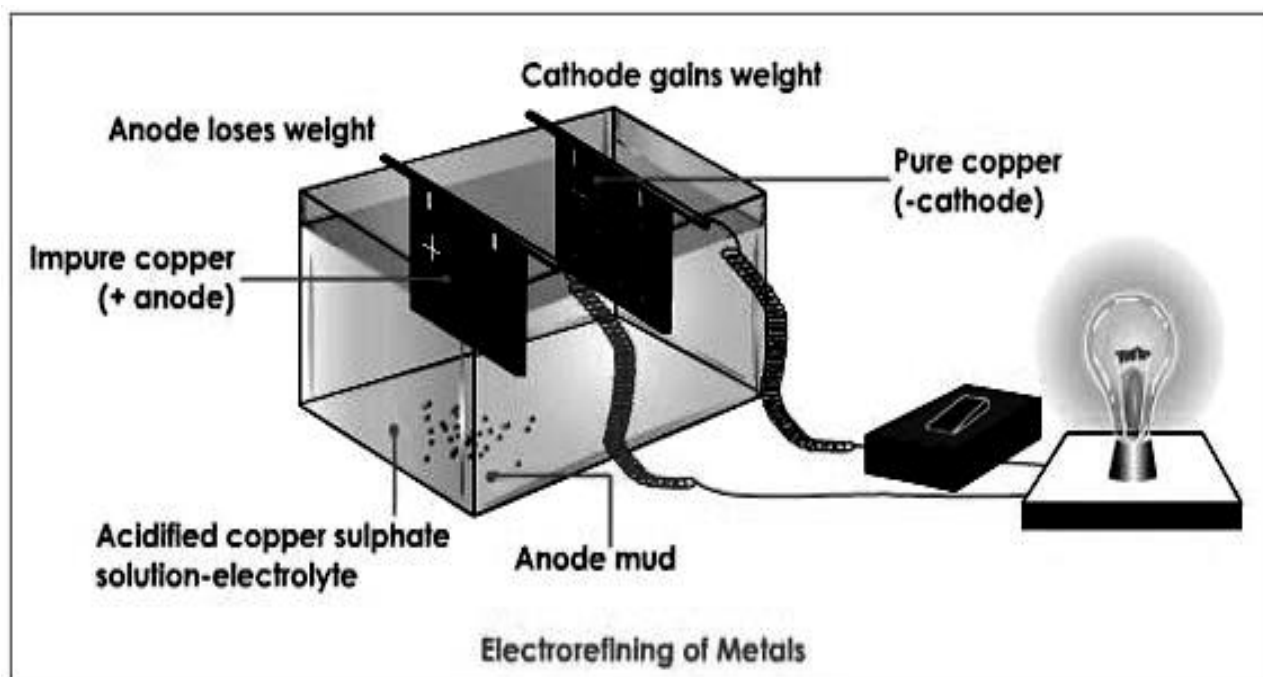
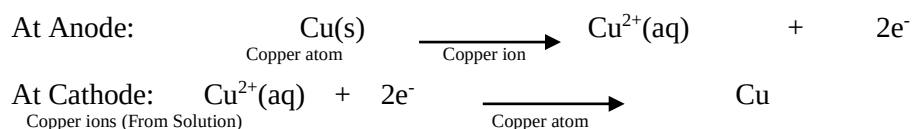
The differences between roasting and Calcination:

Roasting	Calcination
1. Heating in the presence of air 2. Sulphide ores are roasted 3. Sulphur dioxide is released $2\text{ZnS} + 3\text{O}_2 \xrightarrow{\Delta} 2\text{ZnO} + 2\text{SO}_2 \uparrow$ 4. Not used for dehydrating an ore	1. Heating in the absence of air 2. Carbonate ores are calcinated 3. Carbon dioxide is released $\text{CaCO}_3 \xrightarrow{\Delta} \text{CaO} + \text{CO}_2 \uparrow$ 4. Can be used to drive out moisture from an ore

- **REFINING OF METALS:** Most metals obtained by the reduction process are not very pure. These have to be further refined or purified to obtain pure metal. Purification of the metal is the last step in metallurgy. The most widely used method for refining metal is Electrolytic refining.

Electrolytic refining: Electrolysis can be used for both extractions of metal (which cannot be separated by chemical reduction process) as well as for further purification of metals obtained by any other method. In the electro refining process a block of impure metal is made the anode and a thin sheet of pure metal is made the cathode of an electrolytic cell containing an aqueous solution of the metal salt. When electric current of a suitable voltage is passed, impure metal at the anode gets dissolved to deposit the pure metal at the cathode. Metal ions from the anode enter the electrolyte and the impurities are left behind as anode mud near the anode. The anode finally disintegrates while the cathode gains in weight due to the collection of pure metal. This method is used for refining volatile metals like copper, silver, tin, nickel that have boiling points lower than their impurities. e.g., zinc, mercury, copper etc.

Example: the electrolytic refining of copper.



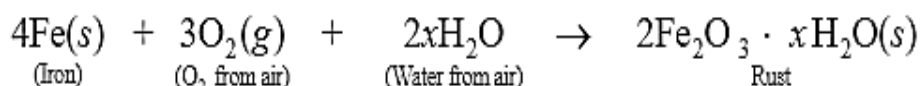
CORROSION OF METALS:

The Natural process of gradual destruction of pure metals by the action of oxygen, moisture or a chemical (such as an acid) on their surface and as a result of chemical reaction or electrochemical reaction between a metal and the surrounding environment to form a stable oxide is called corrosion. When a metal object is kept in damp air for a considerable time, then a powdery substance is formed on its surface. rate of deterioration of metal depends on the environmental conditions, particularly gasses that are in contact with the metal. All small group of metals, called the Noble Metals (rhodium, palladium, silver, platinum, and gold), are much less reactive than others. As a result, they corrode rarely.

The conditions necessary for the corrosion of metals are:

- iv. The presence of atmospheric oxygen.
- v. The presence of moisture (water).
- vi. The presence of a Pure Metal.

Corrosion of Iron Metal: Iron metal corrodes when kept in moist air for a considerable time. When an iron object is kept in moist air for a considerable time, then a red-brown substance called 'rust' ($\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$) is formed on its surface and the process is called rusting of iron. Rust is soft and porous and it gradually falls off from the surface of the iron object and then the iron below starts corroding. So, corrosion of iron is a continuous process that ultimately eats up the whole iron object. A large amount of metal is lost every year because of corrosion. The more reactive a metal is, the faster it is to corrode it.



Corrosion of Aluminium Metal: It can be observed that the aluminium utensils lose their shine and become dull very soon. This is due to aluminium metal's deterioration when exposed to damp air. When a gleaming aluminium metal is exposed to wet air, the oxygen in the air reacts with the metal to generate a thin, dull film of aluminium oxide all over it.

Corrosion of Copper Metal: It can be observed that the copper objects gradually lose their shine due to the formation of a copper oxide layer on them. When the copper metal is exposed to the atmosphere, copper oxidizes which causes the bright copper surfaces to tarnish. After some time, the tarnish turns dark brown or black, and then green.

Types of Corrosion: There are several types of corrosion and the science and understanding of these processes are constantly evolving. Here is a brief overview of some common types of corrosion:

6. **General Attack Corrosion:** This very common form of corrosion attacks the entire surface of a metal structure. It is caused by chemical or electrochemical reactions. While general attack corrosion can cause a metal to fail, it is also a known and predictable issue. As a result, it is possible to plan for and manage general attack corrosion.
7. **Localized Corrosion:** This corrosion attacks only portions of a metal structure.
8. **Galvanic Corrosion:** This can occur when two different metals are located together in a liquid electrolyte such as salt water. In essence, one metal's molecules are drawn toward the other metal, leading to corrosion in only one of the two metals.
9. **Environmental Cracking:** When environmental conditions are stressful enough, some metal can begin to crack, fatigue, or become brittle and weakened.
10. **Microbial corrosion:-** commonly referred to as microbiologically influenced corrosion (MIC) is caused by microorganisms. It applies to both metallic and non-metallic materials with or without oxygen.

How to Prevent Corrosion of Metals: Following are the various methods by which we can protect a metal from corrosion and they are Painting, Greasing or oiling, alloying, galvanizing, electroplating, etc.

Painting: Rusting of iron can be easily prevented by coating the surface with paint which protects iron from air and moisture,

Greasing or oiling: When some grease or oil is applied to the surface of an iron object, then the air and moisture cannot come in contact with it, and hence rusting is prevented.

Alloying: An alloy is a mixture of two or more metals. Alloying is a process where metals like iron or steel are mixed with a less reactive metal like chromium, magnesium, etc for protection against corrosion and to create non-rusting alloys. For e.g. Brass is an alloy which consists of copper is a cheap and non – reactive alloy. Another example of a non-rusting alloy is stainless steel, a mixture of iron and carbon.

Galvanizing: Coating iron or steel with Zinc to prevent corrosion is known as galvanizing. Dipping iron or steel in a liquid form of zinc and using it as the negative cathode zinc is coated on it, the layer is produced by electrolytic deposition. Zinc oxides or corrodes to create a zinc oxide layer that does not flake off like iron oxide rust.

Electroplating: Electroplating is a process where a metal is coated by electrolytic deposition with chromium, silver, or another metal. This process is generally held at room temperature from aqueous electrolytes. It is one of the most popular and common methods to prevent corrosion.

ALLOYS: An alloy is defined as a homogeneous mixture of two or more metals, or a metal and a non metal. i.e. An alloy is prepared by mixing or fusing various metals together in molten state in required proportions and then cooling their mixture to the room temperature. Alloys are more resistant to corrosion and have different properties than its constituent metal. Metals when melted tend to dissolve in one another forming alloys. E.g: Aluminium metal is light but not strong, but an alloy of aluminium with copper, magnesium and manganese is light as well as strong. Since aluminium metal is light but not hard, but an alloy of aluminium with magnesium is called magnalium is light as well as hard.

EXAMPLES OF SOME ALLOYS:

- When a small amount of carbon is mixed with iron, we get an alloy called steel. This alloy of iron at all.
- When iron metal is mixed with other metals like chromium and nickel, we get an alloy called stainless steel, which is strong, tough and does not rust at all.
- When copper is mixed with zinc metal they form an alloy called Brass.
- When copper is mixed with tin metal they form an alloy called Bronze.
- When mercury is mixed with other metal they form an alloy called Amalgam.
- When Gold is mixed with a small amount of silver or copper to make it hard. Otherwise due to soft nature of gold it is not suitable for making jewellery. The purity of gold is expressed in terms to be of 24 carats.

TEXTUAL QUESTIONS:

Q NO 1: Give an example of a metal which :

- (i) is a liquid at room temperature.
- (ii) can be easily cut with a knife.
- (iii) is the best conductor of heat.
- (iv) is a poor conductor of heat.

Answer: (i) Mercury (ii) Sodium (iii) Silver (iv) Lead

Q NO 2: Explain the meanings of malleable and ductile.

Answer:

Malleable: A metal that can be beaten into thin sheets on hammering is called malleable.

Ductile: A metal which can be drawn into thin wires is called ductile.

Q NO 3: Why is sodium kept immersed in kerosene oil ?

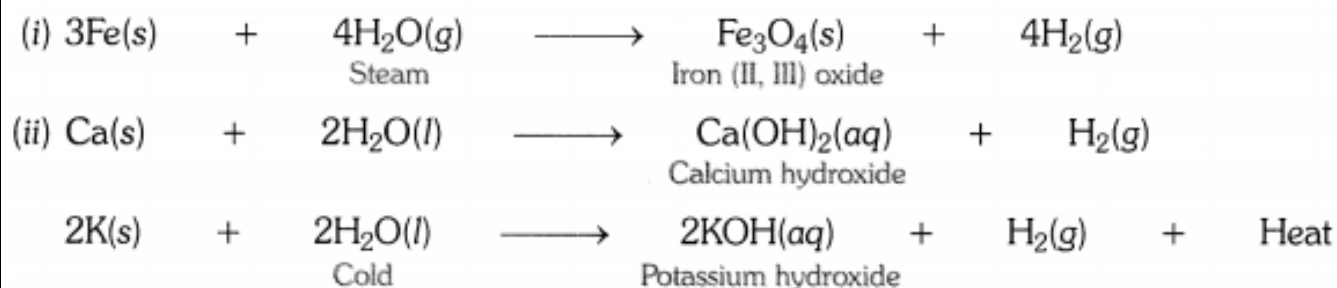
ANS:- Sodium is highly reactive. So it is kept immersed in kerosene oil to prevent its reaction with oxygen, moisture and carbon dioxide of air to prevent accidental fires.

Q NO 4: Write equations for the reactions of

(i) iron with steam.

(ii) calcium and potassium with water.

ANS:



Q NO 5: Samples of four metals A, B, C and D were taken and added to the following solution one by one. The results obtained have been tabulated as follows

Metal	Iron (II) sulphate	Copper (II) sulphate	Zinc sulphate	Silver nitrate
A	No reaction	Displacement		
B	Displacement		No reaction	
C	No reaction	No reaction	No reaction	Displacement
D	No reaction	No reaction	No reaction	No reaction

Use the Table above to answer the following questions about metals A, B, C and D.

- (i) Which is the most reactive metal?
- (ii) What would you observe if B is added to a solution of copper (II) sulphate?
- (iii) Arrange the metals A, B, C and D in the order of decreasing reactivity.

ANS:

(i) B is the most reactive metal because it gives displacement reaction with iron (II) sulphate.

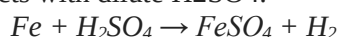
(ii) When metal B is added to copper (II) sulphate solution, a displacement reaction will take place due to which the blue colour of copper (II) sulphate solution will fade and a red-brown deposit of copper will be formed on metal B.

(iii) Metal B is the most reactive because it displaces iron from its salt solution. Metal A is less reactive because it displaces copper from its salt solution. Metal C is still less reactive because it can displace only silver from its salt solution and metal D is the least reactive because it cannot displace any metal from its salt solution. Hence, the decreasing order of reactivity of the metals is $B > A > C > D$.

Q NO 6: Which gas is produced when dilute hydrochloric acid is added to a reactive metal ? Write the chemical reaction when iron reacts with dilute H_2SO_4 .

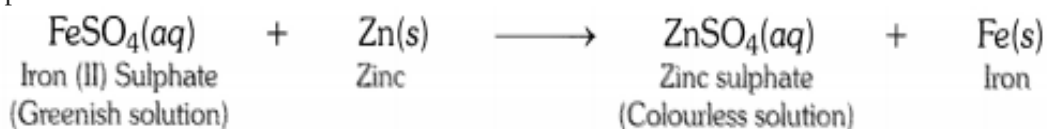
ANS: Hydrogen gas is produced when dilute hydrochloric acid is added to a reactive metal.

Chemical reaction when iron reacts with dilute H_2SO_4 :



Q NO 7; What would you observe when zinc is added to a solution of iron (II) sulphate? Write the chemical reaction that takes place.

ANS: Zinc is more reactive than iron. Therefore, when zinc is added to a solution of iron (II) sulphate, then the greenish colour of iron (II) sulphate solution fades gradually due to the formation of colourless zinc sulphate solution and iron metal is deposited on zinc.



Q NO 8:

(i) Write the electron dot structures for sodium, oxygen and magnesium.

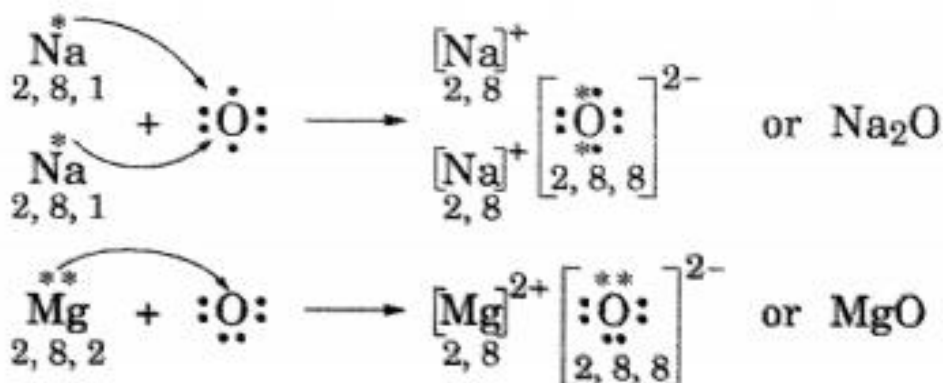
(ii) Show the formation of Na_2O and MgO by the transfer of electrons.

(iii) What are ions present in these compounds?

ANS:

(i) Element	Sodium (Na)	Oxygen (O)	Magnesium (Mg)
Electron dot structure	Na $\cdot$ 2, 8, 1	$\cdot\ddot{\text{O}}\cdot$ $\cdot\ddot{\text{O}}\cdot$ 2, 8, 6	$\ddot{\text{Mg}}$ $\cdot\ddot{\text{Mg}}\cdot$ 2, 8, 2

(ii) Formation of Na_2O and MgO



(iii) What are ions present in these compounds?

In Na_2O , ions present are Na^+ and O^{2-} .

In MgO , ions present are Mg^{2+} and O^{2-} .

Q NO 9: Why do ionic compounds have high melting points?

ANS: The ionic compounds are made up of positive and negative ions. There is a strong force of attraction between the oppositely charged ions, so a lot of heat energy is required to break this force of attraction and melt the ionic compound. Due to this, ionic compounds have high melting points.

Q NO 10: Define the following terms : (i) Mineral, (ii) Ore and (iii) Gangue.

ANS:

(i) **Mineral:** The natural materials in which the metals or their compounds are found in earth are called minerals.

(ii) **Ore:** Those minerals from which the metals can be extracted conveniently and profitably are called ores.

(iii) **Gangue:** The unwanted impurities like sand, rocky material, earth particles, lime stone, mica, etc in an ore are called gangue.

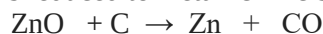
Q NO 11: Name two metals which are found in nature in the free state.

ANS: Gold and platinum

Q NO 12: What chemical process is used for obtaining a metal from its oxide.

ANS:- Reduction process is used for obtaining a metal from its oxide.

For example, zinc oxide is reduced to metallic zinc by heating with carbon.



Besides carbon, highly reactive metals like sodium, calcium, aluminium etc. are used as reducing agents. These displace metals of low reactivity from their oxides.

For example,



Q NO 13: Gold is Metal or Non-metal?

ANS: Gold is a metal found in nature in the free state

Q NO 14: Metallic oxides of zinc, magnesium and copper were heated with the following metals :

	Metal	Zinc	Magnesium	Copper
1.	Zinc oxide	-----	Displacement	-----
2.	Magnesium oxide	-----	-----	-----
3.	Copper oxide	Displacement	Displacement	-----

In which cases will you find displacement reactions taking place?

ANS: A more reactive metal can displace a less reactive metal from its oxide. But out of zinc, magnesium, and copper metals, magnesium is the most reactive, zinc is less reactive whereas copper is the least reactive metal.

Q NO 15: Which of the following pairs will give displacement reactions?

- (a) NaCl solution and copper metal.
- (b) $MgCl_2$ solution and aluminium metal.
- (c) $FeSO_4$ solution and silver metal.
- (d) $AgNO_3$ solution and copper metal.

ANS:- (d) $AgNO_3$ solution and copper metal.

Q NO 16: Which of the following methods is suitable for preventing an iron frying pan from rusting?

- (a) Applying grease
- (b) Applying paint.
- (c) Applying a coating of zinc
- (d) All the above.

ANS: (c) Applying a coating of zinc.

Q NO 17: An element reacts with oxygen to give a compound with a high melting point. This compound is also soluble in water. The element is likely to be

- (a) calcium (b) carbon (c) silicon (d) iron

ANS: (a) Calcium.

Q NO 18: Food cans are coated with tin and not with zinc because

- (a) zinc is costlier than tin
- (b) zinc has a higher melting point than tin
- (c) zinc is less reactive than tin
- (d) zinc is more reactive than tin.

ANS: (d) Zinc is more reactive than tin.

Q NO 19: You are given a hammer, a battery, a bulb, wires and a switch.

- (a) How could you use them to distinguish between samples of metals and non-metals?
- (b) Assess the usefulness of these tests in distinguishing between metals and non-metals.

ANS:

(a) Metals can be beaten into thin sheets with a hammer without breaking. Non-metals cannot be beaten with a hammer to form thin sheets. Non-metals break into pieces when hammered. Metals are malleable, while non-metals are non-malleable. When metals are connected into circuit using a battery, bulb, wires and switch, current passes through the circuit and the bulb glows. When non-metals (like sulphur) are connected, the bulb does not light up at all. Metals are good conductors of electricity.

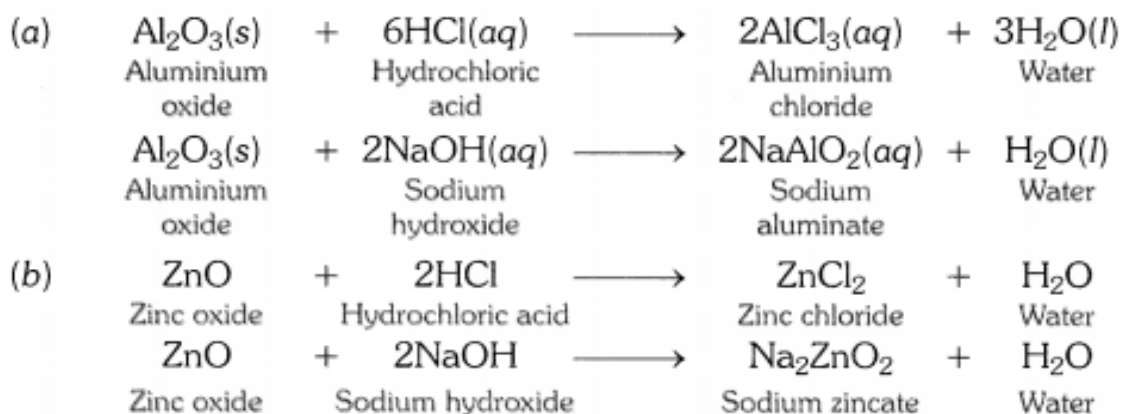
(b) Because of malleability, metals can be casted into sheets. Metals are good conductors of electricity so these can be used for electrical cables.

Q NO 20:- What are amphoteric oxides ? Give two examples of amphoteric oxides ?

OR

Write chemical equations that show aluminium oxide reacts with acid as well as base.

ANS: Those metal oxides which show basic as well as acidic behaviour are known as amphoteric oxides. In other words, metal oxides that react with both acids and bases to form salt and water are called amphoteric oxides. Aluminium oxide and zinc oxide are amphoteric in nature.



Q NO 21: Name two metals which will displace hydrogen from dilute acids and two metals which will not.

ANS:

- (i) Metals above hydrogen in the activity series like sodium and magnesium displace hydrogen from dilute acids.
(ii) Metals below hydrogen in the activity series like copper, silver do not displace hydrogen from dilute acids.

Q NO 22: In the electrolytic refining of a metal M, what would you take as the anode, the cathode and the electrolyte?

ANS:

- Cathode – Pure metal
Anode – Impure metal
Electrolyte – Metal salt solution

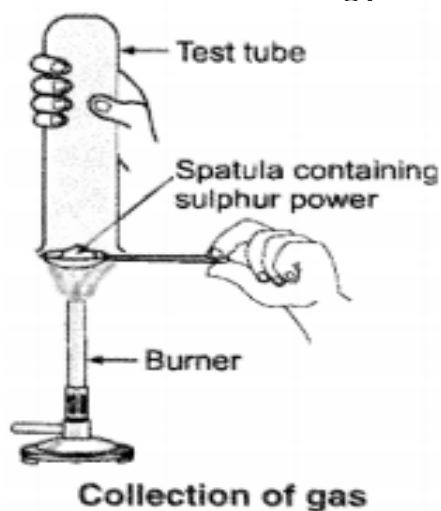
Q NO 23: Pratyush took sulphur powder on a spatula and heated it. He collected the gas evolved by inverting a test tube over it, as shown in the figure.

(a) What will be the action of gas on

(i) dry litmus paper ?

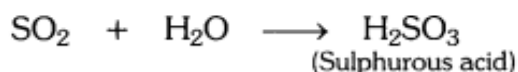
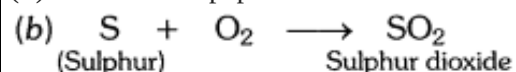
(ii) moist litmus paper ?

(b) Write a balanced chemical equation for the reaction taking place.



ANS:

- (i) Dry litmus paper – no action.
(ii) Moist litmus paper – becomes red.



Q NO 24: Which metals do not corrode easily ?

ANS: Gold and Platinum.

Q NO 25: What are alloys?

ANS:- An alloy is a homogeneous mixture of two or more metals, or a metal and a non-metal. For example, bronze is an alloy of copper and tin.

Q NO 26: State two ways to prevent the rusting of iron.

ANS: Ways to prevent rusting of iron are :

- (a) By painting
- (b) By galvanizing

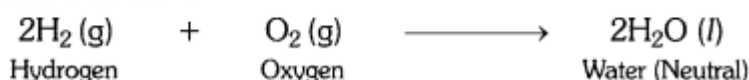
Q NO 27: What type of oxides are formed when non-metals combine with oxygen ?

ANS:- Non-metals combine with oxygen to form acidic oxides or neutral oxides.

Acidic oxide as sulphur dioxide



Neutral oxide as water



Q NO 28: Give reasons:

- (a) Platinum, gold and silver are used to make jewellery.
- (b) Sodium, potassium and lithium are stored under oil.
- (c) Aluminium is a highly reactive metal, yet it is used to make utensils for cooking.
- (d) Carbonate and sulphide ores are usually converted into oxides during the process of extraction.

ANS:

- (a) Platinum, gold and silver are used to make jewellery because these are malleable and ductile. These are highly resistant to corrosion.
- (b) Sodium, potassium and lithium are very reactive and catch fire when exposed to air. This is due to their low ignition temperature and high reactivity.
- (c) Aluminium forms a non-reactive layer of aluminium oxide on its surface. This layer prevents aluminium to react with other substances. That's why aluminium is used to make cooking utensils.
- (d) It is easier to reduce a metal oxide into free metal. Since it is easier to obtain metals from their oxides than from their carbonates or sulphides directly, therefore, the carbonate and sulphide ores are first converted to oxides for extracting the metals.

Q NO 29: You must have seen tarnished copper vessels being cleaned with lemon or tamarind juice. Explain why these sour substances are effective in cleaning the vessels.

ANS: The sour substances such as lemon or tamarind juice contain acids. These acids dissolve the coating of copper oxide or basic copper carbonate present on the surface of tarnished copper vessels and makes them shining red-brown again.

Q NO 30: Differentiate between metal and non-metal on the basis of their chemical properties.

ANS: Difference between metals and non-metals

Metals	Non-metals
(i) Metals form basic oxides or amphoteric oxides.	(i) Non-metals form acidic or neutral oxides.
(ii) Metals replace hydrogen from acids and form salts.	(ii) Non-metals do not replace hydrogen from acids.
(iii) With chlorine, metals form chlorides which are electrovalent.	(iii) With chlorine, non-metals form chlorides which are covalent.
(iv) With hydrogen few metals form hydrides which are electrovalent.	(iv) With hydrogen, non-metals form many stable hydrides which are covalent.

Q NO 31: A man went door-to door posing as a goldsmith. He promised to bring back the glitter of old and dull gold ornaments. An unsuspecting lady gave a set of gold bangles to him which he dipped in a particular solution. The bangles sparkled like new but their weight was reduced drastically. The lady was upset but after a futile

argument the man beat a hasty repeat. Can you play the detective to find out the nature of the solution he has used?

ANS: The dishonest goldsmith dipped the gold bangles in aqua-regia (which contains 1 part of concentrated nitric acid and 3 parts of concentrated hydrochloric acid, by volume). Aqua-regia dissolved a considerable amount of gold from gold bangles and hence reduced their weight drastically. The dishonest goldsmith can recover the dissolved gold from aqua-regia by a suitable treatment.

Q NO 22: Give reasons why copper is used to make hot water tanks and not steel (an alloy of iron).

ANS:

- (i) Copper is a better conductor of heat than steel.
- (ii) Copper does not corrode easily. But steel corrodes easily.
- (iii) Copper does not react with water at any temperature, whereas iron reacts with water on heating.

CARBON & ITS COMPOUNDS

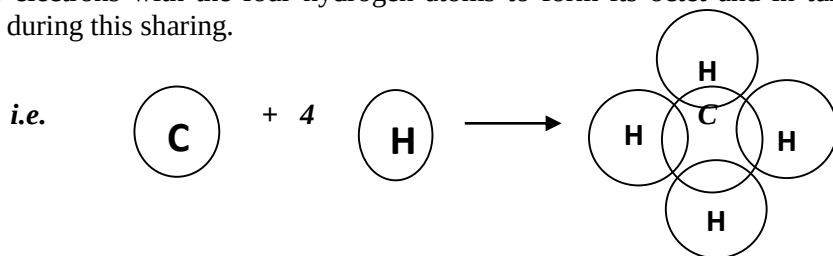
CARBON: Carbon is an element which is of immense significance in both its elemental form and in the combined form. It is a non metal with symbol 'C'. All the living things including human beings, animals, birds, insects, plants etc are made up of carbon compounds like carbohydrates proteins, fats, vitamins, hormones, nucleic acids etc. The compounds of carbon are known as organic compounds. It is the sixth element of the periodic table. The atomic number of carbon is 6. i.e. One atom of carbon contains six electrons with its electronic configuration K-2, L-4. The valency of carbon is 4. The significance of carbon seems to be immense to life in spite of only a small amount of carbon being available in nature. The earth's crust has only 0.02% carbon in the form of minerals (like petroleum, coal, carbonates and hydrogen carbonates) and the atmosphere comprises of only 0.03% of carbon dioxide.

Bonding in Carbon. (Covalent Bond): The atomic number of carbon is 6 and its electronic configuration is K = 2, L = 4. From the electronic configuration of carbon it is clear that its valency is 4. Valency is the combining capacity of an element with other elements to form new compounds. Valency of an element depends upon the number of valence electrons (i.e. number of electrons present in its outermost shell).

From the valency of carbon it is clear that it could either gain four electrons to form C^{4-} ions to make 8 valence electrons in its outermost shell or lose four electrons from its outermost shell to form C^{4+} ions with only two electrons in its first and the last shell. But it is not possible for carbon to form such anion or cation as it would be highly unstable because the nucleus needs a very large amount of energy to hold 10 electrons with only 6 protons or to remove four electrons due to small size, the valence electrons of carbon are strongly held by the nucleus. Hence the only way by which carbon can complete its octet is to share its valence electrons with atoms of other elements so that carbon as well as the other combining element could obtain stability. Such type of sharing of electrons between atoms of different elements is known as covalent bond.

A covalent bond is defined 'as the force of attraction arising due to mutual sharing of electrons between the two atoms.' The combining atoms may share one, two or three pairs of electrons. The compounds formed due to covalent bonding are called covalent compounds. e.g. Formation of methane molecule (CH_4).

The atomic carbon and hydrogen are 6 and 1 respectively. Therefore electronic configuration of carbon is K = 2, L = 4 and that of hydrogen is K = 1. From the electronic configuration of carbon and hydrogen it is clear that hydrogen has 1 electron in its outermost shell and needs four more electrons to complete its octet and on the other hand hydrogen has only 1 electron in its only shell and needs one more to complete its duplet. So in order to complete its octet, carbon shares its four electrons with the four hydrogen atoms to form its octet and in turn the four hydrogen atoms complete their duplets during this sharing.



Properties of covalent compounds: The important properties of covalent compounds are

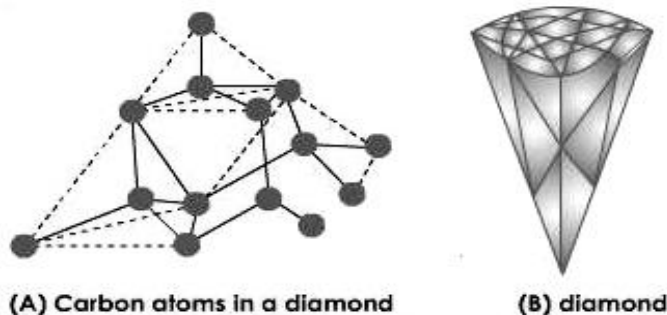
- i. The covalent compounds do not exist as ions but they exist as molecules.
- ii. They exist at room temperature, as liquids or gases. However, a few compounds also exist in the solid state e.g. urea, sugar, etc.
- iii. The melting and boiling points of covalent compounds are generally low.
- iv. Covalent compounds are generally insoluble or less soluble in water and in other polar solvents.
- v. These are poor conductors of electricity in the fused or dissolved state.
- vi. Since the covalent bond is localized in between the nuclei of atoms, it is directional in nature.

ALLOTROPY: When an element exists in different forms with the same chemical properties but different physical properties are known as allotropes of that element and the phenomenon is known as allotropy. The various allotropic forms of carbon can be broadly classified into two classes.

Crystalline forms: Diamond, Graphite & Fullerenes.

Amorphous form: Coal, Coke, Charcoal (or wood charcoal), Animal Charcoal (or bone black), Lamp black, Carbon black, Gas carbon and Petroleum coke.

1. **DIAMOND:** Diamond is the purest form of natural carbon. It occurs as small crystals embedded in rocks known as kimberlite. These are supposed to have been formed by the crystallization of carbon under extreme pressure and temperature in the interior of the earth. Nowadays, synthetic industrial diamonds are being manufactured by subjecting graphite to very high temperatures and pressures. Carbon atoms in diamond have tetrahedral structure. Each atom of carbon is surrounded by four other atoms that together form the tetrahedral structure. South Africa is the largest producer of diamonds in the world and a pioneer in diamond mining. In India too, diamond deposits are found in certain areas especially in Golconda in Andhra Pradesh and Panna in Madhya Pradesh.



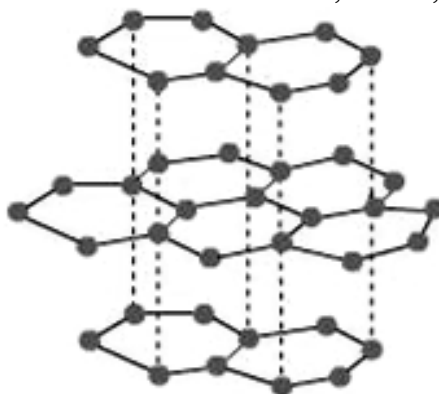
Properties of Diamond:

1. It is the purest form of carbon.
2. It is the hardest substance known.
3. It has very high melting point 3843K and high-density 3.5 g/cm³.
4. It is a bad conductor of electricity, but good conductor of heat.
5. It has a very high refractive index (2.5), it can reflect and refract light. Therefore, it is a transparent substance.
6. It is insoluble in all solvents.
7. It is a colourless solid.
8. If heated strongly without excess of air it changes to Graphite.

Uses of diamond:

- i. It is used in cutting instruments like glass cutters and in rock drilling equipments.
- ii. Diamonds are used for making high precision thermometer.
- iii. Diamonds are used for making jewellery because of their extra brilliance.
- iv. Diamonds are used for making protective window panes for space satellites.
- v. Sharp edged diamonds are used by eye surgeons as a tool to remove the cataract from the eyes with great precision.

2. **GRAPHITE:** Unlike the tetrahedral arrangement of atoms in diamond, the carbon atoms in graphite are arranged in the form of hexagonal rings in layers. Each carbon is bonded to only three other carbon atoms in that layer. Different layers of graphite are held together by rather weak forces. Hence they can slide over one another. This is one reason why graphite scales off easily and can mark impressions on substrates. Because of this property, it is also used as a lubricant. Graphite occurs as a mineral in Sri Lanka, Siberia, Madagascar, etc.



Properties of graphite:

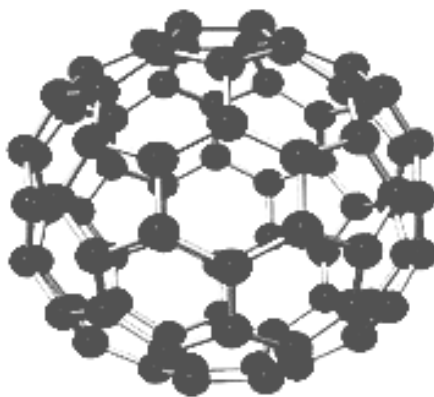
1. It is also the purest form of carbon (90-97%) of carbon)
2. It is soft and good lubricant.
3. It is dark grey having a metallic luster.
4. It is a good conductor of heat and electricity.
5. Its density is 2.25 gcm^{-3}
6. It is insoluble in water and organic solvents.
7. It can mark paper.
8. It burns on strong heating (700°C) to form CO_2 .

Uses of graphite:

- i. Graphite is used in making the 'lead' of pencils.
- ii. It is used in the production of refractory crucibles, which can withstand very high temperature.
- iii. Graphite being a conductor of electricity finds application in making electrodes.
- iv. It is used in making polishes and paints.
- v. Graphite is used as lubricant in machines, which have to be operated at high temperatures. All such machines cannot be lubricated with oils, grease, etc.
- vi. Graphite is extensively used in nuclear reactors, to absorb neutrons. This helps in moderating the nuclear reaction.

3. **FULLERENES:** Fullerenes are allotropes of carbon that were discovered as recently as 1985. They have been found to exist in the interstellar dust as well as in geological formations on earth. They are large cage like spherical molecules with formulae C_{32} , C_{50} , C_{60} , C_{70} , C_{76} , C_{84} etc. The most commonly known fullerene is C_{60} which is named as 'Buckminster fullerene' after the designer of the geodesic dome, American architect Buckminster. C_{60} molecule has marvelously symmetrical structure. It is a fused-ring of aromatic system containing 20 hexagons and 12 pentagons of C atoms. The structure bends around and closes to form a soccer ball shaped molecule. It is therefore, called buckyball also. Fullerene looks different from diamond and graphite. It is a yellow powdery substance, which turns pink on dissolution in solvents like toluene. It polymerizes on exposure to U.V. radiations. Fullerenes are fascinating because they show unusual characteristics and applications like:

- i. They are wonderful lubricants because the balls can roll between the surfaces.
- ii. Alkali compounds of C_{60} (A_3C_{60}) are super conducting materials even at high temperatures of the order of 10-40 K.



The Structure of C-60 Buckminster Fullerene

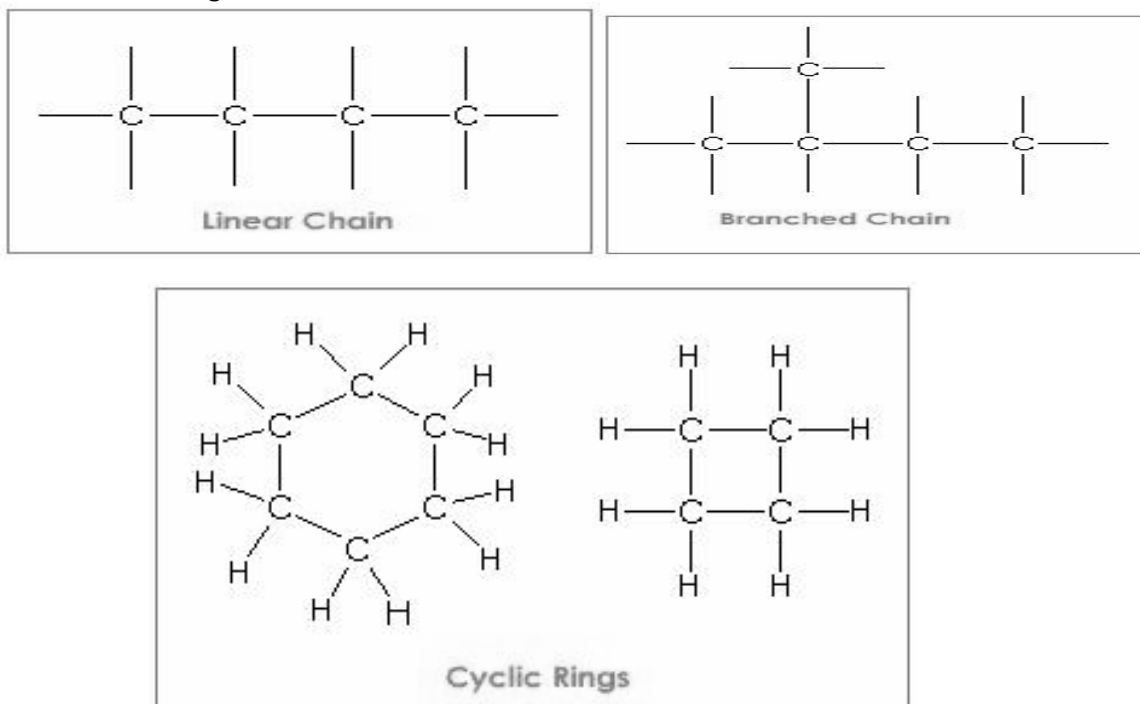
Differences between diamond and graphite

Diamond	Graphite
i. It occurs naturally in Free State.	i. It occurs naturally and is manufactured artificially.
ii. It is the hardest natural substance known.	ii. It is soft and greasy to touch.
iii. It has high relative density (about 3.5).	iii. Its relative density is 2.3.
iv. It is transparent and has high refractive index (2.45).	iv. It is black in colour and opaque.
v. It is non-conductor of heat and electricity.	v. Graphite is a good conductor of heat and electricity.
vi. It burns in air at 900°C to give CO_2 .	vi. It burns in air at $700-800^{\circ}\text{C}$ to give CO_2 .
vii. It occurs as octahedral crystals.	vii. It occurs as hexagonal crystals.
viii. It is insoluble in all solvents.	viii. It is insoluble in all ordinary solvent.

VERSATILE NATURE OF CARBON: Carbon is the element being considered to be a versatile element to give rise to the formation of various new useful compounds. The unique nature of carbon atom and the arrangement of the bond carbon forms with other atoms enable the existence of such a large number of organic compounds. It is because of two main properties of carbon i.e. catenation and tetravalency.

i. **Catenation:** It is the property of self linking of carbon atoms through covalent bonds to form long straight or branched chains and rings of different sizes is called as catenation. It is due to this property of catenation that carbon can combine with other carbon atoms to form straight or branched chains of varying lengths or rings of different sizes. When carbon atoms combine with each other, three types of chains can be formed and are:

- *Straight chain or linear chain*
- *Branched chain*
- *Closed chain or ringed chain.*



The compounds can be branched when the carbon atoms are more than three. The branched chains give rise to a different structure to the parent alkane and are named differently. For example, in pentane (C_5H_{12}) one can see a straight chain (normal pentane), branched chain (Iso-pentane) and ring of cyclopentane (C_5H_{10}).

ii. **Tetravalency:** The valency of carbon is 4 and hence it is capable of bonding four other carbon atoms or atoms of any other mono-valent compound. Due to the small size of, the molecules of carbon atoms can hold its shared pairs of electrons strongly and hence formed the compounds with oxygen, hydrogen, nitrogen, sulphur, chlorine etc. to form stable compounds. Two unique features of carbon stand out due to the property of catenation. i.e. Formation of Strong C-C Covalent Bonds and Formation of C-C Multiple Bonds.

carbon – carbon

carbon = carbon

carbon $\equiv$ carbon

ORGANIC COMPOUNDS: Organic compounds are those chemical compounds having carbon as one of their constituents for e.g: CH_4 , C_2H_6 , C_2H_4 , C_2H_2 , C_2H_5OH . Organic compounds are covalent in nature and they have low melting and boiling points except diamond having low melting point indicates that force of attraction between their molecules are not strong. Some organic compounds conduct electricity whereas most does not.

Organic compounds occur in all living things as in plants and animals also. Initially all the organic compounds were extracted from natural materials obtained from livings and it was thought that the organic compounds could only be formed within a living body under mysterious force called vital force. This vital force theory was disapproved by Freidrich Wohler in 1828 as follows:

Urea is an organic compound prepared by Wohler in the laboratory. This preparation lead to rejection of vital force theory. At present more than five million organic compounds are present and many more new organic compounds are being prepared.

Oxides of carbon like carbon monoxide, carbon dioxide, carbonate, hydrogen carbonates and carbides contain carbon but still they are not considered as organic compounds because their properties are very different from organic compounds. Study about organic compounds is known as organic chemistry. There are many types of organic compounds.

Types of organic compounds:- There are six types of organic compounds and they are:

1. Hydrocarbons:-
2. Haloalkanes
3. Alcohols
4. Aldehydes
5. Ketones
6. Carboxylic acids or organic acids.

1) **Hydrocarbons:** Those Compounds which contains only the atoms of carbon and hydrogen are called hydrocarbons (carbon + hydrogen = hydrocarbon). They are a parent class of organic compounds from which all other classes of hydrocarbon sub groupings of compounds can be made. Examples: Ethane (C_2H_6), hexene (C_6H_{12}), Ethyne (C_2H_2) and benzene (C_6H_6). Natural source of Hydrocarbons is petroleum and natural gas. Hydrocarbons are divided in two groups i.e. saturated and unsaturated hydrocarbons.

1. **Saturated hydrocarbons:** Compounds of carbon and hydrogen whose adjacent carbon atoms contain only one (carbon-carbon) bond are known as saturated hydrocarbons. Their carbon-hydrogen bonds are also single covalent bonds. They are called saturated compounds because all the four bonds of carbon are fully utilised and no more hydrogen or other atoms can attach to it. Thus, they can undergo only substitution reactions. They are also representative of open-chain aliphatic hydrocarbons. These saturated hydrocarbons are called as alkanes. The general formula for alkanes is C_nH_{2n+2} . Where 'n' is the number of carbon atoms in one molecule of one alkane. For example ethane, propane.
2. **Unsaturated hydrocarbons:** Compounds of carbon and hydrogen that contain one double covalent bond between carbon atoms (carbon=carbon) or a triple covalent bond between carbon atoms (carbon $\equiv$ carbon) are called unsaturated hydrocarbons. In these molecules, since all the bonds of carbon are not fully utilized by hydrogen atoms, more of these can be attached to them. Thus, they undergo addition reactions (add on hydrogen) as they have two or more hydrogen atoms less than the saturated hydrocarbons (alkanes). Unsaturated hydrocarbons can be divided into 'alkenes' and 'alkynes' depending on the presence of double or triple bonds respectively.
 - a. **Alkenes:** An unsaturated hydrocarbon in which the two carbon atoms are connected by a double bond is called as an alkene. e.g. ethylene, propylene, butylenes etc. the general formula for alkenes is C_nH_{2n} . Where n is the number of carbon atoms in one molecule of an alkene. For example
 - b. **Alkynes:** Commonly known as acetylene. An unsaturated hydrocarbon in which two carbons are connected by a triple bond is called as alkyne. e.g. Ethyne, propyne, butyne. The general formula for the alkynes is C_nH_{2n-2} . Where n is the number of the carbon atoms in one molecule of an alkyne. For exampleThe compound of alkane or alkene or alkyne can be represented by three types of structures viz:

1. General structure.
2. Electron dot structure.
3. Structural formula.

In alkane two carbon atoms share one pair of electrons among themselves or each carbon atom in alkane share one electron to another carbon to form single covalent bond.

In alkene two carbon atoms share two pair of electrons among themselves or each carbon atom in alkene share two electrons to another carbon to form double bond.

In alkyne two carbon atoms share three pair of electrons among themselves or each carbon atom in alkyne share three electrons to another carbon to form triple bond.

Difference between saturated and unsaturated carbon compounds

Saturated Hydrocarbons	Unsaturated Hydrocarbons
1. These organic compounds contain single carbon-carbon covalent bond. 2. Due to the presence of all single covalent bonds, these compounds are less reactive. 3. Saturated compounds undergo substitution reactions. $\text{CH}_4 + \text{Cl}_2 \longrightarrow \text{CH}_3\text{Cl} + \text{HCl}$ Chloromethane 4. The number of hydrogen atoms is more when compared to its corresponding unsaturated hydrocarbon.	1. These organic compounds contain at least one double or triple covalent bond. $>C=C<$ or $-C\equiv C-$ 2. Due to the presence of double and triple bonds, these compounds are more reactive. 3. Unsaturated compounds undergo addition reactions. $\text{C}_2\text{H}_4 + \text{Cl}_2 \longrightarrow \text{C}_2\text{H}_4\text{Cl}_2$ Ethene 1,2 dichloroethane 4. The number of hydrogen atoms is less when compared to its corresponding unsaturated hydrocarbon.

COAL & PETROLEIUM:

Coal: Coal is a complex mixture of compounds of carbon, hydrogen and oxygen, and some free carbon.

Small amounts of nitrogen and sulphur compounds are also present in coal. It is found in deep coal mines under the surface of the earth. In India, coal is found mainly in Bihar, West Bengal, Orissa and Madhya Pradesh.

Formation of coal: Coal was formed by the decomposition of large land plants and trees buried under the earth millions of years ago. It is believed that millions of years ago, due to earthquakes and volcanoes etc. the forests were buried under the surface of the earth and covered with sand, clay and water. Due to high temperature and high pressure inside the earth, and in the absence of air, wood was converted into coal. The slow chemical process of the conversion of wood into coal under the surface of the earth is called as carbonization. It is a very slow process and may take thousands of years to take place. The different types of coal are Peat – 60% carbon, Lignite (soft coal) – 70% carbon, Bituminous (Household coal) – 80% carbon and Anthracite (Hard coal) – 90% carbon. The different varieties of coal differ in their carbon content, volatile matter and moisture.

Petroleum: The crude oil petroleum is a complex mixture of several solid, liquid and gaseous hydrocarbons mixed with water, salt and earth particles. It is a mixture of compounds. Petroleum is lighter than water and insoluble in water. Petroleum is a natural product obtained from oil wells.

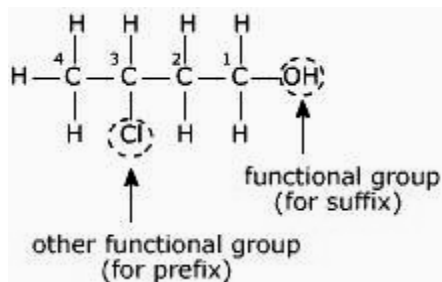
Formation of petroleum: Petroleum oil was formed by the decomposition of the remains of extremely small plants and animals (micro-organisms) buried under the sea millions of years ago. It is believed that millions of years ago, the microscopic plants and animals which lived in seas died and their bodies sank to the bottom of the sea and were soon covered with mud and sand, the chemical effects of pressure, heat and bacteria converted the remains of microscopic plants and animals into petroleum oil. This conversion took place in the absence of oxygen. The petroleum thus formed got trapped between two layers of impervious rocks (non porous rocks) forming an oil trap. In India petroleum is found in Assam, Gujarat, and Mumbai high and in the river basins of Godavari and Krishna.

Functional group:- Functional group is defined as an atom or group of atoms which largely determine the properties of the organic compounds (carbon compounds). While making compounds carbon forms bonds with other elements like halogens, oxygen, sulphur etc. In a hydrocarbon chain one or more hydrogens can be replaced by these elements, such that the valency of carbon remains satisfied. In such compounds the elements, the elements replacing hydrogen are called as heteroatoms. These heteroatoms determine specific properties to the compounds, regardless of the length and nature of the carbon atom chain and hence are known as functional groups.

IUPAC Nomenclature of Carbon Compounds: "Nomenclature is the system of assigning a proper name to a particular carbon compound on the basis of certain rules". With the large growth of carbon compounds, it was necessary to name these compounds in a more systematic way. A committee called the 'International Union for Pure and Applied Chemistry' (IUPAC) put forward a system of giving proper scientific names to carbon based compounds. The names derived by their rules are the names followed all over the world and in short are called IUPAC names. In this system the name of a carbon compound has three main parts as mentioned below:

- Word Root:** This denotes the number of carbon atoms present in a given molecule. For e.g., C₁-Meth, C₂-Eth, C₃-Prop, C₄-But.
- Suffix:** The suffix denotes the type of bonds or the functional group present in the carbon chain, e.g.

Type of bond	Functional group
'ane' (single bond)	'ol' for alcohols (-OH)
'ene' (double bond)	'al' for aldehydes (-CHO)
'yne' (triple bond)	'oic acid' for carboxylic acid (-COOH)
- Prefix:** This denotes the presence of other functional groups and their position.
For e.g., the following compound can be named as:



Word root: But (C₄)

Prefix: 3, chloro

Suffix: -ol

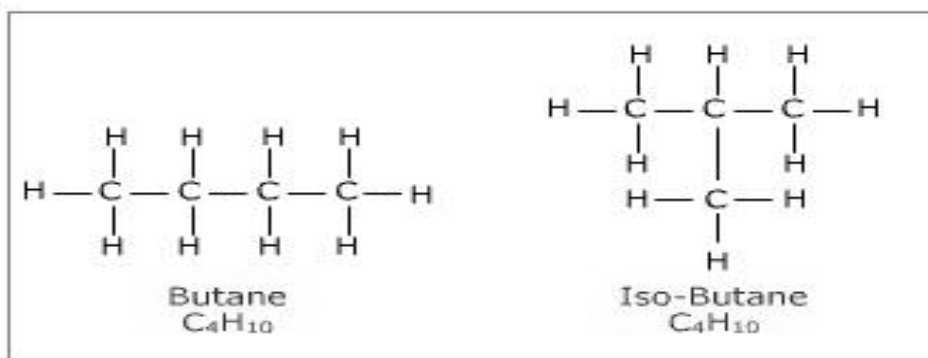
Name: 3-chloro butanol

Carbon atoms are numbered from the side of the functional group (-OH in this case).

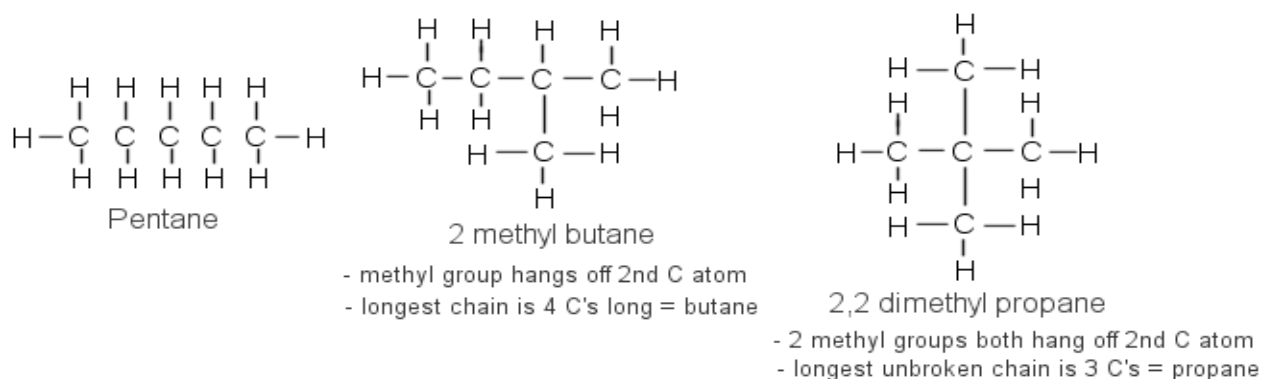
Isomerism & Isomers: Organic compounds are the compounds containing carbon. This carbon in organic compounds can be arranged in many ways to form different structures & hence different compounds. E.g: In organic chemistry, the same molecular formula C₄H₁₀ represents two compounds i.e normal butane & iso butane by arranging carbon in other place.

So, isomers are the organic compounds having same molecular formula but different structures & the phenomenon by which organic compounds show same molecular formula but different structures is known as isomerism.

e.g. Isomerism in butane, where there are following two possible structures for the same molecular formula C₄H₁₀.



Isomerism in pentane, where there are following three possible structures for the same molecular formula C_5H_{12}



Isomerism is only possible in hydrocarbons having four or more than four carbon atoms. So, isomerism is not possible in methane, ethane & propane because they contain less than four carbon atoms. More the number of carbon atoms in a hydrocarbon chain, more the possible isomers can be formed.

E.g:- Pentane can form three isomers, whereas hexane can form five isomers.

Homologous series: All organic compounds are made up of a progressively building chain of carbon atoms with a number of compounds having the same functional groups. Such a series of similarly constituted compounds are called a homologous series. Or we can also say that a homologous series is a group of organic compounds having similar chemical properties in which the successive compounds differ by CH_2 group. Each step of homologous series is called homologues.

Example of homologous series of Alkanes:

Example of homologous series of Alkenes:

Example of homologous series of Alkynes:

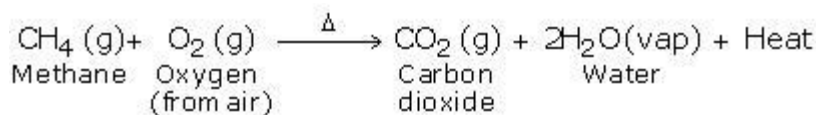
Characteristics of homologous series:

- i. All the members conform to a general molecular formula and have a similar functional group. e.g. General formula for alkanes is C_nH_{2n+2} .
- ii. Each consecutive member differs in the molecular formula by a unit of ' CH_2 '.
- iii. All the members of the series exhibit similar chemical properties, but the extent of the reactions varies with increasing relative molecular mass.
- iv. The physical properties, such as solubility, melting point, boiling point, specific gravity etc. show a gradual change with the increase in their relative molecular masses.
- v. The molecular mass of any two members of analogous series differ by 14 units.

Chemical properties of carbon compounds: The important chemical properties of carbon compounds are

- I. *Combustion Reaction:-*
- II. *Substitution Reaction:-*
- III. *Addition Reaction:-*
- IV. *Substitution Reaction:-*

- I. **Combustion Reaction:-** Combustion means the burning of a carbon compound in air to give CO_2 , Water(vapour), heat & light. It is a process that is highly exothermic i.e., produces a lot of heat. Most of the carbon compounds like the hydrocarbons when burnt in air or oxygen produce large amounts of heat, together with carbon dioxide and water vapor formation. Hence they are used as fuels. For example, methane burns with a blue flame in air.



Alkanes when burn in air, forms a blue, non-sooty flame because the percentage of carbon in alkane is comparatively low which gets oxidized completely by the presence of O₂ that is why they are known as excellent fuels. If O₂ level is not sufficient for burning of carbon compounds. Their incomplete combustion will take place followed by a sooty flame (sooty=black smoke).

The gas stove we use at our homes have tiny holes for air so that sufficient O₂ is available for the complete burning of fuel to produce non-sooty blue flame. Thus when ever flame in a gas stove is blue, then the fuel is burning completely. Sometimes the holes or inlets of gas stove gets blacked followed by reduced O₂ supply then the fuel is not burning completely due to less O₂ supply & we experience this by blackened utensils. So, In a very limited supply of air methane gives carbon black.



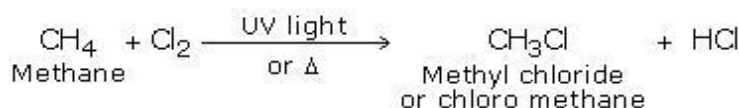
II. Substitution Reaction: The reactions in which one atom or group of atoms in a molecule is replaced or substituted by another atom or group of atoms is called a substitution reaction.

Substitution reaction are only given by only saturated hydrocarbons (alkanes) & not by unsaturated hydrocarbons (alkenes & alkynes).

In substitution reaction by alkanes the hydrogen of the alkane molecule is replaced by some other atom like chlorine & resulting in the formation of the derivatives of hydrocarbon.

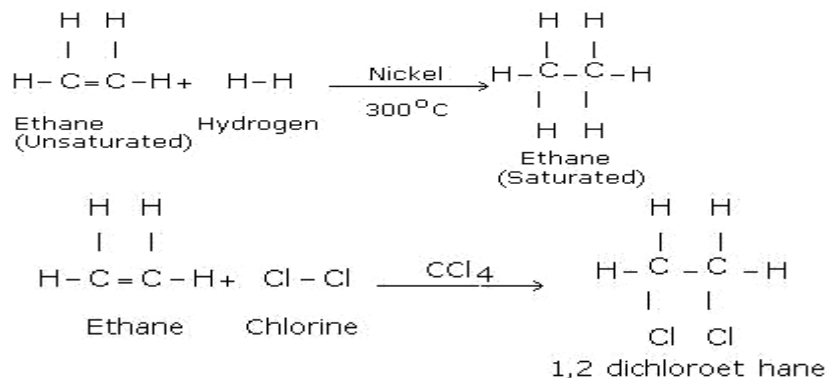
The substitution reaction of alkane with chlorine take place in the presence of sunlight or by heating to a temperature of 250°C to 300°C.

For example,



III. Addition Reaction: The reaction means the reaction in which an unsaturated hydrocarbon (alkene & alkyne) react with other substance to form a single product. Addition reactions are only given by unsaturated compounds (alkenes & alkynes) and not by alkanes.

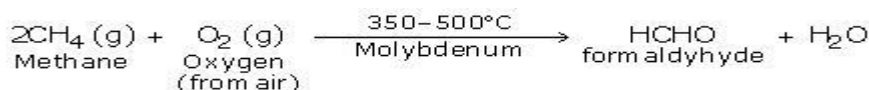
The addition of halogen atom (Cl₂, Br₂ or I₂) to alkene or alkyne in presence of a catalyst (nickel), the double bond or triple bond breaks up & halogen is added there & this process is known as halogenation & addition of hydrogen to alkene or alkyne follows the same process as in case of halogen and its addition is called as hydrogenation.



The process of hydrogenation has a great importance & is used in preparation of vegetable ghee (Vanaspati ghee) from vegetable oil.

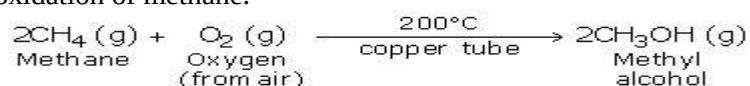
For e.g: The vegetable oils like ground nut oil, cotton seed oil & mustard oil are unsaturated compound containing double bonds. Addition of hydrogen to vegetable oils (unsaturated) leads to formation of vegetable ghee or vanaspati ghee & this reaction is known as hydrogenation of oils.

IV. Oxidation Reaction:- Carbon Compounds undergo oxidation reactions when burnt in air or oxygen. For example, when methane is mixed with oxygen and heated in presence of molybdenum oxide, it gets oxidized to methanal or formaldehyde.



Oxidation of carbon compounds is used as for producing other carbon compounds with different functional groups like alcohol, carboxylic acid, ethers etc. Oxidation is achieved by using an oxygen atmosphere or

oxidizing agents like alkaline KMnO_4 or acidified $\text{K}_2\text{Cr}_2\text{O}_7$. Methanol, an industrial alcohol, for instance, is prepared by the oxidation of methane.



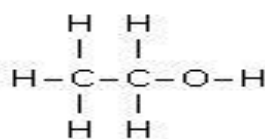
Acetic acid is manufactured by the oxidation of fermented liquors (10-15% alcohol) in air along with the presence of mycoderma aceti. A 3-7% solution of acetic acid is obtained and it is called vinegar.



Ethanol or Ethyl Alcohol ($\text{C}_2\text{H}_5\text{OH}$)

Ethanol is prepared on a large scale from sugar cane. Sugar cane is grinded and is obtained sugar cane juice. This sugar cane juice contains more dissolved sugar in it. This sugar cane juice is used to obtain sugar by the process of crystallization. After the crystallization of sugar from concentrated sugar cane juice, a thick, dark brown liquid called molasses is left behind. This remained molasses still contains about 30% of sugar & this 30% can't be separated by crystallization. So, instead of wastage of this 30% sugar present in molasses, ethanol is prepared by the process of fermentation by using some enzymes.

Structural formula

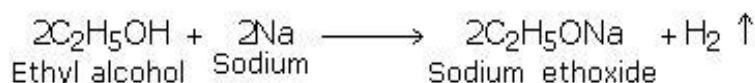


Physical Properties of Ethanol:

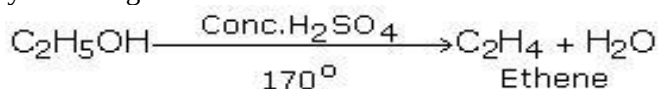
- Ethanol is colorless liquid and has a distinct smell with burning taste.
- Its boiling point is 78°C and its freezing point is -114°C .
- It is soluble in water and almost all the organic solvents.
- It is highly intoxicating in nature.
- It is combustible and burns with a blue flame.
- Ethanol is lighter than water & is soluble in it.
- Ethanol containing water is called rectified spirit (Commercially alcohol or diluted) & 100% pure alcohol is called absolute alcohol (concentrated).
- Ethanol has no effect on any litmus paper because of having neutral nature.

Chemical Properties of Ethanol:

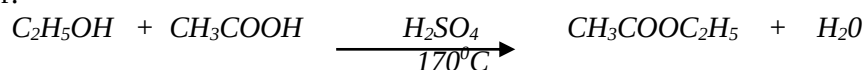
- **Reaction of Ethanol with sodium metal:** When a piece of sodium is dropped in ethanol, sodium ethoxide & bubbles of hydrogen gas is produced. This reaction is a test for presence of alcohol.



- **Reaction of Ethanol with Concentrated Sulphuric Acid / Dehydration:** At 170°C ethanol undergoes dehydration (removal of water) when reacted with concentrated H_2SO_4 to form ethane. Concentrated Sulphuric Acid in this reaction acts as dehydration agent.

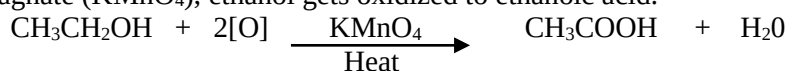


- **Reaction of Ethanol with carboxylic acid:** When ever ethanol reacts with carboxylic acid (ethanoic acid) at 140°C in presence of concentrated sulphuric acid to form a sweet smelling substance called ester (ethyl ethanoate) & water.

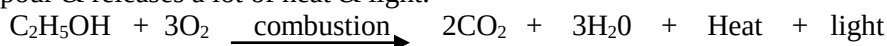


This reaction also indicates the presence of ethanol. The reaction in which a carboxylic acid combines with an alcohol to form an ester is called esterification. Nowadays this ester (sweet smelling substance) is used in cold drinks, sweets, ice creams, artificial perfumes as artificial flavours because they are said to have fruity smell.

- **Reaction of Ethanol with nascent oxygen:-** when ethanol is heated with nascent oxygen in presence of potassium permanganate (KMnO_4), ethanol gets oxidized to ethanoic acid.



- **Reaction of Ethanol with molecular Oxygen / (combustion):** Ethanol is highly inflammable & when reacts with molecular oxygen in presence of heat catches fire easily & burns with a blue flame to form carbon-dioxide, water vapour & releases a lot of heat & light.



Having combustion property, therefore it is also used as fuel in cars in Brazil. Ethanol is mixed with petrol to be used as a fuel in cars. Ethanol is a clean fuel because it doesn't produce any poisonous gas like carbon-monoxide that is why ethanol is mixed to petrol to reduce the production of carbon monoxide gas.

Uses of Ethanol:

- Ethanol is used as an antiseptic to sterilize wounds & syringes in hospitals.
- Ethanol is used in alcoholic drinks (beverages) like whisky (contains 35% of alcohol & rest water) & beer (contains 6% of alcohol & rest water).
- Ethyl Alcohol is used as a solvent for many organic solutes which are insoluble in water.
- It is used in the preparation of perfumes, Soaps & Medicines.
- It is used in the manufacturing of gasohol, which is 90% mixture of petrol (gasoline) and 10% ethanol. It helps to save gasoline.
- Ethanol is used in making tinctures and medical syrups (such as cough syrups).
- It is used as a solvent for paints, varnishes, dyes etc.
- It is used in the production of many organic compounds.

Ethanoic acid (Acetic acid- CH_3COOH)

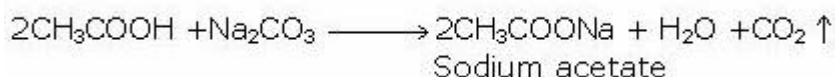
Ethanoic acid or Acetic acid is one of the commonest organic acids and has been known for quite a long time in the form of vinegar. Dilution of about 92% of water to ethanoic acid forms vinegar. It is the second member of homologous series of carboxylic acids. It is also present in a number of fruit juices. In the combined state it occurs in many oils and essential oils. The chemical formula of Ethanoic acid is CH_3COOH .

Physical properties of Ethanoic acid:

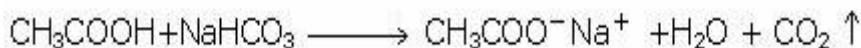
- Acetic acid is a colourless, Sour, corrosive liquid with a pungent smell at ordinary temperatures.
- When ethanoic acid is cooled, it freezes & forms a colourless glacier like solid mass. Due to this property, ethanoic acid is called glacial ethanoic acid or glacial acetic acid.
- It boils at 118°C and its specific gravity is 1.08 at 273K .
- It is miscible with water, alcohol and ether in all ratio's.
- It is a good solvent for phosphorus, sulphur, iodine and inorganic compounds.

Chemical properties of Ethanoic acid:

- Ethanoic acid turns blue litmus paper into red litmus paper. This property shows that ethanoic acid is acidic in nature. This reaction is a test for ethanoic acid.
- Reaction of ethanoic acid with sodium carbonate:-** ethanoic acid when reacts with sodium carbonate to form sodium ethanoate (acetate), carbon dioxide gas & water.



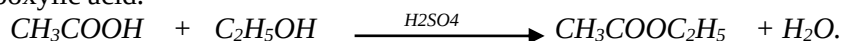
- Reaction of Ethanoic acid with sodium hydrogen carbonate:-** When ever ethanoic acid reacts with sodium hydrogen carbonate to form sodium ethanoate, effervescence of CO_2 & H_2O .



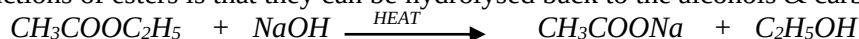
- Reaction of Ethanoic acid with Sodium Hydroxide:-** Sodium hydroxide is a base & ethanoic acid is an acid that means when ever acid reacts with a base, they form salt & water.



- Reaction of Ethanoic acid with alcohols:-** Ethanoic acid when reacts with alcohols like ethanol, formation of ester (sweet smelling substance) will be formed. This reaction takes place in presence of concentrated sulphuric acid. Another name of ester is ethyl ethanoate or ethyl acetate. Formation of ester shows the presence of carboxylic acid.



Reaction of ethanoic acid with ethanol to form an ester is called esterification. One of the most important reactions of esters is that they can be hydrolysed back to the alcohols & carboxylic acid.



By heating the ester with NaOH it gets hydrolysed (break down) to form alcohol & carboxylic acid. The hydrolysis of esters by heat with NaOH is known as saponification (soap formation).

Uses of Ethanoic acid:

- Ethanoic acid is used in the manufacture of dyes, perfumes, rayons, casein, rubber, propanone, medicines & esters.
- It is used to coagulate rubber from latex.. It is used for coagulation.
- Ethanoic acid in the form of dilute (vinegar) is used as a food preservative like in pickles & tomato sauce.
- In the form of salts in medicine and paints.
- In dilute form is used as vinegar and in the concentrated form as a solvent.

SOAPS & DETERGENTS

Dirty hands or dirty body part don't get cleaned by the use of water only. But if we use some substances along with water which makes cleaning very easy. Now, we can say that any substance which has cleaning action in water is called a detergent or detergent is a substance mixed with water which removes dirt. Two types of detergents are present

- Soapy detergent
- Non-soapy detergent.

Nowadays soapy detergent are called as soaps & non-soapy detergents are called as synthetic detergents or detergents. Even though both are used for cleaning.

Soaps (soapy detergent): A soap is also called as soapy detergent which is composed of sodium salt or potassium salt with a long hydrocarbon chain along with a functional group of carboxylic acid (fatty acid). Soap has cleaning property in water. a molecule of soap has a long non-ionic hydrocarbon part & an ionic part of COO^-Na^+ . Soaps are classified on the basis of number of carbon atoms present in hydrocarbon part (non-ionic part) some examples of soaps are.

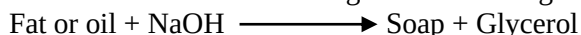
- Sodium stearate soap/ $\text{C}_{17}\text{H}_{35}\text{COO}^-\text{Na}^+$:** Sodium stearate soap is composed of sodium salt ion & a long chain saturated hydrocarbon along with fatty acid. $\text{C}_{17}\text{H}_{35}\text{COO}^-\text{Na}^+$ is also called as stearic acid that is why it is named so.
- Sodium palmitate soap/ $\text{C}_{15}\text{H}_{33}\text{COO}^-\text{Na}^+$:** Sodium palmitate soap is composed of sodium salt ion & a long chain saturated hydrocarbon alongwith fatty acid. $\text{C}_{15}\text{H}_{33}\text{COO}^-\text{Na}^+$ is also called as palmitic acid that is why it is named so.

A soap is actually a salt made upof a strong base namely sodium hydroxide & a weak acid called carboxylic acid. So, net ratio is basic, being basic in nature a soap solution turns red litmus paper to blue litmus paper.

MANUFACTURE OF SOAP:

Soaps can be prepared from animal fats or vegetable Oil. Animal fats & vegetable oils contain long hydrocarbon chain along with carboxylic acid & glycerol (an alcohol).

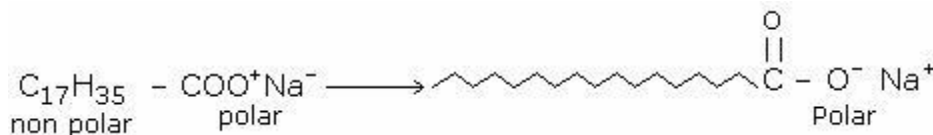
Animal fats or vegetable oil are heated with conc. Sodium hydroxide. This heat splits sodium hydroxide which mixes with animal fat or vegetable oil to form sodium salt of long hydrocarbon along with fatty acid called soap & bi-product is glycerol which came out from vegetable oil during heating.



The process of preparation of soap by spilitting (hydrolysis) of fat or oil with the use of sodium hydroxide (alkali) is called saponification.

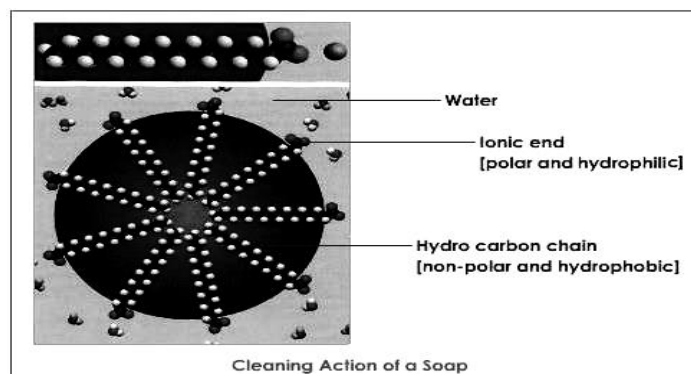
STRUCTURE OF SOAP MOLECULE:

A soap molecule is a tadpole shaped structure made upof two parts, a long hydrocarbon part & a short ionic part (COO^-Na^+). Hydrocarbon paret is hydrophobic that means water repelling & ionic part is hydrophilic that is water loving or attracting. The hydrocarbon part of soap molecule is insoluble in water but is soluble in oil or Greece. Ionic part of soap molecule is soluble in water but is insoluble in oil or Greece.



Cleansing Action of Soaps: When soap is shaken in water it becomes a colloidal solution by shaking or agitating soap water foaming will take place. When a dirty cloth is put in water containing dissolved soap, then the hydrocarbon part of the soap molecules attach to the oil or Greece (dirt) particles present on dirty cloth. (dirt is made upof oil/Greece & dust). Whereas the ionic part (COONa) of the soap molecule are directed outwards & attaches with water. a number of soap or detergent molecule surround or encircle the dirt in a spherical structure called micelle. When the dirty cloth is agitated in soap solution, the oily or greasy particles present on its surface are trapped tightly by soap micelle & gets dispersed in water due to which soap water becomes dirty but the cloth gets cleaned. After that cloth is rinsed in clean water number of times for removal of remained micelle.

Soap is used during cleaning because it binds with Greece or oil particles & puts them out from surface.



Limitations of Soap: Soap is not suitable for washing clothes with hard water (hard water means in which are present calcium & magnesium ions), because soap doesn't give lather easily in hard water. soap molecules in hard water reacts with calcium & magnesium ions present in hard water & form insoluble precipitate called Scum. Instead of reacting with oil or fat molecule. So, by the formation of scum a large amount of soap is wasted & large amount of soap is needed for washing clothes when the water is hard.

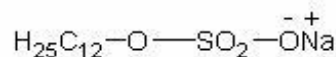
Soft water (water without calcium & magnesium ions) however doesn't contain any ion & therefore lather formation easily takes place.

Detergent: Detergents are also called as soapless soaps or synthetic detergents because they act like soaps in having cleaning property but don't contain the usual soaps like sodium stearate etc. detergents are better effective cleaning agents than soaps even in hard water because they lather more than soaps & don't react with calcium & magnesium ion of hard water & no formation of scum takes place.

A detergent molecule also consists of two parts one part is non-polar long chain of hydrocarbon & benzene which is hydrophobic. Another part is ionic part (SO_3^-Na^+) which is hydrophilic. Cleaning action of a detergent is similar as that of a soap.



Sodium p-dodecyl benzene sulphonate



Sodium lauryl sulphate or
Sodium n-dodecyl sulphate

Advantages of Detergents:

- Synthetic detergents clean effectively and lather well even in hard water and salt water (sea water). There is no scum formation.
- Since detergents are the salts of strong acids they do not decompose in acidic medium. Thus detergents can effectively clean fabric even if the water is acidic.
- Synthetic detergents are more soluble in water than soaps.
- They have a stronger cleansing action than soaps.
- As detergents are derived from petroleum they save on natural vegetable oils, which are important as essential cooking medium.

Disadvantages of Detergents:

- Detergents are surface-active agents and cause a variety of water pollution problems.
- Many detergents are resistant to the action of biological agents and thus are not biodegradable.
- They have a tendency to produce stable foams in rivers that extend over several hundred meters of the river water.
- They tend to inhibit oxidation of organic substances present in wastewaters because they form a sort of envelope around them.

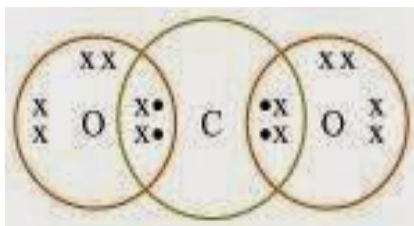
Differences between Soaps and Detergents

<i>Soaps</i>	<i>Detergents</i>
- These are prepared from vegetable oils and animal fats. - They cannot be used effectively in hard water as they produce scum (insoluble precipitates) - These cannot be used in acid solutions. - Their cleansing action is not as strong as that of detergents. - These are biodegradable.	- They are prepared from hydrocarbons of petroleum or coal. - These do not produce insoluble precipitates in hard water. They are effective in soft, hard or salt water. - They can be used even in acid solutions. - Their cleansing action is by surfactants, which is a strong cleansing action. - Some of these are not biodegradable.

TEXTUAL QUESTIONS

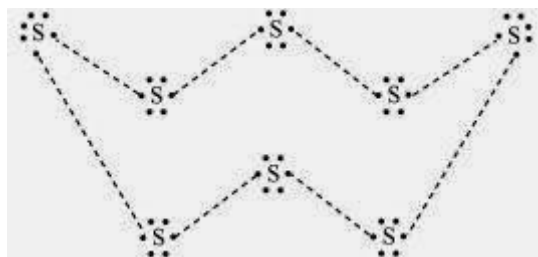
Q NO 1: What would be the electron dot structure of carbon dioxide which has the formula CO_2 ?

ANS:



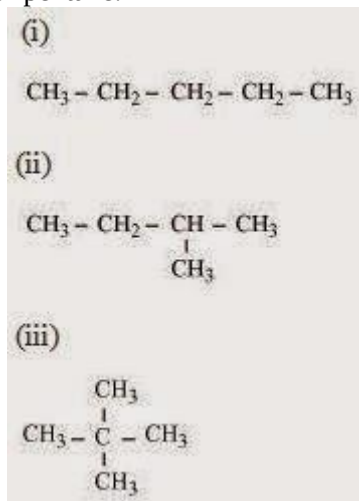
Q NO 2: What would be the electron dot structure of a molecule of sulphur which is made up of eight atoms of sulphur? (Hint - the eight atoms of sulphur are joined together in the form of a ring.)

ANS:



Q NO 3: How many structural isomers can you draw for pentane?

ANS: Three structural isomers are possible for pentane.



Q NO 4: What are the two properties of carbon which lead to the huge number of carbon compounds we see around us?

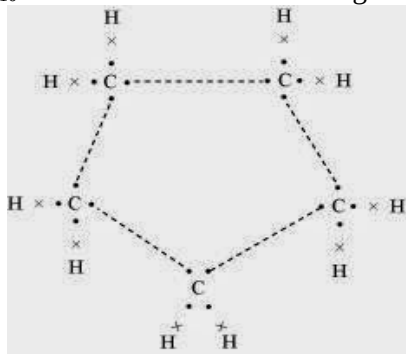
ANS: The two features of carbon that give rise to a large number of compounds are as follows:

→ Catenation - It is the ability to form bonds with other atoms of carbon.

→ Tetravalency - With the valency of four, carbon is capable of bonding with four other atoms.

Q NO 5: What will be the formula and electron dot structure of cyclopentane?

ANS: The formula for cyclopentane is C_5H_{10} . Its electron dot structure is given below.



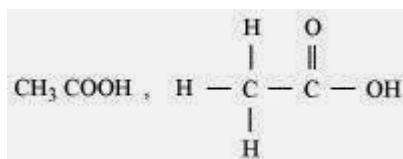
Q NO 6: Draw the structures for the following compounds.

(a) Ethanoic acid (b) Bromopentane*

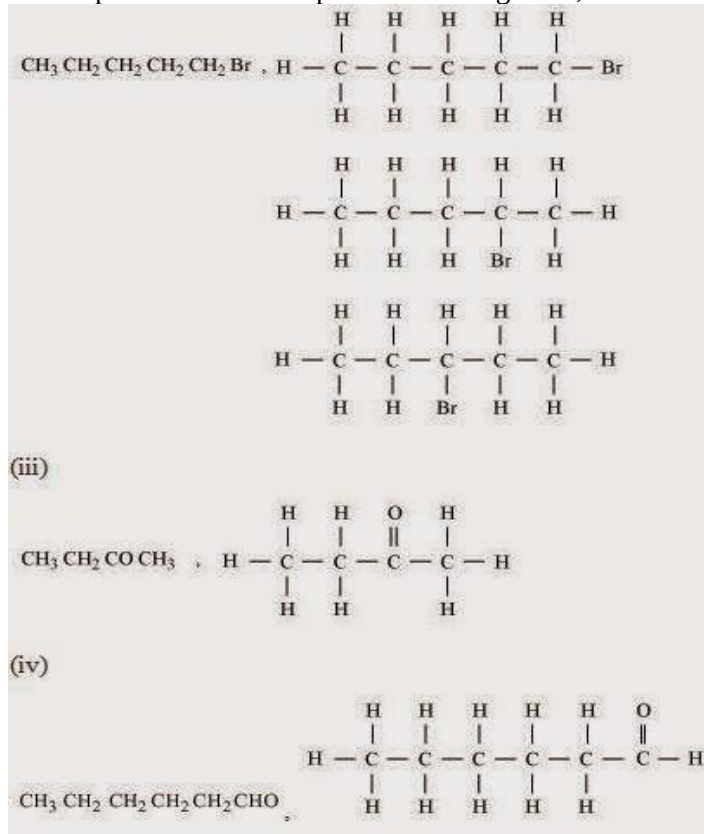
(c) Butanone (d) Hexanal

Are structural isomers possible for bromopentane?

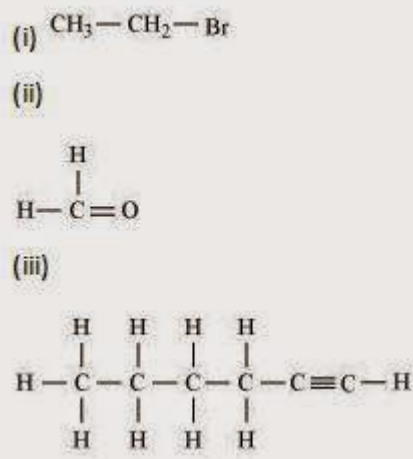
ANS:



(ii) There are many structural isomers possible for bromopentane. Among them, the structures of three isomers are given.



Q NO 7: How would you name the following compounds?



ANS:

(i) Bromoethane

(ii) Methanal (formaldehyde)

(iii) Hexyne

Q NO 8: Why is the conversion of ethanol to ethanoic acid an oxidation reaction?

ANS: $\text{CH}_3\text{CH}_2\text{OH} + (\text{Alkaline KMnO}_4) \rightarrow \text{CH}_3\text{COOH}$

Since, in this reaction one oxygen is added to ethanol, hence it is an oxidation reaction.

Q NO 9: A mixture of oxygen and ethyne is burnt for welding. Can you tell why a mixture of ethyne and air is not used?

ANS: $2\text{HC} \equiv \text{CH} + 5\text{O}_2 \rightarrow 4\text{CO}_2 + 2\text{H}_2\text{O} + \text{Heat}$

When ethyne is burnt in air, it gives a sooty flame. This is due to incomplete combustion caused by limited supply of air. However, if ethyne is burnt with oxygen, it gives a clean flame with temperature 3000°C because of complete combustion. This oxy-acetylene flame is used for welding. It is not possible to attain such a high temperature without mixing oxygen. This is the reason why a mixture of ethyne and air is not used.

Q NO 10: How would you distinguish experimentally between an alcohol and a carboxylic acid?

ANS: We can distinguish between an alcohol and a carboxylic acid on the basis of their reaction with carbonates and hydrogen carbonates. Acid reacts with carbonate and hydrogen carbonate to evolve CO_2 gas that turns lime water milky.

Metal Carbonate/Metal Hydrogencarbonate + Carboxylic acid

Salt + Water + Carbon dioxide

Alcohols, on the other hand, do not react with carbonates and hydrogen carbonates.

Q NO 11: What are oxidizing agents?

ANS: Oxidizing agents are the substances that gain electrons in redox reaction and whose oxidation number is reduced.

Q NO 12: Would you be able to check if water is hard by using a detergent?

ANS: Detergent gives lather with hard and soft water both, while a soap gives lather with soft water only. Thus, it is not possible to check if water is hard; by using a detergent.

Q NO 13: People use a variety of methods to wash clothes. Usually after adding the soap, they 'beat' the clothes on a stone, or beat it with a paddle, scrub with a brush or the mixture is agitated in a washing machine. Why is agitation necessary to get clean clothes?

ANS: A soap molecule has two parts namely hydrophobic and hydrophilic. With the help of these, it attaches to the grease or dirt particle and forms a cluster called micelle. These micelles remain suspended as a colloid. To remove these micelles, it is necessary to agitate clothes.

Q NO 14: Ethane, with the molecular formula C_2H_6 has

(a) 6 covalent bonds.

(b) 7 covalent bonds.

(c) 8 covalent bonds.

(d) 9 covalent bonds.

► (b) 7 covalent bonds.

Q NO 15: Butanone is a four-carbon compound with the functional group

(a) carboxylic acid.

(b) aldehyde.

(c) ketone.

(d) alcohol.

► (c) ketone.

Q NO 16: While cooking, if the bottom of the vessel is getting blackened on the outside, it means that

(a) the food is not cooked completely.

(b) the fuel is not burning completely.

(c) the fuel is wet.

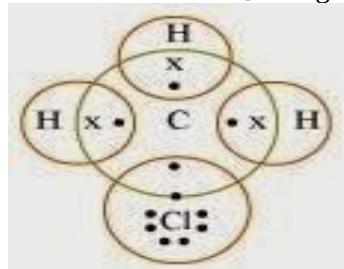
(d) the fuel is burning completely.

► (b) the fuel is not burning completely.

Q NO 17: Explain the nature of the covalent bond using the bond formation in CH_3Cl ?

ANS:

The structure of CH_3Cl is given below:

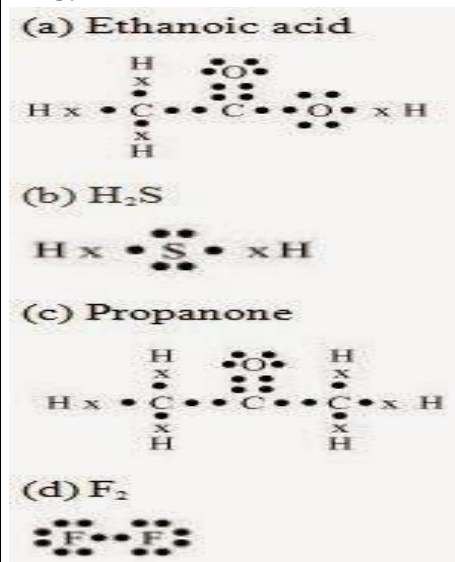


Carbon has four valence electrons. It shares 1 electron each with 3 hydrogen atoms and 1 electron with chlorine. The bond between C and Cl atoms is covalent but due to higher value of electro-negativity of Cl, the C–Cl bond is polar in nature.

Q NO 18: Draw the electron dot structures for

- (a) Ethanoic acid.
- (b) H_2S .
- (c) Propanone.
- (d) F_2 .

ANS:



Q NO 19: What is a homologous series? Explain with an example.

ANS: A homologous series is a series of carbon compounds that have different numbers of carbon atoms but contain the same functional group.

For example, methane, ethane, propane, butane, etc. are all part of the alkane homologous series. The general formula of this series is $\text{C}_n\text{H}_{2n+2}$

Methane CH_4

Ethane CH_3CH_3

Propane $\text{CH}_3\text{CH}_2\text{CH}_3$

Butane $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$

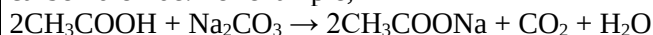
It can be noticed that there is a difference of $-\text{CH}_2$ unit between each successive compound.

Q NO 20: How can ethanol and ethanoic acid be differentiated on the basis of their physical and chemical properties?

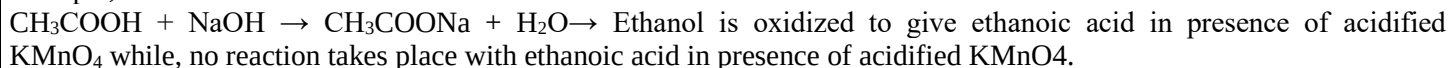
ANS: Ethanol and Ethanoic acid be differentiated on the basis of their following properties:

→ Ethanol is a liquid at room temperature with a pleasant smell. Ethanoic acid has a melting point of 17°C . Since it is below the room temperature so, it freezes during winter. Moreover, ethanoic acid has a smell like vinegar.

→ Ethanol does not react with metal carbonates while, ethanoic acid reacts with metal carbonates to form salt, water and carbon dioxide. For example,



→ Ethanol does not react with NaOH while ethanoic acid reacts with NaOH to form sodium ethanoate and water. For example,



Q NO 21: Why does micelle formation take place when soap is added to water? Will a micelle be formed in other solvents such as ethanol also?

ANS: Soap molecule has two ends. One end is hydrophilic and another end is hydrophobic. When soap is dissolved in water and clothes are put in the soapy solution, soap molecules converge in a typical fashion to make a structure; called micelle. The hydrophobic ends of different molecules surround a particle of grease and make the micelle; which is a spherical structure. In this, the hydrophilic end is outside the sphere and hydrophobic end is towards the centre of the sphere. This is why micelle formation takes place when soap is added to water.

Since ethanol is not as polar as soap, so micelles will not be formed in other solvents such as ethanol.

Q NO 22: Why are carbon and its compounds used as fuels for most applications?

ANS: Carbon and its compounds give large amount of heat on combustion due to high percentage of carbon and hydrogen. Carbon compounds used as fuel have optimum ignition temperature with high calorific values and are easy to handle. Their combustion can be controlled. Therefore, carbon and its compounds are used as fuels.

Q NO 23: Explain the formation of scum when hard water is treated with soap.

ANS: Hard water often contains salts of calcium and magnesium. Soap molecules react with the salts of calcium and magnesium and form a precipitate. This precipitate begins floating as an off-white layer over water. This layer is called scum. Soaps lose their cleansing property in hard water because of formation of scum.

Q NO 24: What change will you observe if you test soap with litmus paper (red and blue)?

ANS: Since soap is basic in nature, it will turn red litmus blue. However, the colour of blue litmus will remain blue.

Q NO 25: What is hydrogenation? What is its industrial application?

ANS: Hydrogenation is the chemical reaction between hydrogen and other compounds in the presence of catalyst. Hydrogenation is used mainly to reduce saturated hydrocarbons. Hydrogenation is an addition reaction.

For Example: When ethene is heated with the catalyst nickel it is reduced to ethane.

Industrial application:

→ >Hydrogenation is used in many industrial applications. For example; in Petrochemical Industry, hydrogenation is used to convert alkenes into alkanes (paraffins) and cycloalkanes.

→ It is also used to prepare vegetable ghee from vegetable oils.

Q NO 26: Which of the following hydrocarbons undergo addition reactions:

C_2H_6 , C_3H_8 , C_3H_6 , C_2H_2 and CH_4 .

ANS: Unsaturated hydrocarbons undergo addition reactions. Being unsaturated hydrocarbons, C_3H_6 and C_2H_2 undergo addition reactions.

Q NO 27: Give a test that can be used to differentiate chemically between butter and cooking oil.

ANS: Butter contains saturated fats. Therefore, it cannot be hydrogenated. On the other hand, oil has unsaturated fats.

That is why it can be hydrogenated to saturated fats (solids).

Q NO 28: Explain the mechanism of the cleaning action of soaps.

ANS: The dirt present on clothes is organic in nature and insoluble in water. Therefore, it cannot be removed by only washing with water. When soap is dissolved in water, its hydrophobic ends attach themselves to the dirt and remove it from the cloth. Then, the molecules of soap arrange themselves in micelle formation and trap the dirt at the centre of the cluster. These micelles remain suspended in the water. Hence, the dirt particles are easily rinsed away by water.